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# **BUILDING UP MATHEMATICS**



**Secondary Education**  
Third Year  
Literature and Humanities Section

**Nazih Mokaddem**

*this manual is the Literature and Humanities version  
of the manual of the Sociology and Economics section*

Educational Center for Research and Development



**Republic of Lebanon**

Ministry of National Education, Youth and Sports

**BUILDING UP  
MATHEMATICS**

**Secondary Education**

Third Year

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Educational Center for Research and Development

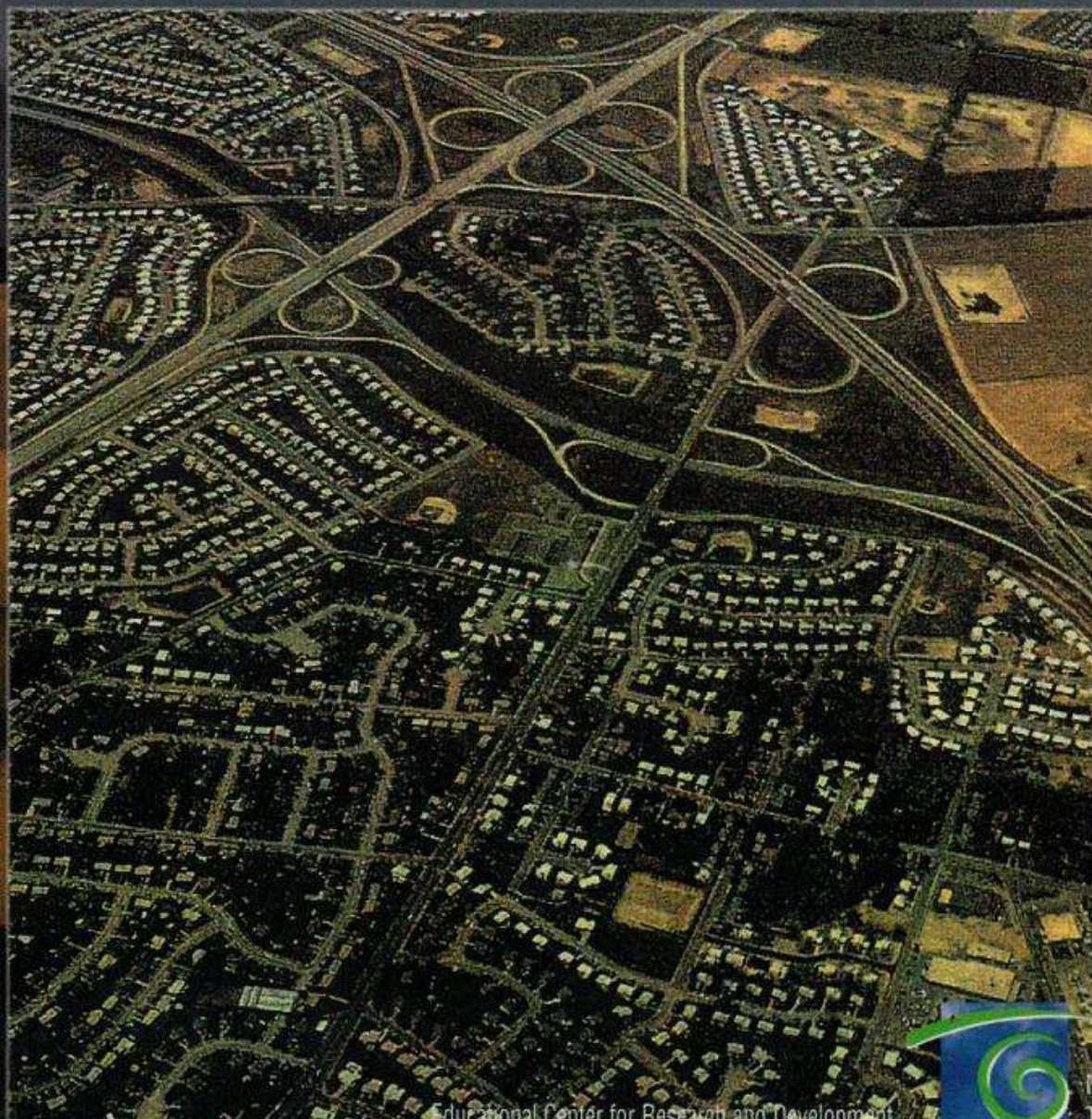


National  
Textbook

**New Curricula**

# *Building up* **MATHEMATICS**

Secondary Education  
Third Year  
Literature and Humanities Section



Educational Center for Research and Development



National  
Textbook

New Curricula



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# HOW TO USE THIS MANUAL

## I. Introduction

Gives a general idea of the theme and fixes the objectives.

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INTRODUCTION TO MATHEMATICAL LOGIC

INTRODUCTION

Logic is the set of the rules, which manages the mental approach in the search of the truth. The roots of logic may be traced to Aristotle, who systematized and codified the subject in a way that was not significantly surpassed for over two millennia. Modern logic, however, stems largely from the work of the German philosopher Gottlob Frege in the late nineteenth century, and has developed tremendously during the twentieth century. Today logic has applications in many areas besides philosophy, including mathematics, linguistics, engineering, and computer science.

Among the various definitions of logic, we will retain the one which summarizes best the contributions of this discipline to mathematics: "Logic is above all the discipline which makes correct inference" (L. Vax). The inference being the reasoning by deduction, an operation that consists to infer from one or many true statements, in each case or many right statements. In other words, logic brings to mathematics rules for correct proofs.

However, sometimes our logical reasoning is faulty and errors can result. Therefore, it is important to identify the laws underlying logical derivations. Basic to logical derivations are propositions, which are statements that are either true or false. Propositions can be combined by various connectives to form compound propositions. There are certain propositions, called tautologies, that are always true. Tautologies give rise to logical implications and logical equivalences. Logical implications are basic to sound reasoning, and logical equivalences provide the means to manipulate propositions algebraically.

The objectives of this chapter are the following:

1. to identify a proposition;
2. to recognize and use the basic logical operations;
3. to use the truth table.

## II. Activities

(Introduction to the subject)  
Prepare the field to construct new concepts by using pre-known information.

ACTIVITIES

**Activity 1. Only sentences**

The philosophy teacher writes on the board the two following groups of sentences:

I -

- "The man is an alive human being"
- "Lebanon is in Europe"
- "Number 5 is even"
- "Joubran Khalil Joubran is a Lebanese philosopher"

## III. Course

(Panoramic idea)  
Gathers up definitions and properties and indicates the main uses and applications.

TEXT

1 GENERAL INFORMATION ON PROPOSITIONS

**Definition**

Any statement that is either true or false is called a **proposition**.

To the question "is this statement true?" One must be able to answer without ambiguity by yes or no.

## IV. FOCUS

(Notions to remember)  
Summarizes the course, indicates the efficient methods and gives useful advice.

FOCUS

• A proposition is a statement that can be true or false.

• To determine the truth value of the negation of a proposition, we use the connective "not". The compound propositions we apply the following rules:

1.  $\neg p$  is true if, and only if,  $p$  is false.
2.  $p \wedge q$  is true if, and only if,  $p$  and  $q$  are both true.
3.  $p \vee q$  is true if, and only if, at least one of the two propositions  $p$  or  $q$  is true.
4.  $p \rightarrow q$  is true if, and only if,  $p$  is false or  $q$  is true. (It is false only when  $p$  is true and  $q$  is false.)
5.  $p \leftrightarrow q$  is true if, and only if,  $p$  and  $q$  are both true or both false. (It is false when one is true and the other is false.)

• A tautology is a proposition which is always true and a contradiction is a proposition which is always false.

• To show that a proposition is true, one often uses, besides the preceding rules, the following principles:

1. The deduction principle:  $p \rightarrow (q \rightarrow r) \rightarrow (p \rightarrow q) \rightarrow r$ .
2. The contraposition rule:  $(p \rightarrow q) \rightarrow (q \rightarrow p)$ .
3. The reasoning by contradiction:  $p \rightarrow (p \rightarrow q) \rightarrow q$ .
4. The contrapositive principle:  $(p \rightarrow q) \rightarrow (q \rightarrow p)$ .

# PREFACE

The idea that Mathematics, Sciences, and Technology have close and profitable connections is widely acknowledged. It is known that economics use the Mathematics at a large scale, and that it was at the origin of what is known currently as Financial Mathematics. For a few decades, the use of Mathematics in the social sciences becomes an increasing extensive curve. These sciences contribute, in their turn, to the development of certain branches of Mathematics, for example, the role played by sociologists in the development of statistics.

Statistics, Probabilities, Financial Mathematics, Analysis, Graphical interpretation, Equations and Logic are the titles of this handbook. These various topics expose the study of Mathematics rigorously, but stressing at the same time the practical and useful aspect of the exposed methods. Our objective is to make the student acquire the mathematical needs so as to continue the studies in various fields, for example, the fields of philosophy, law, history, psychology, communication etc..., and also to practice a number of careers in administration, banks, education, craft industries and other various companies. Our major concern is to present this basic knowledge and to relate it to the daily life problems, hoping that students feel the important role the Mathematics play in the comprehension and analysis of the problems revealed in society, and in making the good decisions to solve these problems.

We have also tried to incorporate in each chapter complete review for the topics treated in former classes; we have proceeded thus to avoid, as much as possible, the use of other handbooks. Part of this review is mentioned in the course, which explains the great emphasis put on this issue, the other part is reviewed in exercises and problems, where the student requires total knowledge of the concerned topic.

We finally wish we have made the task of the student and the teacher as easy and pleasant as possible.

**The authors.**

# The National Textbook Project

By issuing the textbooks for the third year of each educational cycle, the Educational Center for Research and Development will have completed the third and last installment of books called for by the New Curricula. We are placing these books in the hands of students with the great hope that we are moving, step by step, toward the goal of acquiring sound learning, using sophisticated educational means and up-to-date methodology that encourage and reinforce individual thinking and research, acquisition of skills, development of ethical and national attitudes, the feeling of national belonging as well as the feeling of belonging to humanity at large.

The on-going revolution in information, communication and educational-aids technology has undoubtedly limited the role of the textbook and lowered the rank it used so recently to occupy. However, in our society and in many other societies, the textbook remains the basic means of education, and it is our duty to exert our utmost effort and care to come up with the best product as to form and content. Yet we should not lose sight of the fact that the textbook is not sufficient by itself but should rather be used as a stepping stone to access other sources of information. What is important is to keep a clear vision and maintain the right course toward our objective. The means should not turn into the end and the student should always remain the focus of the learning/teaching process.

No one ignores or denies the fact that textbook writing requires very high academic and educational qualifications and very wide field experience. The authors committees undeniably possess such qualities. Yet the textbooks of the last two years contained some negative aspects. Such is the nature of human work, no matter how good the intentions or how great the effort extended. Here constructive criticism constitutes a real contribution to raising the standard of authorship, minimizing errors and filling gaps. We say that, with all due appreciation and respect to all those who have contributed to the success of this project.

The Educational Center for Research and Development is embarking this year on a process of evaluating the New Curricula and related textbooks, teacher training courses and student achievement. This is a natural and necessary step now that the new system has been put into effect. This process aims at identifying the curricular objectives that have been achieved as well as those that have not been achieved, with a view to proceeding with the positive aspects and correcting the negative ones.

As part of this correction process, we plan to review the versions that have been issued in order to secure good textbooks for our students, who always deserve the best.

March 13, 2000

President, Educational Center for  
Research and Development

**Nemer FRAYHA**

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## ❖ Examples

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The following four statements are propositions:

"Beirut is the capital of Lebanon"

"Number 3 is smaller than number 5"

"The area of Lebanon is larger than that of France"

"All the rational numbers are integers"

The first two propositions are true and the last two are false.

Not to confuse a proposition with a judgment (or given proposition) which is true or false according to the observer.  
Ex: - "I am intelligent";  
- "This baby is the most beautiful in the world".

## Remark

The propositions are various. Among the simplest let us quote:

- Propositions for existence:

"There is a school in the village"

- Propositions for membership:

"Nabil is a doctor"

- Propositions for inclusion:

"The German are Europeans"

- Propositions for comparison

"Souad is younger than Souha"

The set of reference used in the statements are implicitly regarded as being well defined.

## Definition

A non-declaratory sentence is a sentence whose contents can be declared neither true nor false.

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## ❖ Examples

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The interrogative sentences: "Where are you going".

The command sentences: "Leave to Italy!".

The wish sentences: "Have a nice trip!.."

- II - "Where do you live?"  
"Read novels"  
"I hope that he behaves well."  
"What a boy"  
Can you find the essential difference between the sentences of each one of these two groups?

It is to find the difference on the level of the contents and not of the form of these sentences.

## Activity 2.

Here are three statements which we suppose true concerning our friend Fadi :

- Fadi is a trader or a poet;
- Fadi is courageous and generous;
- Fadi is poor.

By supposing that the following statements are true:

- Any businessman is rich,
- Every trader is a businessman

Can you determine the qualities of Fadi?

## Activity 3. She is right!.... It is logical

Salam says to Mounir " if  $m = n$  then  $m + 1 = n + 1$ ".

Mounir approves. Salam says then "if  $3 = 4$  then  $5 = 6$ ". Mounir protests and says that Salam is wrong since  $5 \neq 6$ . The Math teacher interferes and says that Salam was right in both cases. Can you explain why?

# TEXT

## I GENERAL INFORMATION ON PROPOSITIONS

### Definition

Any statement that is either true or false is called a **proposition**.

To the question "is this statement true?" One must be able to answer without ambiguity by yes or no.

## HOW TO USE THIS MANUAL

### V. EXERCISES

(Preliminary steps)

Help to assimilate the results, to master the new techniques and to solve direct application problems.

#### EXERCISES

1. 8 students obtained in an exam the following grades: 37, 65, 52, 49, 50, 69, 70, 62. Put these grades in increasing order and find the range of the data.
2. Fouad has paid, during the last six months, the following telephone bills: 35 500 LL, 33 400 LL, 55 000 LL, 29 300 LL, 45 300 LL. Find the range of this data.
3. The minimum temperatures registered during 8 consecutive days were: 20, 15, 18, 19, 20, 23, 22, 21. Find:  
a) the sum of the deviations from the mean;  
b) the absolute deviation from the mean.
4. The following given represents the waiting time (in minutes) of 10 clients in a bank: 10, 8, 7, 15, 14, 5, 10, 17, 9, and 11. Find:  
a) the range;  
b) the absolute deviation from the mean;  
c) the variance;  
d) the standard deviation.
- 5.

### VI. Self evaluation

(An applicable test)

Allows the students to control by themselves their learning and their capacities to use the new results and techniques.

#### SELF-EVALUATION

1. Answer the following propositions by TRUE or FALSE.

|                                                                                                                              | TRUE | FALSE |
|------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 1. The range of a statistical data is the difference between the standard deviation and the average.                         |      |       |
| 2. Two different data sets can have the same average.                                                                        |      |       |
| 3. Small standard deviation indicates that the values are widely dispersed.                                                  |      |       |
| 4. The sum of the deviations from the mean is null.                                                                          |      |       |
| 5. We can calculate the absolute deviation from the mean, the variance and the standard deviation without using the average. |      |       |
| 6. A statistical data of equal modalities has a zero standard deviation.                                                     |      |       |

### VII. Problems

(Analysis and syntheses)

Help to use the new notions simultaneously with the old ones to solve application problems taken from diversified quotidian situations.

#### PROBLEMS

1. We give the class centers and the relative frequencies of two different distributions in the following tables:

| Data A       |           | Data B       |           |
|--------------|-----------|--------------|-----------|
| Class center | Frequency | Class center | Frequency |
| 18           | 43        | 2            | 1         |
| 23           | 36        | 5            | 6         |
| 28           | 25        | 8            | 11        |
| 33           | 14        | 11           | 23        |
| 38           | 8         | 14           | 16        |
| 43           | 16        | 17           | 7         |
| 48           | 29        | 20           | 4         |
| 53           | 33        | 23           | 1         |
| 58           | 41        |              |           |

Calculate the average, the variance and the standard deviation of each of the two distributions.

### VIII. Just for fun

(To go further)

Gives the possibility to think about notions which prolong the course and which are useful and pleasant to know.

#### JUST FOR FUN

Two teachers of a class of 30 students, exchange their experiences after 40 years. The French teacher expresses his satisfaction: the rate his students did very well in their French exams, because the class average is 12/20 and the success rate is 80%. The Greek teacher expresses his surprise: the class average of the students from his class is 12/20. The average is 12/20 but the success rate is only 40%. Knowing that the passing mark in each subject is six marks (= 10/20), give two hypotheses of grades showing these two statistics. Justify your answer by using the statistical techniques.

## INTRODUCTION

Logic is the set of the rules, which manages the mental approach in the search of the truth. The roots of logic may be traced to Aristotle, who systemized and codified the subject in a way that was not significantly surpassed for over two millennia. Modern logic, however, stems largely from the work of the German philosopher Gottlob Frege in the late nineteenth century, and has developed tremendously during the twentieth century. Today logic has applications in many areas besides philosophy, including mathematics, linguistics, engineering, and computer science. Among the various definitions of logic we will retain the one which summarizes best the contribution of this discipline to mathematics: "Logic is above all the discipline which treats correct inference" (L. Vax). The inference being the reasoning by deduction, an operation that consists to refer to one or many true statements, to reach one or many right statement(s). In other words, logic brings to mathematics rules for correct proofs.

However, sometimes our logical reasoning is faulty and errors can result. Therefore, it is important to identify the laws underlying logical derivations. Basic to logical derivations are propositions, which are statements that are either true or false. Propositions can be combined by various connectives to form compound propositions. There are certain propositions, called tautologies, that are always true. Tautologies give rise to logical implications and logical equivalencies. Logical implications are basic to sound reasoning, and logical equivalencies provide the means to manipulate propositions algebraically.

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## ACTIVITIES

### Activity 1. Only sentences

The philosophy teacher writes on the board the two following groups of sentences:

- I - "The man is an alive human being"  
"Lebanon is in Europe"  
"Number 5 is even"  
"Joubran Khalil Joubran is a Lebanese philosopher"

## II

## LOGICAL OPERATION

### A- Conjunction

To the propositions  $p$  and  $q$  we associate the proposition  $r = p \wedge q$  (read  $p$  and  $q$ ) which is, by definition, true if, and only if,  $p$  and  $q$  are true at the same time and false in other cases.

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| $T$ | $T$ | $T$          |
| $T$ | $F$ | $F$          |
| $F$ | $T$ | $F$          |
| $F$ | $F$ | $F$          |

This table shows that if at least one of the two propositions  $p$  or  $q$  is false then  $p \wedge q$  is false.

#### ❖ Examples

1°)  $p$  : "Karine is a student"  
 $q$  : "Karine is a Lebanese"  
 $p \wedge q$  : "Karine is a Lebanese student"

2°)  $p$  : " $a > 4$ ";  $q$  : " $a \leq 8$ "  
 $p \wedge q$  : " $4 < a \leq 8$ "

3°)  $p$  : " $ABC$  is a triangle"  
 $q$  : "angle  $BAC$  is right"  
 $p \wedge q$  : " $ABC$  is a right triangle at  $A$ "

### B- Disjunction

To the two propositions  $p$  and  $q$  we associate the proposition  $r = p \vee q$  (read  $p$  or  $q$ ) which is by definition, true if, and only if, at least one of the propositions  $p$  or  $q$  is true and false if both  $p$  and  $q$  are false.

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| $T$ | $T$ | $T$        |
| $T$ | $F$ | $T$        |
| $F$ | $T$ | $T$        |
| $F$ | $F$ | $F$        |

This table shows that:  $p \vee q$  is false if, and only if,  $p$  and  $q$  are both false.

## Truth value

We associate with each proposition  $p$  the letter  $T$  if it is true and the letter  $F$  if it is false. The letter  $T$  or  $F$  is called the truth value of the proposition  $p$ .

A table in which one indicates the values of truth of a proposition obtained from one or several other propositions is known as a truth table.

## Remark

By definition even the value of the proposition obeys the two following principles:

**Principle of the third excluded:** A proposition is either true or false and nothing else.

**Principle of non-contradiction :** A proposition cannot be true or false at the same time.

A hypothetical proposition does not obey these two principles. It can be true for some false for others and even neither true nor false for the rest.

## Negation of a proposition

To the proposition  $p$  we associate the proposition  $\neg p$  (read not  $p$ ) called the **negation** of  $p$ . It is defined using the truth values in the following way:

$\neg p$  is true if  $p$  is false

$\neg p$  is false if  $p$  is true.

| $p$ | $\neg p$ |
|-----|----------|
| $T$ | $F$      |
| $F$ | $T$      |

Truth table of the proposition  $\neg p$

### ❖ Examples

1°) If  $p$  is the proposal "Nadine is Lebanese",  $\neg p$  is the proposition "Nadine is not Lebanese".

2°) If  $p$  is the proposition " $a < 5$ ",  $\neg p$  is the proposition " $a \geq 5$ ".

Attention: " $a > 5$ " is different than  $\neg p$ .

## Reasoning by contradiction

### Definition

A proposition is a **contradiction** if it is always false

Ex :  $q = p \wedge (\neg p)$

| $p$ | $\neg p$ | $p \wedge (\neg p)$ |
|-----|----------|---------------------|
| $T$ | $F$      | $F$                 |
| $F$ | $T$      | $F$                 |

### Demonstration by contradiction

To show that a proposition  $p$  is true, we show that if the negation of  $p$  ( $\neg p$ ) is true one has a contradiction.

In other words we show that  $(\neg p \Rightarrow q)$  is true and  $q$  is a contradiction.

An example of reasoning by absurdity is the following:  
Let us admit that any writer is cultivated. However, Karim is not cultivated so Karim is not a writer. Because if Karim were a writer he would be cultivated, which is contradictory to the fact that he is not.

### ❖ Example

It is known that:

Lina and Mona have together more than \$100.

Lina and Fadia have together less than \$50.

Knowing that one of the three girls has \$100, find the girl.

It is of course Mona since if it is one of the others we get a contradiction.

## D- Equivalence (biconditional proposition)

If  $p$  and  $q$  are two propositions the proposition  $(p \Rightarrow q) \wedge (q \Rightarrow p)$  is denoted  $(p \Leftrightarrow q)$  (read  **$p$  is equivalent to  $q$**  or  $p$  if, and only if  $q$ ). It is true if, and only if,  $p$  and  $q$  are simultaneously true or simultaneously false.

To show that  $(p \Leftrightarrow q)$  is true it is enough to show that if  $p$  is true then  $q$  is true and that if  $q$  is true then  $p$  is also true.

| $p$ | $q$ | $p \Rightarrow q$ | $q \Rightarrow p$ | $p \Leftrightarrow q$ |
|-----|-----|-------------------|-------------------|-----------------------|
| $T$ | $T$ | $T$               | $T$               | $T$                   |
| $T$ | $F$ | $F$               | $T$               | $F$                   |
| $F$ | $T$ | $T$               | $F$               | $F$                   |
| $F$ | $F$ | $T$               | $T$               | $T$                   |

$(p \Leftrightarrow q)$  is false if, and only if, one of the propositions  $p$  or  $q$  is true and the other is false.

## Definition

The proposition " $p \Rightarrow q$ " or  $(p \Rightarrow q)$  (which is read  **$p$  implies  $q$**  or **if  $p$  then  $q$** ) is by definition the proposition " $(\neg p) \vee q$ ".

The truth table of this new proposition is thus:

| $p$ | $q$ | $\neg p$ | $(\neg p) \vee q$ | $(p \Rightarrow q)$ |
|-----|-----|----------|-------------------|---------------------|
| $T$ | $T$ | $F$      | $T$               | $T$                 |
| $T$ | $F$ | $F$      | $F$               | $F$                 |
| $F$ | $T$ | $T$      | $T$               | $T$                 |
| $F$ | $F$ | $T$      | $T$               | $T$                 |

The truth table of the proposition  $p \Rightarrow q$  shows that it is false if, and only if,  $p$  is true, and  $q$  is false at the same time.

To show that the proposition  $(p \Rightarrow q)$  is true it is enough to show that if the proposition  $p$  is true then the proposition  $q$  is true.

Instead of saying that the proposition  $(p \Rightarrow q)$  is true one can say that if the proposition  $p$  is true then the proposition  $q$  is true or more simply that the proposition  $p$  implies the proposition  $q$ . One can then write  $p \Rightarrow q$  without quotation marks or brackets.

Notice the analogy with the conditional sentence which stipulates that if a proposition  $p$  is true then the proposition  $q$  is true.

Ex: - If I study then I pass.  
- If  $a > 4$  then  $a > 2$ .

## Principle of deduction

The principle of deduction is schematized as follows:  
if  $(p \Rightarrow q)$  is true and if  $p$  is true then  $q$  is true.

### ❖ Examples

1°) We know that every human being is mortal; however I'm a human, therefore I am mortal.

2°) If any element of  $F$  is an element of  $E$  ( $F \subset E$ ) and if  $a$  is an element of  $F$  ( $a \in F$ ), then  $a$  is an element of  $E$  ( $a \in E$ ).

## ❖ Examples

1°)  $p$ : "Charles is a singer"

$q$ : "Charles is a composer"

$p \vee q$ : "Charles is a singer or a composer"

Notice that  $p \vee q$  is true if Charles is a singer, if Charles is a composer, or if Charles is a singer and composer.

2°) Let  $E$  be a set and  $F$  and  $G$  two subsets of  $E$  and let  $a \in E$ .

$p$ : ( $a \in F$ );  $q$ : ( $a \in G$ )

$p \vee q$ : [ $(a \in F) \vee (a \in G)$ ] which is ( $a \in F \cup G$ ).

Notice that  $p \vee q$  is true if  $a \in F \cap G$  which presupposes that  $F \cap G$  is non empty.

Concerning these two examples one can carry the following reasoning:

1°) It is known that Charles is a singer or a composer. Then, if Charles is not a singer he is a Composer.

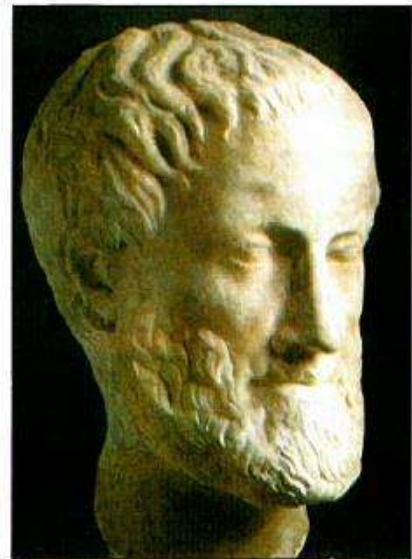
2°) If the proposition [ $(a \in F) \vee (a \in G)$ ] is true and if  $a \notin F$ , then  $a \in G$ .

## Remark

To the two propositions  $p$  and  $q$  concerning the same object, the philosophers associate another proposition known as exclusive disjunction of  $p$  and  $q$  denoted by " $p$  or  $q$ " which presupposes that  $p$  and  $q$  cannot be carried out at the same time (to be true at the same time). This kind of disjunction leads to the following reasoning (which is not valid for mathematical disjunction):

It is known that Riad is a doctor or a lawyer. However, Riad is a lawyer, therefore he is not a doctor.

These two concepts coincide if the two propositions  $p$  and  $q$  cannot be true at the same time. Ex: The following reasoning: (it is known that it is daylight or it is night. However, it is night, therefore it is not daylight) is valid since it cannot be daylight and night at the same time.



Aristotle (384 - 322 av. J.C.)

## C- Implication (conditional proposition)

In logical arguments the "If...then" construct is very important.

This construct expresses the conditional.

## Fundamental examples

1°) If  $p$  is a proposition, then  $p \vee (\neg p)$  is a tautology

Thus if  $a$  is an element of the set  $E$  and  $F$  a subset of  $E$  then the proposition

" $(a \in F) \vee (a \notin F)$ " is a tautology.

Ex: - "The number 95 is divisible by 3 or non divisible by 3".

- "Ziad may or may not sit for his bac at the end of the year".

| $p$ | $\neg p$ | $p \vee (\neg p)$ |
|-----|----------|-------------------|
| $T$ | $F$      | $T$               |
| $F$ | $T$      | $T$               |

A proposition  $p$  is a tautology if, and only if  $\neg p$  is a contradiction.

2°) The proposition " $p \Leftrightarrow \neg(\neg p)$ " is a tautology.

Indeed if  $p$  is a proposition, then  $p$  and  $\neg(\neg p)$  have simultaneously the same values of truth.

Ex: - Let  $E$  be a set,  $a \in E$  and  $A \subset E$ .

The proposition " $(a \in A) \Leftrightarrow (a \notin A)$ " is a tautology.

- The proposition "Marc is alive if, and only if, he is not dead" is a tautology.

| $p$ | $\neg p$ | $\neg(\neg p)$ | $p \Leftrightarrow \neg(\neg p)$ |
|-----|----------|----------------|----------------------------------|
| $T$ | $F$      | $T$            | $T$                              |
| $F$ | $T$      | $F$            | $T$                              |

## 3°) Contraposition

### Definition

If  $p$  and  $q$  are two propositions, we call the **contrapositive** of the proposition  $(p \Rightarrow q)$  the proposition  $(\neg q \Rightarrow \neg p)$ .

### Property

The proposition " $(p \Rightarrow q) \Leftrightarrow [(\neg q) \Rightarrow (\neg p)]$ " is a tautology.

We show this property using a truth table.

To show that a proposition of the form  $(p \Rightarrow q)$  is true, we show that the proposition  $(\neg q \Rightarrow \neg p)$  is true.

Ex: Let  $a$  and  $b$  be two strictly positive real numbers. To prove that the proposition  $[(a < b) \Rightarrow (a^2 < b^2)]$  is true (which is the same to show that  $a < b \Rightarrow a^2 < b^2$ ), one shows that the proposition  $[(a^2 \geq b^2) \Rightarrow (a \geq b)]$  is true (which is the same to show that  $a^2 \geq b^2 \Rightarrow a \geq b$ ).

However,  $a^2 \geq b^2 \Rightarrow a^2 - b^2 \geq 0 \Rightarrow (a - b)(a + b) \geq 0$

$\Rightarrow a - b \geq 0 \Rightarrow a \geq b$  because  $a + b > 0$

The same,  $ABC$  being a triangle, we can show that if  $ABC$  is a right triangle then it is not equilateral while proceeding as follows:

$ABC$  is equilateral  $\Rightarrow$

$\hat{A} = \hat{B} = \hat{C} = 60^\circ \Rightarrow ABC$  is not a right triangle neither at  $A$  nor at  $B$  and nor at  $C$ .

## Remark

To a proposal of the form  $(p \Rightarrow q)$  we associate, other than contraposition  $(\neg q \Rightarrow \neg p)$ , the **converse** proposal  $(q \Rightarrow p)$  and the **inverse** proposal  $(\neg p \Rightarrow \neg q)$ .

Ex : "If Amin is a writer then he is cultivated"

"If Amin is not cultivated then he is not a writer"

"If Amin is cultivated then he is a writer"

"If Amin is not a writer then he is not cultivated"

If we suppose that  $(p \Rightarrow q)$  is true, its contrapositive is clearly true (since it is equivalent) but its converse and its inverse are not necessarily true. Moreover, if the converse or the inverse is true then  $p$  and  $q$  are equivalent. The inverse being in fact the contrapositive of the converse.

## 4°) Principle of deduction

If  $p$  and  $q$  are two propositions then the proposition  $r = "[p \wedge (p \Rightarrow q)] \Rightarrow q$ " is a tautology.

## 5°) Syllogism rule (or transitivity)

If  $p$ ,  $q$  and  $r$  are three propositions then the proposition  $s = "[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)"$  is a tautology.

The two properties are shown using the truth table; the second comprising  $2^3 = 8$  lines corresponding to the 3-lists of the set  $\{T, F\}$

Instead of saying that the proposition  $(p \Leftrightarrow q)$  is true, one can say that the proposition  $p$  is true if, and only if, the proposition  $q$  is true or that the propositions  $p$  and  $q$  are equivalent. One can then write  $p \Leftrightarrow q$  without the quotation marks or the brackets.

Notice the analogy with the biconditional sentence which stipulates that a proposition  $p$  is true if, and only if, the proposition  $q$  is true.

Ex:

- I will be proud of you if, and only if you would be successful..

- A triangle  $ABC$  is equilateral if, and only if,  $\hat{A}=\hat{B}=\hat{C}$ .

### ❖ Examples

1°) The two propositions:

"Salim is the son of Farid" and "Farid is the father of Salim" are equivalent. One can also say Salim is the son of Farid if, and only if, Farid is the father of Salim.

Notice that the proposal " Salim is the only son of Farid" is not equivalent to " Farid is the father of Salim" since if the first is true then the second is true but if the second is true then the first is not necessarily true.

2°) If  $A$  and  $B$  are two subsets of a set  $E$ , then the proposition  $(A = B)$  and  $[(A \subset B) \wedge (B \subset A)]$  are equivalent.

### Remark

Two equivalent propositions are not necessarily identical.

Ex: In a triangle  $ABC$ , we have:

$AB = AC$  if, and only if,  $\hat{B}=\hat{C}$ .

## E- Tautology

### Definition

A proposition is a **tautology** if it is always true.

A tautology is, in fact a proposition for which the last column of its truth table contains only  $T$ .

❖ A proposition is a statement that one can say is either true or false

❖ To determine the truth value of the negation of a proposition or the composite of two or several propositions we apply the following rules:

1.  $\neg p$  is true if, and only if,  $p$  is false;
2.  $p \wedge q$  is true if, and only if,  $p$  and  $q$  are both true;
3.  $p \vee q$  is true if, and only if, at least one of the two propositions  $p$  or  $q$  is true;
4.  $(p \Rightarrow q)$  is true if, and only if,  $p$  is false or if  $p$  and  $q$  are both true. In that case one can say that  $p$  implies  $q$  and we can write  $p \Rightarrow q$  (without the brackets);
5.  $(p \Leftrightarrow q)$  is true if, and only if,  $p$  and  $q$  are both true or both false. In that case one can say that  $p$  and  $q$  are equivalent and we can write  $p \Leftrightarrow q$  (without the brackets);

❖ A tautology is a proposition which is always true and a contradiction is a proposition which is always false.

❖ To show that a proposition is true, one often uses, beside the preceding rules, the following principles:

1. The deduction principle:  
 $p \wedge (p \Rightarrow q) \Rightarrow q$ ;
2. The syllogism rule:  
 $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ ;
3. The reasoning by contradiction:  $p$  is true if  $\neg p$  leads to a contradiction;
4. The contrapositive proof: to show that  $p \Rightarrow q$ , we show that  $\neg q \Rightarrow \neg p$ .

# EXERCISES

1 Are the following statements propositions:

- i) "Tarek is on journey";
- ii) "George is Lebanese";
- iii) "Good morning the children!";
- iv) " $3 + 5 = 9$ ";
- v) "I would like to pass the bac";
- vi) " $5 < 4$ ";
- vii) "A square is a quadrilateral whose sides are equal";
- viii) "The number 9 is a multiple of 3 and multiple of 2";
- ix) "Did you spend good holidays?";
- x) "Where did you spent your vacation?";
- xi) "I was at the village for one month".

2 Are the following propositions true or false:

- i) "Lebanon is an Asian or an African country";
- ii) "Italy is a European and a Mediterranean country";
- iii) "Number 15 is divisible by 3 or divisible by 5";
- iv) "Number 15 is a multiple of 5 and a multiple of 2";
- v) "Number 8 is odd or a multiple of 4";
- vi) "Either Great Britain or Spain is an Asian country".

3 A **propositional form** (or **function**) is a sentence implying an unknown (or a variable  $X$ ) belonging to a reference set  $E$  which becomes a proposition when we substitute for  $X$  a well defined element of  $E$ .

Among the following sentences determine the propositions, the propositional forms and the hypothetical propositions (judgements).

- i) "Number 2 is even";
- ii) "Number  $x$  is odd";
- iii) "Lebanon is the most beautiful country on earth";

iv) "Salem is more intelligent than Rima";

v) "Number 5 is smaller than 3" (" $5 < 3$ ");

vi) "Number  $x$  is greater than 10" (" $x > 10$ ");

vii) "Michel is a French tourist";

viii) "Number 5 is smaller than 8 and greater than 6";

ix) "M.  $X$  is a Lebanese minister";

x) "The triangle  $\Delta$  is equilateral";

xi) "Bachir is handsome and rich";

xii) "Riad is not very courageous";

xiii) "The village  $Y$  is in Spain";

xiv) "Zahle is not a Lebanese village".

Show that the proposition

4 " $(p \Rightarrow q) \Leftrightarrow [(\neg q) \Rightarrow (\neg p)]$ " is a tautology.

a) Show that the proposition

5 " $\neg(p \wedge q) \Leftrightarrow (\neg p) \vee (\neg q)$ " is a tautology.

b) Express in a different form the following propositions:

"It is not true that the mother of Marc is French and his father is English";

"It is false to say that Sami is a Lebanese lawyer".

c) Let  $p$ : " $3 < a < 5$ ". Determine  $\neg p$ .

a) Show the proposition:

6 " $\neg(p \vee q) \Leftrightarrow (\neg p) \wedge (\neg q)$ " is a tautology.

b) Express differently the following propositions:

"It is not true that the mother of Marc is French or his father is English".

"It is false to say that Sami is a Lebanese or a lawyer".

c) Let  $p$ : " $a \in ]-\infty, 3[ \cup [5, +\infty[$ ".



However the moon orbits the earth, therefore, the earth orbits the sun.

vi) I am proud of myself if, and only if my father appreciates me. Either I am brilliant in sports or I am not proud of my self.

If I work sufficiently I cannot be brilliant in sports.

Therefore, if my father appreciates me, I do not work sufficiently.

vii) If I like math, then I study. I do not study and I miss my examination.

Therefore, if I miss my examination, then I do not like math.

viii) I succeed my exam or it is difficult.

If I succeed, my examination must be prepared.

But it was not prepared, therefore, it is difficult.

## JUST FOR FUN

The preferred entertainment of a group of boys and girls are: reading, sports, and music. Can you determine the gender and the preferred pastime of a person of that group knowing that this person:

- ❖ Likes to play football or to read French novels or to play the guitar.
- ❖ Does not like to read.

And knowing that:

- ❖ All the persons who play music like to read.
- ❖ All the girls play music.

- 14 With the proposition " $p \Rightarrow q$ " we associate the converse " $q \Rightarrow p$ ", the 'inverse' " $\neg p \Rightarrow \neg q$ " and the contrapositive:

" $\neg q \Rightarrow \neg p$ ".

Consider the following proposition::

"If  $ABCD$  is a square, then  $ABCD$  is a rectangle".

- Determine its converse, its inverse, and its contrapositive.
- Determine the truth value of each of the 4 propositions..

- 15 Write in a symbolic language and study the validity of the following reasoning:

- If it rains Hamid will be sick. But it did not rain. Thus, Hamid was not sick.
- If it rains Hamid will be sick. But Hamid was not sick. Thus, it did not rain.

- If it rains, Hamid will be sick. However, Hamid was sick, thus it rained.

- 16 A direct proof is a proof in which we directly pass from the assumptions (or premises) to the conclusion.

Show that the following proposition

" $[(p \Rightarrow \neg q) \wedge q] \Rightarrow \neg p$ "

is a tautology in using:

- A truth table.
- A direct proof.

Deduce the validity of the following reasoning:

If Rima studies, she does not miss her examination. However, Rima missed her examination, Thus Rima did not study.

## SELF-EVALUATION

- 1 Among the following sentences, determine those, which are propositions.

- " Said is an engineer or a lawyer".
- "Farewell the beautiful life!".
- "The mountain is more beautiful than the sea coast."
- "Lina is a Lebanese student in France."
- "Mr.  $X$  is the new math teacher."

- 2 Determine the following propositions::

$[p \vee q \Leftrightarrow (\neg p \Rightarrow q)]$

and  $[\neg(p \Rightarrow q) \Leftrightarrow p \wedge (\neg q)]$  are tautologies

- 3 Study the validity of the following statements:

- If I walk too much, I do not study enough.

If I do not study enough, I miss my examination.

However, I missed my examination, therefore I walked too much.

- Rachid is tired or sick  
If Rachid is tired, then he is bearable.

However, he is unbearable, therefore Rachid is sick.

# PROBLEMS

- 1 we suppose that one has a well defined criterion to come to a conclusion about the validity of the following propositions:

$p$ : " he is beautiful";

$q$ : " he is big";

$r$ : " he is intelligent";

Write in a symbolic form the following propositions:

i) " If he is big, then he is ugly";

ii) "If he is big and intelligent, then he is beautiful";

iii) " If he is not intelligent but big, then he is not beautiful";

iv) " It is sufficient to be big and intelligent to be beautiful";

v) " He is not big or he is intelligent"(2 ways);

vi) " He is big but intelligent"(2 ways);

vii) " He is young, but intelligent"(2 ways);

viii) " He is beautiful if, and only if, he is intelligent";

ix) " It is false to say that he is ugly if, and only if, he is big" (2 ways);

x) " It is enough to be big and intelligent to be beautiful".

- 2 Study the following reasoning:

i) If Ghassan is a student, he is poor.

If he is poor, he does not eat.

If he does not eat, he dies young.

Therefore, if Ghassan is a student he dies young.

ii) If Nabih marries Leyla, he will be happy.

If Nabih does not pass his examination, he will be unhappy.

Therefore, if Nabih marries Leyla, he passes his examination.

iii) If Sabah is not sick, she smiles. Sabah is sick.

Hence, she is not going to smile.

iv) If the teacher is gentle, the student will not miss his examination. If the student is lazy, he will miss his examinations.

The student is lazy. Therefore, the professor is not gentle.

v) If the club gains, the trainer will be happy.

However, the trainer is happy or he is sick, but not both.

Therefore, if the trainer is sick, the club will not win.

- 3 Study the following reasoning and discuss the validity of the conclusions:

i) If Roma is not in Spain, then Beirut is not in Lebanon.

However Beirut is in Lebanon, therefore Roma is in Spain.

ii) If 6 is odd, then 3 does not divide 5.

3 is not prime or 3 divides 5.

But 3 is prime, therefore 6 is even.

iii) If it is the birthday of my mother, I offer her flowers. It is the birthday of my mother or I work late.

I did not offer flowers to my mother, therefore I worked late.

iv) If I walk, I cannot study. I study or I pass the math.

I walk, therefore I pass the math.

v) If the earth does not orbit the sun then the moon does not orbit the earth.

$$= "(a < 3) \vee (a \geq 5)".$$

Determine  $\neg p$ .

- 7 Show that the proposition  $"[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)"$  is a tautology.  
Deduce the validity of the following reasoning: If it rains, I do not leave. If I do not leave, I am bored. If I am bored, I feel morose. If I feel morose, I do not love anyone. Therefore, if it rains, I do not love anyone..

- 8 Let  $p$ : "Jacques is a tourist"  $q$ : "Jacques is French". Translate into symbolic language the following propositions:  
i) "Jacques is a French tourist";  
ii) "Jacques is a tourist but not French";  
iii) "Jacques is French but not a tourist";  
iv) "Jacques is neither French nor a tourist";  
v) "Jacques is not French or he is not a tourist";  
vi) "It is wrong to say that Jacques is a French tourist".

- 9 Translate into symbolic language and reformulate in the usual language (but in a different way) the following propositions:  
i) "It is not true that the students study and fail in the exam";  
ii) "It is not true that the teacher is qualified and that the students are studious";  
iii) "It is false to say that the student is attentive in the class or that he reads his newspaper";  
iv) "It is false to say that Maxime doesn't like math or doesn't study philosophy".

- 10 Show using two different methods that the following propositions are true :

$$P_1: "If 3 = 4 then 1 = 1"$$

$$P_2: "If a + a = a then 2a + 2a = a" \text{ where } a \in \mathbb{N}^*.$$

- 11 Determine the truth value of each of the following propositions:  
a) "If Beirut is the capital of France then Paris is the capital of Lebanon";  
b) "If Beirut is the capital of Lebanon then London is the capital of France";  
c) "If the earth orbits the sun then the moon orbits the earth";  
d) "If the sun orbits the earth then the moon orbits the earth."

- 12 Determine the truth value of each of the following propositions:  
i) "There are 8 days in the week if, and only if, there are 20 days in the month";  
ii) "There are 7 days in the week if, and only if, we have two Sundays in a week";  
iii) " There are 7 days in the week if, and only if, we have one Friday in the week";

- 13 Prove using two different methods that the following propositions are true:  
i) "4 is odd if , and only if, 3 is even";  
ii) " $3 > 4$  if, and only if,  $5 < 4$ ".

We will try to observe this same rule for subsets of  $\mathbb{N}$ .

a) Let  $I$  be the set of odd natural numbers. Does the preceding rule allow us to associate an element of  $I$  to two elements of  $I$ ?

$$\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$$

$$I = \{1, 3, 5, 7, \dots\}$$

b) Let  $P$  be the set of even natural integers and  $x, y$  any elements of  $P$ . To the couple  $(x, y)$  we associate the element  $s = x + y$  of  $\mathbb{N}$ . Are the following statements true?

$$P = \{0, 2, 4, 6, \dots\}$$

$$1 \spadesuit x + y \in P;$$

$$2 \spadesuit x + y = y + x;$$

$$3 \spadesuit \text{For all elements of } P: (x + y) + z = x + (y + z);$$

$x + y = y + x$  expresses the commutativity of the  $+$  operation

$$4 \spadesuit \text{There exists an element } e \text{ of } P \text{ such that, for all } x \in P: \\ x + e = e + x = x.$$

$(x + y) + z = x + (y + z)$  expresses the associativity of  $+$  operation.

c) Does it exist for all  $x \in P$  an element  $x'$  of  $P$  such that  $x + x' = e$ ?

## Activity 2.

Let us consider the sets  $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$  and  $\mathbb{R}$  provided with the operation  $+$  and  $\times$ .

It is known that for each of these sets, 0 is an identity element for addition and 1 is an identity element for multiplication.

For all  $x$  in  $\mathbb{R}$ :

$$x + 0 = 0 + x = x$$

$$x \times 1 = 1 \times x = x.$$

a) Does there exist for all element  $x$  of  $\mathbb{N}$ , and element  $x'$  of  $\mathbb{N}$  such that  $x + x' = 0$ ?

b) Same question for  $x$  and  $x'$  in  $\mathbb{Z}$ ?,  $\mathbb{Q}$ ?,  $\mathbb{R}$ ?

c) Does it exist for all elements  $x$  of  $\mathbb{Z}$ , an element  $x'$  of  $\mathbb{Z}$  such that  $x \times x' = 1$ ?

d) Same question for  $x$  and  $x'$  in  $\mathbb{Q}$ ?,  $\mathbb{R}$ ?,  $\mathbb{Q}^*$ ?,  $\mathbb{R}^*$ ?

## I BINARY OPERATION

### Definition

A binary operation on a set  $E$  is a rule which associates with any pair  $(x, y)$  of  $E \times E$  a unique element  $z$  of  $E$ .  $z$  is then called the composite of  $x$  and  $y$  by this operation.

Certain operations have been given particular names. In fact:

- In  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$ , the usual operation denoted  $+$  is called addition, the result of this operation, denoted  $a + b$ , is called sum of  $a$  and  $b$ ;
- The operation denoted  $\times$  is called multiplication, the result of this operation, denoted  $a \times b$ , is called product of  $a$  and  $b$ ;
- In  $P(E)$ , the "union" is the operation denoted  $\cup$ , and the result is  $A \cup B$  (union of  $A$  and  $B$ ), and the "intersection" is the operation denoted  $\cap$  and the result is  $A \cap B$  (intersection of  $A$  and  $B$ ).

the signs frequently used to denote an operation are the followings:  $+$ ;  $\times$ ;  $;$ ;  $\cup$ ;  $\cap$ ;  $*$ .....

If the operation is denoted for example by  $*$  the composite of  $x$  and  $y$  is denoted  $x * y$ .

The elements of  $P(E)$  are the parts of  $E$  and if  $A$  and  $B$  are two parts of  $E$ , then  $A \cup B$  and  $A \cap B$  are well defined elements of  $P(E)$ .

### Properties of the binary operation

Consider a binary operation denoted  $*$ , defined on a set  $E$ .

### Definition (Commutativity)

we say that the operation  $*$  is **commutative** if, and only if, for all elements  $x$  and  $y$  in  $E$ , we have:

$$x * y = y * x$$

### ❖ Example

In the sets  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$ , the usual operations  $+$  and  $\times$  are commutative; on the contrary, the subtraction denoted  $-$ , is a binary operation in  $\mathbb{Z}$ , but it is not commutative.

$$\begin{aligned} 3 + 5 &= 5 + 3 = 8 \\ 4 \times 7 &= 7 \times 4 = 28 \\ 8 - 5 &= 3 \text{ and } 5 - 8 = -3 \\ \text{From where} \\ 8 - 5 &\neq 5 - 8. \end{aligned}$$

### Definition (associativity)

we say that the operation  $*$  is **associative** if, and only if, no matter what the elements  $x, y, z$  are in  $E$  we have:

$$(x * y) * z = x * (y * z)$$

In this case we denote these two composite  $x * y * z$  (without the parentheses)

### ❖ Example

The addition and the multiplication on  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  are associative, on the contrary, subtraction on  $\mathbb{Z}$  is not.

$$\begin{aligned} (12 - 3) - 4 &= 9 - 4 = 5 \\ 12 - (3 - 4) &= 12 - (-1) = 13 \\ \text{From where } (12 - 3) - 4 &\neq 12 - (3 - 4). \end{aligned}$$

### Definition (identity element)

An element  $e$  of  $E$  is an **identity element** or **neutral element** for the operation  $*$  if, and only if, for all  $x$  in  $E$ , we have:

$$x * e = e * x = x$$

When the  $+$  operation is defined on  $E$ , its identity element is zero and when the operation is  $\times$ , its identity element is the unit element

In  $\mathbb{R}$ : 0 is a neutral element for addition and 1 is a neutral element for multiplication.

If  $E$  is the set of vectors in a plane then the operation which consists to get the sum of two vectors  $\vec{u}$  and  $\vec{v}$  of  $E$  is denoted  $+$  and its neutral element denoted  $\vec{0}$ .

## 2 BINARY OPERATIONS - GROUPS

### INTRODUCTION

#### Prerequisites

❖ Notion about sets.

The operations of addition and multiplication on the sets  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  are examples of binary operations. The same is true for the union and intersection on the set  $P(E)$  (set of subsets of  $E$ ).

When we define on a set one or more binary operations, characterized by certain properties, we say that these operations confer to the set a structure. The name of this structure depends obviously on the properties of the operations.

However, in  $\mathbb{N}$  provided by the operation  $+$ , the equation  $a + x = b$  does not have always a solution (except when  $a \leq b$ ), while this same equation has always solution in  $\mathbb{Z}$  (if  $a \in \mathbb{Z}$ ,  $b \in \mathbb{Z}$  then  $x = b - a \in \mathbb{Z}$ ). In the same way  $ax = b$  ( $a \neq 0$ ), which does not have a solution in  $\mathbb{Z}$  unless if  $b$  is a multiple of  $a$ , admits always a solution in  $\mathbb{Q}^*$  (if  $a \in \mathbb{Q}^*$ ,  $b \in \mathbb{Q}^*$  then  $x = \frac{b}{a} \in \mathbb{Q}^*$ ).

We introduce in this chapter the structure of a group of which we already know several examples like the set  $\mathbb{Z}$  provided with the operation  $+$ ;  $\mathbb{Q}^*$  provided with the operation  $\times$ ;  $\mathbb{R}$  provided with the operation  $+$ , etc... we also have some sets whose elements are not numbers.

The objectives of this chapter are:

- To define a binary operation;
- To identify the properties of the binary operation;
- To introduce the structure of a group.

### ACTIVITIES

#### Activity 1.

It is known that the addition, in the set of positive integers  $\mathbb{N}$ , is a rule that associates to two elements  $x$  and  $y$  of  $\mathbb{N}$  an element  $z$  of  $\mathbb{N}$  called sum of  $x$  and  $y$ , denoted by  $x + y$ .

### Definition (inverse element)

If a set  $E$  has a neutral element  $e$  for the  $*$  operation, we say that the element  $x$  of  $E$  has an inverse element  $x'$  for the  $*$  operation if

$$x * x' = x' * x = e$$

#### Example

-In the set  $\mathbb{R}$ , every element has an inverse element for addition. The inverse of  $a$  is denoted by  $-a$  and is called the negative of  $a$ .

-In  $\mathbb{N}$ , only 0 has an inverse for addition. The negative of 0 is 0.

-In  $\mathbb{R}$ , all the elements except 0 have an inverse for multiplication. The inverse of  $a$  is denoted by  $\frac{1}{a}$  or  $a^{-1}$ .

In  $\mathbb{Z}$ , only 1 and  $-1$  have inverse elements for the multiplication. The inverse of 1 is 1 and the inverse of  $-1$  is  $-1$ .

## II STRUCTURE OF A GROUP

### Definition (group)

Let  $E$  be the set affected by the binary operation  $*$ .

We say  $(E, *)$  is a **group** or  $E$  has a group structure for the operation  $*$  if, and only if:

- 1-the operation  $*$  is associative;
- 2-There exists in  $E$  an identity element for the operation  $*$ ;
- 3-Every element  $x$  of  $E$  has an inverse in  $E$  for the  $*$  operation.

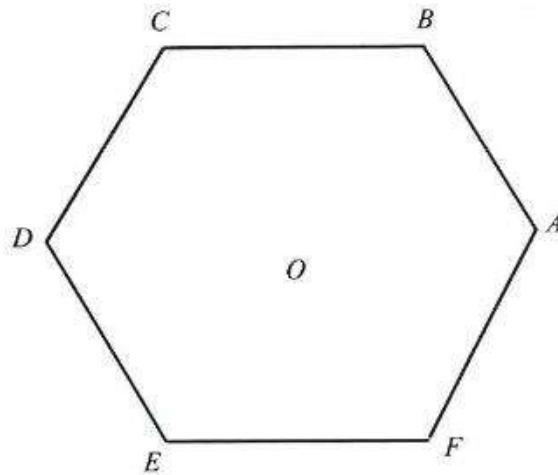
If the  $*$  operation is also commutative then the group  $(E, *)$  is a commutative group or an **abelian groupe**.



Evariste Galois  
(1811 - 1832)

## JUST FOR FUN

Let A, B, C, D, E and F be 6 points of a plane making a regular hexagon of center O. Suppose we construct a hexagon made of cardboard having the same dimensions of ABCDEF and that we fix its center at the point O so we can rotate it about its axis.



- Starting from this situation, find a group?
- Specify the elements of this group: its binary operation, the properties of its operations, its identity element, and the pairs of inverse elements
- Using a table, construct the composite of pairs of elements of the group?
- In a similar way, can you determine groups associated with other geometrical figures?

## M. C. Q.

- 1 Let  $E$  be a set with a binary operation  $*$  and  $e \in E$ . Choose the best answers.  $e$  is an identity element for the binary operation  $*$  if:

- a) for all  $a$  in  $E$ , we have:  $a * e = a$ ;
- b) There exist  $a \in E$  such that  $a * e = a$  and  $e * a = a$ ;
- c) For all  $a$  in  $E$ , we have:  $a * e = a$  et  $e * a = a$ .

- 2 Let  $*$  be a binary operation defined on a set  $E$ .

Choose the correct statements.

- a)  $*$  is commutative if there exists  $a$  and  $b$  in  $E$  such that  $a * b = b * a$ .
- b)  $*$  is commutative if  $a * b = b * a$  for all  $(a, b) \in E \times E$ .
- c)  $*$  is not commutative if there exists  $a$  and  $b$  in  $E$  such that  $a * b \neq b * a$ .

- 3 Choose the correct statements

- a)  $(\mathbb{Z}, \times)$  is a group.
- b)  $(\mathbb{R}, \times)$  is a group.
- c)  $(\mathbb{R}^*, \times)$  is a group.
- d)  $(\mathbb{Q}^*, \times)$  is a group.

## SELF-EVALUATION

- 1 Let  $I$  be the set of odd natural integers and  $P$  the set of even natural integers.

- a) Does the usual multiplication define a binary operation on  $I$ ? On  $P$ ?
- b) Do these two sets have an identity element for this operation?
- c) Show that these two sets with this operation are not groups.

- 2 Consider the  $*$  operation which for 2 numbers  $a$  and  $b$ , associates the number

$$a * b = \frac{a + b}{2}$$

- a) Calculate  $0 * 1$ ;  $0 * 2$ ;  $0 * (-1)$ ;  $0 * (-\frac{1}{2})$ ;  $1 * 1$ ;  $1 * (-1)$ ;  $\frac{1}{2} * \frac{1}{3}$

- b) Does  $*$  define a binary operation in  $\mathbb{N}$ ? in  $\mathbb{Z}$ ?

- c) Show that  $*$  define a binary operation on  $\mathbb{Q}$  which is commutative but not associative and does not have an identity element.

# PROBLEMS

1 Consider the set  $E$  and the set  $F = P(E)$  of subsets of  $E$ .

a) • Show that the union  $A \cup B$  of two elements defines on  $F$  a binary operation.

• Show that this operation is commutative, associative and that there exists an identity element.

b) • Show that the intersection  $A \cap B$  of two elements defines on  $F$  a binary operation.

• Show that this operation is commutative, associative and that there exists an identity element.

2 We define on the set of natural integers  $\mathbb{N}$  the operation  $*$  such that

$$a * b = 2a + b$$

for all  $(a, b) \in \mathbb{N} \times \mathbb{N}$ .

a) Draw the table part of this operation, relative to the elements 0, 1, 2, 3, 4.

b) Is this operation commutative?

c) Is it associative?

d) Does it have an identity element?

3 Let  $a$  and  $b$  be any two real numbers and consider the binary operation  $\top$  defined in  $\mathbb{R}$  by  $x \top y = ax + by$  for all  $(x, y) \in \mathbb{R} \times \mathbb{R}$ .

How should we choose  $a$  and  $b$  in order that this operation is commutative? Associative?

4 Consider the operation  $*$  defined in  $\mathbb{R}$  by  $x * y = x + y + xy$  for all  $(x, y) \in \mathbb{R} \times \mathbb{R}$ .

a) Is this operation commutative? Associative?

b) Verify that 0 is the identity element for this operation.

c) Verify that for all  $x$  in  $\mathbb{R}$  we have:

$$(-1) * x = x * (-1) = -1$$

d) Show that all real numbers different than  $(-1)$  have an inverse element.

e) Deduce that  $\mathbb{R} \setminus \{-1\}$  with  $*$  is an abelian group.

5 Let  $E$  be a set with a binary operation  $*$ . We suppose that this operation is associative and has an identity element  $e$ .

a) Show that, if for an element

$a$  exists an element  $a'$  such that

$a * a' = e$  and an element  $a''$  such

that  $a'' * a = e$ , then  $a' = a''$  and  $a$

admits an inverse  $a^{-1} = a' = a''$ .

b) Show that, if for every  $a$  there exist two elements  $a'$  and  $a''$  such that  $a * a' = e$  and  $a'' * a = e$ , then

$(E, *)$  is a group.

6 Let  $E$  be a set with an associative binary operation  $\top$  for which there exists an element  $e$  of  $E$  such that:

- for all  $x$  in  $E$  we have  $x \top e = x$ ,

- for all  $x$  in  $E$ , there exists  $y \in E$  that verifies  $x \top y = e$ .

a) Show that  $e$  is an identity element of  $(E, \top)$ .

b) Show that all elements of  $E$  admit an inverse and consequently  $(E, \top)$  is a group.

## Examples

1-  $(\mathbb{Z}, +)$ ;  $(\mathbb{Q}, +)$  and  $(\mathbb{R}, +)$  are commutative groups. 0 is the identity element and the negative of  $a$  is  $-a$ .

2-  $(\mathbb{Q}^*, \times)$  and  $(\mathbb{R}^*, \times)$  are commutative groups. The neutral element is 1 and the inverse of  $a$  is  $\frac{1}{a}$ .

$(\mathbb{N}, \times)$ ;  $(\mathbb{Z}, \times)$   $(\mathbb{Q}, \times)$  and  $(\mathbb{R}, \times)$  are not groups

- The only element having an inverse in  $\mathbb{N}$  is 1.

- The only elements having an inverse in  $\mathbb{Z}$  are 1 and  $-1$ .

- In  $\mathbb{Q}$  (respectively in  $\mathbb{R}$ ), all the elements except 0 have their inverses in  $\mathbb{Q}$  (respectively in  $\mathbb{R}$ ).

## Properties

In a group  $(E, *)$ , all equations  $a * x = b$  ( $a$  and  $b$  being two elements of  $E$  and  $x$  the unknown) have one and only one unique solution  $x = a' * b$  where  $a'$  is the inverse of  $a$ .

In fact we have:  $a * x = b$ , hence  $a' * (a * x) = a' * b$  or  $(a' * a) * x = a' * b$ ,

Therefore  $e * x = a' * b$  or  $x = a' * b$ .

## FOCUS

Let  $E$  be a set affected by a binary operation denoted by  $*$ .

- This operation is said to be **associative** if for all  $a, b$  and  $c$  in  $E$  we have  $(a * b) * c = a * (b * c)$ . this element is then denoted by  $a * b * c$ .

- This operation is said to be **commutative** if for all elements  $a$  and  $b$  in  $E$  we have  $a * b = b * a$ .

- An element  $e$  of  $E$  is called an **identity element** for the  $*$  operation if for all elements  $a$  in  $E$  we have  $a * e = e * a = a$ .

- An element  $a$  in  $E$  is said to have an inverse if there exists an element  $a'$  in  $E$  such that

$$a * a' = a' * a = e.$$

- If  $F$  is a subset of  $E$ , then in order to ensure that the  $*$  operation defines a binary operation on  $F$ , we verify that

for all elements  $a$  and  $b$  in  $F$  we have  $a * b \in F$ . ( $*$  is closed on  $F$ ), but to verify that  $*$  is not a binary operation on  $F$  it is enough for one pair  $(a, b)$  of  $F \times F$  to have  $a * b \notin F$ .

- $(E, *)$  is a **group** if  $*$  is associative, has an identity element, and if every element of  $E$  has an inverse.

- The group  $(E, *)$  is said to be abelian or commutative if in addition  $*$  is commutative.

- In a group  $(E, *)$ , all equations

$a * x = b$  has a unique solution

$x = a' * b$  where  $a'$  is the inverse of

$a$  ( $x = -a + b$  if the operation is addition).

# EXERCICES

1 Consider the operation  $\top$  that associates for any two numbers  $a$  and  $b$ , their difference:  $a \top b = a - b$ .

a) Does  $\top$  define a binary operation in  $\mathbb{N}$ ?

b) Does  $\top$  define a binary operation in  $\mathbb{Z}$ ?

If yes, is this operation commutative? associative? does it admit an identity element?

2 Consider the  $*$  operation that associates to any two numbers  $a$  and  $b$  the number  $a + b - 2$ .

a) Does  $*$  define a binary operation on  $\mathbb{N}$ ?

b) Does  $*$  define a binary operation on  $\mathbb{Z}$ ?

If yes, is the operation commutative? Associative? Does it admit an identity element?

c) Is  $(\mathbb{Z}, *)$  a group?

3 We define the operation  $\perp$  on the set  $E = \{a, b, c\}$  in the following table. Write the composites of all pairs  $(x, y)$  of  $E \times E$ .

| $\perp$ | $a$ | $b$ | $c$ |
|---------|-----|-----|-----|
| $a$     | $b$ | $a$ | $c$ |
| $b$     | $c$ | $b$ | $a$ |
| $c$     | $a$ | $c$ | $b$ |

Is the operation commutative?

4 What are the elements having an inverse in:

a)  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  for the usual  $+$  operation?

b)  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  for the usual  $\times$  operation?

5 a) Does the  $+$  operation define a group structure on the following sets:  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$ ?

b) Does the operation  $\times$  define a group structure on the following sets:  $\mathbb{N}^*$ ,  $\mathbb{Z}^*$ ,  $\mathbb{Q}^*$ ,  $\mathbb{R}^*$ ?

6 Let  $(G, *)$  be a group and let  $a$  and  $b$  be two elements of  $G$  such that  $a * b = b$ . Show that  $a$  is an identity element of this group.

7 Let  $(G, *)$  be a group. Show that if  $a$ ,  $b$  and  $c$  are three elements of  $G$  such that  $b$  and  $c$  are distinct, we can not get  $a * b = a * c$ .

8 Let  $E = \{a, b, c\}$  and let  $*$  and  $\top$  be two binary relations represented by the following two tables.

| $*$ | $a$ | $b$ | $c$ |
|-----|-----|-----|-----|
| $a$ | $a$ | $b$ | $c$ |
| $b$ | $b$ | $a$ | $b$ |
| $c$ | $c$ | $a$ | $b$ |

| $\top$ | $a$ | $b$ | $c$ |
|--------|-----|-----|-----|
| $a$    | $a$ | $b$ | $c$ |
| $b$    | $b$ | $c$ | $c$ |
| $c$    | $c$ | $a$ | $a$ |

Show that these two operations do not define a group..

9 We consider the  $*$  operation defined on the set  $E = \{a, b, c, d\}$  by the given table.

| $*$ | $a$ | $b$ | $c$ | $d$ |
|-----|-----|-----|-----|-----|
| $a$ | $a$ | $b$ | $c$ | $d$ |
| $b$ | $b$ | $a$ | $d$ | $c$ |
| $c$ | $c$ | $d$ | $a$ | $b$ |
| $d$ | $d$ | $c$ | $b$ | $a$ |

Show that  $(E, *)$  is an abelian group.

The mode →

| Age | Frequency |
|-----|-----------|
| 18  | 6         |
| 19  | 11        |
| 20  | 8         |
| 21  | 3         |
| 22  | 2         |

### Remark

\* There are some statistical sequences without a mode; it is the case when all the modalities have the same frequency.

\* A statistical sequence can have more than one mode; it is the case when many modalities have the same maximum frequency.

Ex : Let the sequences :

$S_1$  3 ; 3 ; 4 ; 6 ; 8 ; 8 ; 8 ; 11 ; 11 ; 17

$S_2$  2 ; 4 ; 6 ; 7 ; 8 ; 11 ; 13

$S_3$  1 ; 2 ; 2 ; 3 ; 3 ; 3 ; 4 ; 5 ; 7 ; 7 ; 7 ; 10

$S_1$  has 8 as a mode;

$S_2$  does not have any mode;

$S_3$  has 2 modes: 3 and 7

## 2. Case of a statistical sequence where the observations are grouped in classes

### a) Determination of the mode class

In the case of a grouping in classes, we determine first the mode class and then the mode.

### Definition 2

The **mode class** is the class having the greatest frequency.

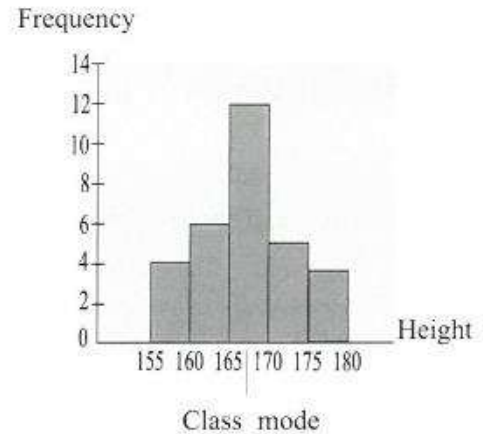
### ❖ Example

Let the statistical sequence which gives the height of the 30 students of the previous example

class mode →

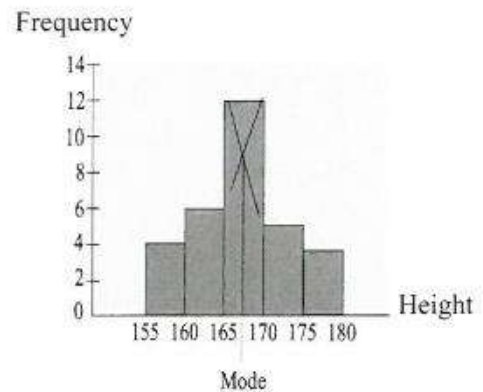
| Height    | Frequency |
|-----------|-----------|
| [155;160[ | 4         |
| [160;165[ | 6         |
| [165;170[ | 12        |
| [170;175[ | 5         |
| [175;180] | 3         |

$$n=30$$



### b) Determination of the mode

**The mode** is contained in the class mode. There are two methods, which allow to determine it, but these two methods give an approximated value because it depends on the cutting in the adopted classes.



**The center of the mode class** can be considered to be an approximated value of the mode.

The mode can be determined **graphically**. It is the abscissa of the point of intersection of the two segments  $[AC]$  and  $[BD]$  constructed in the rectangle relevant to the class mode.

### Example

In the previous example, we can obtain an approximated value in two ways:

- By taking the center of the class mode;

$$x = \frac{165 + 170}{2} = 167.5;$$

- Or by a graphical determination.

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 44 | 51 | 51 | 53 | 50 | 52 | 44 | 55 | 54 | 52 |
| 50 | 44 | 37 | 50 | 49 | 55 | 52 | 57 | 49 | 53 |
| 51 | 55 | 46 | 49 | 49 | 45 | 43 | 54 | 53 | 53 |



1. Arrange the data in increasing order and construct the table of frequencies.
2. Calculate the mean, the median and the mode of this sequence.
3. Regroup these data in classes of amplitude 5 starting with  $[35; 40[$ .
4. Calculate the frequency and the cumulative frequency of each class. In which classes the median and the mode can be found?

## Activity 2.

In a technical study in the domain of genetic engineering, a gene, that controls the growth of trout, was introduced in 21 farming pools. The full growth of the trout is normally reached in 18 months. After the introduction of this gene, the full growth is reached in each farming pool at the end of a given period (in months) as in the following table.

|      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|
| 15.9 | 7.6  | 11.3 | 15.4 | 13.9 | 10.6 | 11.6 |
| 13.1 | 13.1 | 10.6 | 10.8 | 13.2 | 12.2 | 9.7  |
| 10.8 | 9.4  | 9.8  | 11.4 | 14.8 | 16.5 | 16.8 |

- A-
1. Determine the median and the mode of this sequence.
  2. Group these observations in 5 classes of the same amplitude, in a way that the lower boundary is 7.6..
  3. Draw the table of the increasing frequencies and increasing cumulative frequencies of these classes.
- B-
1. Determine the centers  $x_1, x_2, x_3, x_4, x_5$  of these classes.  
We suppose that  $x_i$  represents the class  $i$  and has the same frequency  $n_i$  as this class with  $i \in \{1, 2, 3, 4, 5\}$

2. Find the arithmetic mean of this new sequence  $(x_i)$
3. Indicate the classes containing a median and a mode..

## Discrete variable and continuous variable. Classes.

We are interested, in this chapter, in quantitative variables. These variables are of two types: discrete or continuous. The discrete variables are those taking isolated values. The continuous variables can take any value of an interval.

In the case of a continuous variable, or of a discrete variable that takes a large number of values, the observations are generally grouped by classes or intervals. In this case, a discrete character can express it by replacing each class by its center, which means that some definitions and formulas are the same in the two cases.

The classes or intervals can have equal amplitude or not, depending on the situation. We limit ourselves in this chapter to classes of the same amplitude.

Examples of a discrete variable: the number of children in a family, the number of scored goals by a football team.

Examples of a continuous variable: the height of an individual, the time length of a TV program..

## I THE MODE - THE MODE CLASS

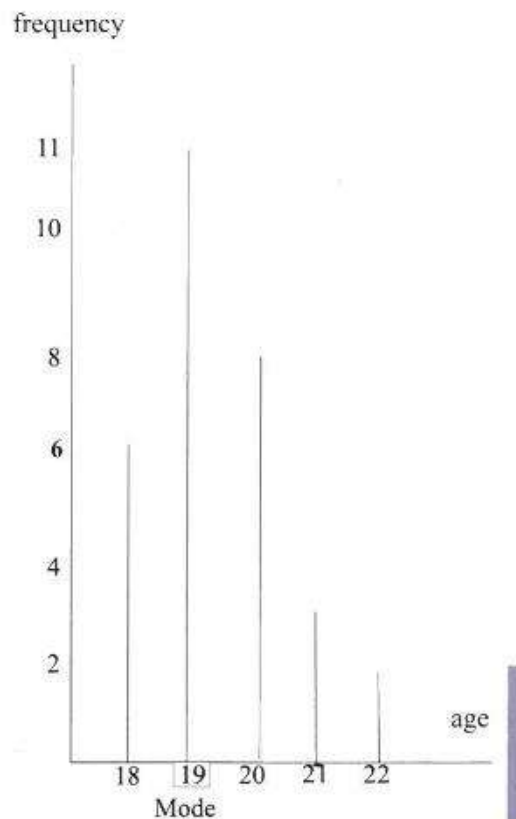
### 1. Case of a discrete variable where the observations are not grouped in classes

#### Definition 1

The **mode** of a statistical sequence is the value of the variable, which has the highest frequency. In a graphical representation in a stick diagram, it is represented by the highest stick.

#### ❖ Example

Let the statistical distribution of 30 students of a grade according to their age.



## INTRODUCTION

A set of data can be described in a table or a histogram of frequencies. But most of the time, we seek to characterize it by some parameters, which represent and summarize the available information. These parameters are called measures or indicators of position.

The mode, the mean and the median, which are measures of position, have been already introduced in the case of a discrete variable. The aim of this chapter is to develop those notions and to extend them to the case where the values of the variable are grouped by classes, whether it consists of a continuous variable, or a discrete variable taking a large number of values. As well, we try to justify the choice of the measure that represents a statistical data in a given context.

The objectives of the chapter are to:

2. Recognize the class median.
3. Recognize the class mode (s).
4. Calculate and determine graphically (if possible), the median and the mode (s).
5. Identify and calculate the mean.
6. Describe the advantages and the disadvantages of using each of the studied measure to represent statistical data.

### Prerequisites

- ❖ Statistical sequence in the case of a discrete variable; graphical representation.
- ❖ Grouping of data in classes in the case of a continuous variable, or discrete variable taking a large number of values; graphical representation.
- ❖ Position measures in the case of a discrete variable.

## ACTIVITIES

### Activity 1.

The following data represent the number of the daily sold pizzas, in a restaurant, at breakfast hour and during 30 days.

## 2. Case the observations are grouped in classes

The definition 3 remains valid when the observations are grouped in classes, the  $x_i$  being in this case the class centers.

The mean is obtained by applying the following formula:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k n_i x_i$$

where  $n$  is the total number of observations,  $x_i$  is the center of the class  $i$ ,  $n_i$  is the number of observations in the class  $i$  and  $k$  the number of classes.

### ❖ Example

(consider again the example giving the height of the 30 students.)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k n_i x_i = \frac{5010}{30} = 167.$$

| class     | $x_i$ | $n_i$  | $n_i x_i$               |
|-----------|-------|--------|-------------------------|
| [155;160[ | 157.5 | 4      | 630                     |
| [160;165[ | 162.5 | 6      | 975                     |
| [165;170[ | 167.5 | 12     | 2010                    |
| [170;175[ | 172.5 | 5      | 862.5                   |
| [175;180] | 177.5 | 3      | 532.5                   |
|           |       | $n=30$ | $\Sigma n_i x_i = 5010$ |

## III MEDIAN AND MEDIAN CLASS

### Definition 4

We call **median** of a statistical sequence any value which, without necessarily being a given of the sequence, divides it into two sequences of equal frequencies.

## II THE MEAN

### 1. Case where the observations are not grouped in classes

Let  $x_1, x_2, \dots, x_n$  be a statistical sequence.

#### Definition 3

The **arithmetic mean** ( $x_i$ ) of a statistical sequence is the ratio of the sum of the observed values to the total number of observations  $n$ .

Many types of means are used in statistics: arithmetic mean, geometric, harmonic... but when saying mean alone, it means the arithmetic mean.

The mean is generally denoted by  $\bar{x}$  (read **x bar**). The mean is given by one of the two following formulas:

$$(1) \quad \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$(2) \quad \bar{x} = \frac{1}{n} \sum_{i=1}^k n_i x_i$$

where  $n$  is the total number of observations,  $x_i$  is the modality  $i$  and  $n_i$  its frequency;  $k$  the number of modalities.

#### Examples

❖ Let the sequence : 2, 4, 5, 6, 7, 12.

Apply (1)

$$\bar{x} = \frac{2+4+5+6+7+12}{6} = 6.$$

❖ Consider again the example giving the age of the 30 students and apply (2) :

|           |    |    |    |    |    |
|-----------|----|----|----|----|----|
| Age       | 18 | 19 | 20 | 21 | 22 |
| Frequency | 6  | 11 | 8  | 3  | 2  |

$$\bar{x} = \frac{18 \times 6 + 19 \times 11 + 20 \times 8 + 21 \times 3 + 22 \times 2}{30}$$

$$\bar{x} = 19.467.$$

#### Remark

We apply the formula (2) when there is a distribution of frequencies for a number  $k$  of modalities  $x_i$  distinct one from another and for each  $x_i$  corresponds  $n_i$  observations.

In this case the mean is said to be moderated.

## 2. Determination of the median and the median class

### a) Observations not grouped in classes

We distinguish the following cases:

#### i) non grouped data

##### 1° The number $n$ of observations is odd

The median corresponds to the value which occupies the middle of the statistical sequence.

##### ❖ Example

Let the group of data be: 2, 3, 7, 1, 5, 9, 11.

We arrange these data in an increasing order.

We obtain : **1 2 3 5 7 9 11**. The median is equal to 5.

There are 3 observations before 5 and 3 observations after 5.

##### 2° $n$ is even

We take generally the mean of the two values, which occupy the middle of the statistical sequence.

##### ❖ Example

Let the sequence **1 3 5 7 9 11**. The median is :  $\frac{5+7}{2} = 6$ .

### ii) Grouped data according to the variable values (distribution of frequencies) and $n$ any integer.

We draw the table of cumulative frequencies.

If  $\frac{n}{2}$  is between two cumulative frequencies

the median will be the modality corresponding to the higher of the two frequencies.

In the opposite example,  $n=30$ ,  $\frac{n}{2}=15$ ;  $\frac{n}{2}$  is between

the values of the cumulative frequencies relative to the ages 18 and 19; the median is then equal to 19.

#### Distribution of frequencies

Ex. Giving the students' ages

|   | Age | $n_i$ | cum-freq |
|---|-----|-------|----------|
|   | 18  | 6     | 6        |
| M | 19  | 11    | 17       |
|   | 20  | 8     | 25       |
|   | 21  | 3     | 28       |
|   | 22  | 2     | 30       |

$\frac{n}{2}$

#### IV

### COMPARISON BETWEEN ARITHMETIC MEAN, MODE AND MEDIAN

(The three position measures are not necessarily values belonging to the considered statistical sequence)

|               | Advantages                                                                                                                                                                                                                                                                            | Disadvantages                                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mode</b>   | <ol style="list-style-type: none"> <li>1. It does not take into consideration the absurd values.</li> <li>2. It is easily calculated.</li> <li>3. It gives the modalities or classes having the greatest frequency.</li> </ol>                                                        | <ol style="list-style-type: none"> <li>1. It does not use all the information.</li> <li>2. In some cases, there is no mode.</li> <li>3. Its use is limited.</li> </ol>                |
| <b>Mean</b>   | <ol style="list-style-type: none"> <li>1. All the observed data are taken into consideration.</li> <li>2. The mean is widely used.</li> <li>3. It is used in the advanced statistical studies.</li> </ol>                                                                             | <ol style="list-style-type: none"> <li>1. The mean is sensible to the extreme values.</li> <li>2. Its calculation takes time when the number of observations is very high.</li> </ol> |
| <b>Median</b> | <ol style="list-style-type: none"> <li>1. It is not affected by the extreme values.</li> <li>2. It can be calculated even if the last class is open.</li> <li>3. It is easily calculated.</li> <li>4. It gives the value existing in the middle of a statistical sequence.</li> </ol> | <ol style="list-style-type: none"> <li>1. It does not use all the data.</li> <li>2. Its use is very limited.</li> </ol>                                                               |

#### ❖ Example

❖ 200, 400, 600, 600, 600, 800, 1000 represent ( in \$) the salaries of the employees working in a company.

The mode, the median and the mean of the sequence are respectively:

Mode = 600      Median = 600      Mean = 600

Those indicators of position indicate, in this case, that the salary the most frequent (mode), the salary existing in the middle of the sequence (median) and the value which mathemati-

## Remark

If  $n$  is even and  $\frac{n}{2}$  is equal to one of the cumulative frequencies (corresponding to  $x_i$  for example) the median is equal to  $\frac{x_i + x_{i+1}}{2}$ .

### b) Observations grouped in classes

## Definition 5

We call **median class** the class, which contains the median. There are two methods to determine the median: one from the median class, the other graphically.

### 1° Determination of the median class

Whatever  $n$  is, the steps to follow are the following

- ❖ We calculate the cumulative frequencies;
- ❖ We locate  $\frac{n}{2}$  in the column of cumulative frequencies;
- ❖ If  $\frac{n}{2}$  is located between two cumulative frequencies the class median will be then the one corresponding to the bigger of the two frequencies.

Ex. Giving the height of the students

| class            | Freq.     | Cum. Freq |
|------------------|-----------|-----------|
| [155;160[        | 4         | 4         |
| [160;165[        | 6         | 10        |
| <b>[165;170[</b> | <b>12</b> | <b>22</b> |
| [170;175[        | 5         | 27        |
| [175;180]        | 3         | 30        |

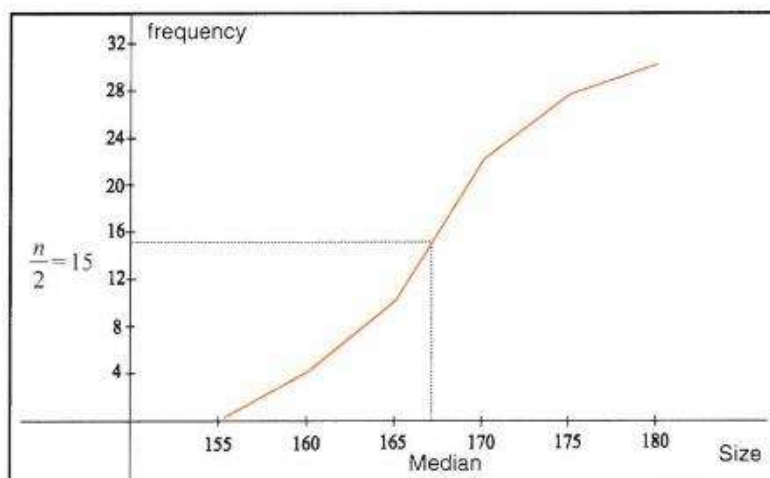
### 2° Determination of the median

#### a) From the median class.

We apply the formula in the margin.

#### b) Graphically from the curve of the cumulative frequencies.

The median is the abscissa of the point on the curve of cumulative frequencies having an ordinate equal to  $\frac{n}{2}$ .



c) If  $n$  is even and  $n/2$  is equal to one of the cumulative frequencies, the median is the upper boundary of the class which corresponds to that frequency.

$$n=30, \frac{n}{2}=15, 10 < 15 < 22$$

**[165;170[ is the median class**

The precise value of the median can be obtained using the following formula:

[ $a, b$ ] being the median class:

$$M = a + \frac{(\frac{n}{2} - N)c}{n_M}$$

where  $a$  is the lower boundary of the median class;  $n$  is the total number of observations;  $N$  the cumulative frequency which precedes the median class;  $c$  the amplitude of the classes and  $n_M$  the frequency of the median class.

In the above example:

$$M = 165 + \frac{(30/2 - 10) \times 5}{12} = 167.08.$$

| Age at the marriage date | Number of women |
|--------------------------|-----------------|
| [ 17 ; 18 [              | 27              |
| [ 18 ; 19 [              | 35              |
| [ 19 ; 20 [              | 50              |
| [ 20 ; 21 [              | 58              |
| [ 21 ; 22 [              | 172             |
| [ 22 ; 23 [              | 48              |
| [ 23 ; 24 [              | 17              |
| [ 24 ; 25 [              | 18              |
| [ 25 ; 26 [              | 13              |
| [ 26 ; 27 ]              | 12              |

- 7 The following table gives the distribution of the obtained marks of statistics exam of the third secondary year students:

| Note         | Frequency |
|--------------|-----------|
| [ 40 ; 50 [  | 2         |
| [ 50 ; 60 [  | 8         |
| [ 60 ; 70 [  | 10        |
| [ 70 ; 80 [  | 25        |
| [ 80 ; 90 [  | 7         |
| [ 90 ; 100 ] | 3         |

Calculate the median, the mode and the arithmetic mean.  
Interpret each of the obtained results (determine the mode graphically and by calculation).

e)

| Value | Frequency |
|-------|-----------|
| 104   | 6         |
| 112   | 14        |
| 115   | 21        |
| 119   | 13        |
| 123   | 22        |
| 127   | 6         |
| 132   | 9         |

f)

| Value | Frequency |
|-------|-----------|
| 246   | 2         |
| 291   | 4         |
| 295   | 3         |
| 304   | 8         |
| 307   | 9         |
| 319   | 2         |

5

Determine the class mode and the class median of the following distributions and calculate the exact value of the median and the arithmetic mean..

a)

| Class         | Frequency |
|---------------|-----------|
| [ 0 ; 25 [    | 4         |
| [ 25 ; 50 [   | 3         |
| [ 50 ; 75 [   | 6         |
| [ 75 ; 100 [  | 3         |
| [ 100 ; 125 [ | 5         |
| [ 125 ; 150 ] | 9         |

b)

| Class        | Frequency |
|--------------|-----------|
| [ 40 ; 50 [  | 2         |
| [ 50 ; 60 [  | 4         |
| [ 60 ; 70 [  | 7         |
| [ 70 ; 80 [  | 9         |
| [ 80 ; 90 [  | 5         |
| [ 90 ; 100 ] | 3         |

6

The following table gives the age at which 450 women got married.

Calculate the median, the mode and the arithmetic mean.

Interpret each of the obtained results (determine the mode graphically and by calculation).

# EXERCISES

1

Find the mode of each of the following set of numbers :

- a) 29, 39, 29, 18, 29, 56, 48
- b) 13, 17, 19, 20, 20, 13, 25, 27, 13, 20
- c) 512, 546, 318, 729, 854, 253

2

Find the arithmetic mean of each of the following set of numbers :

- a) 12, 15, 10, 8, 14, 20
- b) 25, 12 ; 42, 58 ; 76, 19 ; 32 ; 81, 11 ; 26, 41 ; 19.76 ; 59.32 ; 71.18 ; 21.03
- c) 86, 103, 118, 117, 126, 158, 149

3

Find the median of each of the following set of numbers :

- a) 8, 7, 4, 3, 1
- b) 2, 3, 4, 7, 9, 12
- c) 12, 15, 17, 19, 35, 42, 58
- d) 28, 68, 7, 15, 47, 59, 13, 74, 32, 25

4

Find the mode, the median and the arithmetic mean of the following distributions :

a)

| Value | Frequency |
|-------|-----------|
| 3     | 4         |
| 5     | 2         |
| 9     | 1         |
| 12    | 3         |

b)

| Value | Frequency |
|-------|-----------|
| 9     | 3         |
| 12    | 5         |
| 15    | 1         |
| 18    | 1         |

c)

| Value | Frequency |
|-------|-----------|
| 12    | 4         |
| 13    | 2         |
| 15    | 5         |
| 19    | 3         |
| 22    | 1         |
| 23    | 5         |

d)

| Value | Frequency |
|-------|-----------|
| 25    | 1         |
| 26    | 2         |
| 29    | 5         |
| 30    | 4         |
| 32    | 3         |
| 33    | 5         |

cally balances all the values of the sequence (arithmetic mean) are all equal to 600.

❖ If the salary of the manager of this company is 20,000 instead of 1,000 we find:

Mode = 600 Median = 600 Mean = 3200

The two sets of data differ by a unique value (20,000 instead of 1,000). The mean has been strongly affected and gives an incorrect idea of the typical salary of this set.

In the domain of the mode, when we seek to produce clothes of a unique size, we chose the size, which corresponds to the class mode of women. Those clothes correspond to that class and not to all women.

## FOCUS

Let a statistical sequence of  $n$  values be:  $x_1, x_2, \dots, x_n$ . These values being distinct or not. The mode, the mean, and the median are measures of position or central value. To calculate these measures, we distinguish three cases:

- Data not grouped..
- Data grouped according to the values of the variable in this case, for each value  $x_i$  corresponds a frequency  $n_i$ .
- Data grouped in classes; in this case, we represent the classes  $i$  by their centers  $x_i$  affected by the same frequency  $n_i$ .

### Observation non grouped in classes

**1. Mode :** It is the value  $x_i$  of the variable having the greatest frequency

### Observation grouped in classes

In this case, the modalities are classes.

#### 1. Mode and mode class

- ❖ **The mode class** is the class at which corresponds the greatest frequency.
- ❖ **The mode** can be determined graphically.

## 2. Mean

(1) Non grouped data :

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

(2) Data grouped according to the values of the variable :

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k n_i x_i$$

$n$  is the total number of observations,  $n_i$  the frequency of  $x_i$  and  $k$  the number of modalities.

## 2. Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k n_i x_i$$

$x_i$  is the center of the class  $i$  and  $n_i$  its frequency

## 3. Median

### i) Non grouped data

❖  **$n$  is odd** The mean corresponds to the value, which occupies the middle of the statistical sequence.

❖  **$n$  is even.** We take generally the mean of the two values, which correspond to the middle of the statistical sequence

### ii) Case where the data are grouped according to values of the variable.

❖ If  $n/2$  is between two cumulative frequencies, the median will be the modality, which corresponds to the greatest of the two frequencies.

❖ If  $n$  is even and  $n/2$  is equal to one of the cumulative frequencies, corresponding to  $x_i$ , the median is equal to

$$\frac{x_i + x_{i+1}}{2}$$

## 3. Median and median class

The **median class** is the class, which contains the median.

❖ To obtain the median class, we draw first the table of cumulative frequencies.. If  $n/2$  is between two cumulative frequencies, the median class will be the one which corresponds to the greatest of the two frequencies.

❖ The median can be determined graphically as follows:

- We draw the curve of the cumulative frequencies;
- We take on the curve the point of ordinate  $n/2$ ; the abscissa is the median of the sequence.

2

Let the set of data be :

10      12      8      14      40

What is the position measure which represents the best, this set ?

3

Let the set of data be:

4      8      6      6      6      8      10      9

5      7      20      8      11

Which measure you use to find a high concentration ?

## PROBLEMS

1

The following table gives the weight, in kg, of 40 students chosen randomly among the students of a secondary school:.

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 69 | 71 | 52 | 55 | 58 | 58 | 62 | 67 |
| 82 | 71 | 75 | 85 | 75 | 82 | 62 | 72 |
| 72 | 85 | 92 | 75 | 85 | 58 | 72 | 79 |
| 72 | 72 | 79 | 79 | 79 | 58 | 69 | 62 |
| 79 | 79 | 69 | 85 | 79 | 85 | 67 | 71 |

a) Regroup the data in classes of amplitude 5, starting by [ 50 - 55[

1° Find the cumulative frequencies.

2° Calculate the arithmetic mean, the mode and the median.

3° Determine graphically the mode.

b) Regroup the data in classes of amplitude 7, starting by [ 50 - 57[.

1° Solve again the questions 1, 2 and 3 of (a)

2° What do you notice ?

2

Consider the following two sets of data relative to the waiting time of patients to be emergently hospitalized.

a) Find for each sequence, the arithmetic mean, the mode and the median. What do you notice?

b) Consider the two sequences as one sequence, solve again question (a). What do you notice?

c) Take the two sequences together and regroup the data in classes of amplitude equal to 10.

| Sequence 1 in minutes : March |    |    |
|-------------------------------|----|----|
| 39                            | 41 | 53 |
| 40                            | 79 | 50 |
| 100                           | 45 | 49 |
| 72                            | 40 | 60 |
| 47                            | 88 | 63 |
| 74                            | 32 | 87 |
| 62                            | 60 | 42 |
| 69                            | 49 | 63 |
| 71                            | 50 | 62 |
| 85                            | 77 | 68 |

| Sequence 2 in minutes : June |    |    |
|------------------------------|----|----|
| 85                           | 69 | 81 |
| 59                           | 46 | 52 |
| 76                           | 49 | 66 |
| 56                           | 46 | 92 |
| 30                           | 51 | 26 |
| 26                           | 59 | 53 |
| 51                           | 31 | 89 |
| 97                           | 79 | 52 |
| 34                           | 43 | 40 |
| 56                           | 29 | 29 |

- d) Find the cumulative frequencies, the mode class and the median class.  
e) Find by calculation the mode and the median of the new sequence.

**3** The following data are the profits realized by 50 commercial banks in the United States during the 1966 year.

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 12.2 | 9.9  | 11.2 | 12.5 | 9.8  | 11.5 | 11.8 | 11.1 | 12.3 | 10.1 |
| 11.4 | 9.2  | 12.8 | 9.8  | 12.6 | 9.9  | 10.3 | 12.6 | 14.4 | 13.1 |
| 10.2 | 10.3 | 11.6 | 10.4 | 13.4 | 10.4 | 10.9 | 8.4  | 14.6 | 11   |
| 12.3 | 11.4 | 9.2  | 12.8 | 12.9 | 11.2 | 10.9 | 10.1 | 10.9 | 13.6 |
| 11.2 | 13.2 | 10.2 | 16   | 13.6 | 10.9 | 11.4 | 11.6 | 11.7 | 13   |

- a) Regroup the data in classes of amplitude equal to 1.  
b) Find the mode class and the median class.  
c) Determine by calculation the mode, the median and the arithmetic mean.  
d) Determine graphically the mode.  
e) Compare the 3 obtained measures. What do you notice?

## JUST FOR FUN

Find a sequence having two modes and its mean is equal to the median.

- 8 The monthly salaries of the employees of a big company are given in the following table :

| Salary (en \$)    | Number of persons |
|-------------------|-------------------|
| [ 5000 ; 10000 [  | 2                 |
| [ 10000 ; 15000 [ | 7                 |
| [ 15000 ; 20000 [ | 13                |
| [ 20000 ; 25000 [ | 16                |
| [ 25000 ; 30000 [ | 15                |
| [ 30000 ; 35000 [ | 9                 |
| [ 35000 ; 40000 [ | 6                 |
| [ 40000 ; 45000 [ | 10                |
| [ 45000 ; 50000 [ | 7                 |
| [ 50000 ; 55000 ] | 5                 |

Calculate the median, the mode and the arithmetic mean.  
Interpret each of the obtained results (determine the mode graphically and by calculation).

## SELF-EVALUATION

1

| Answer by True or False the following questions :                                                        | True | False |
|----------------------------------------------------------------------------------------------------------|------|-------|
| a) Every distribution has one and only one mode.                                                         |      |       |
| b) The median is an element always belonging to the statistical sequence.                                |      |       |
| c) All the values of a statistical sequence are taken into consideration in finding the arithmetic mean. |      |       |
| d) The number of observations does not have any importance in finding the median.                        |      |       |
| e) For a statistical sequence, the mean, the mode and the arithmetic mean are equal.                     |      |       |

## Activity 2.

Let  $f$  be the function  $x \mapsto \frac{x+1}{x}$ .

1°) Determine the domain of definition of  $f$ .

2°) Determine  $\lim_{x \rightarrow +\infty} f(x)$ ,  $\lim_{x \rightarrow -\infty} f(x)$ ,  $\lim_{\substack{x \rightarrow 0 \\ x > 0}} f(x)$  and  $\lim_{\substack{x \rightarrow 0 \\ x < 0}} f(x)$

and deduce the asymptotes to the curve  $C_f$  of  $f$ .

3°) Determine the derivative function  $f'$ .

4°) Draw the table of variation of  $f$ .

5°) Draw a table of values of  $f$ .

6°) Draw, in an orthonormal system, the asymptotes, then draw the curve  $C_f$ .

7°) In writing  $f(x) = 1 + g(x)$ , can you obtain  $C_f$  starting from  $C_g$ ?

The table of variation of  $f$  is a table where we indicate, according to the values of  $x$ , the sign of  $f'(x)$  and the sense of variation of  $f$  (increasing or decreasing of  $f$ ).

A table of values of  $f$  is a table which contains the particular values of  $x$  and their corresponding  $f(x)$  values.

## TEXT

### I GENERALITIES

#### Definition

A **rational function** is a function  $f$  of the form

$$x \mapsto f(x) = \frac{P(x)}{Q(x)}$$

where  $P$  and  $Q$  are polynomial functions,  $Q$  is not identically null.

This function is defined for all values of  $x$  such that

$$Q(x) \neq 0.$$

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

$$Q(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0.$$

$Q$  is identically null if for all  $x$ ,  $Q(x) = 0$ .

#### ❖ Examples

1°) The polynomial functions  $x \mapsto P(x)$  are particular rational functions, their denominator being the polynomial identically equal to 1.

2°) The **homographic functions** are of the form

$$x \mapsto f(x) = \frac{ax+b}{cx+d}.$$

$Q$  is identically equal to 1 if for all  $x$ ,  $Q(x) = 1$ .

## INTRODUCTION

The study of the rational functions of the form  $x \mapsto f(x) = \frac{P(x)}{Q(x)}$

where  $P$  and  $Q$  are polynomial functions is the continuation of the study of the function already started in former classes. The particular cases of the polynomial functions of degree  $\leq 2$ , whose graphical representations are straight lines and parabolas, and the homographical functions of the form

$x \mapsto f(x) = \frac{ax+b}{cx+d}$ , whose graphical representations are hyperbolas,

were treated in the first and second secondary years. Due to the importance of these functions, we propose, in this chapter, to proceed to review their study. The aim is to use the general method of study of these functions, to apply it to the study of the polynomial functions of the 3rd degree and the simple rational functions.

The field of application of the rational functions in general and of the polynomial functions of degree  $\leq 2$  and the homographical functions in particular is very wide and it concerns various disciplines. Let us quote as examples: physics, economics, and sociology.

The objectives of this chapter are:

1. To study and represent graphically the simple rational functions
2. To solve graphically an equation of the form  $\frac{P(x)}{Q(x)} = m$  where  $m$  is a real number.

### Prerequisites

**Study and graphical representation of :**  
**❖ Polynomials of degree  $\leq 2$ .**

## ACTIVITIES

### Activity 1.

1°) Plot, in an orthonormal system, the following two parabolic equations:

$$y = x^2 - 1 \text{ and } y = 7 - x^2.$$

2°) a) Determine the coordinates of their points of intersection..

b) Determine the coordinates of the points of intersection of the two curves with the coordinate axes.

### III

## STUDY OF THE POLYNOMIAL FUNCTION

$$f: x \mapsto x^3 - 3x - 1$$

1°) The domain of definition of  $f$  is  $\mathbb{R}$

2°)  $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} x^3 = +\infty$

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} x^3 = -\infty$$

3°)  $f'(x) = 3x^2 - 3 = 0 \Leftrightarrow 3x^2 = 3 \Leftrightarrow x^2 = 1 \Leftrightarrow$   
 $x = 1 \text{ or } x = -1$   
 $f'(x) < 0$  for  $-1 < x < 1$   
 $f'(x) > 0$  for  $x < -1$  or  $x > 1$ .

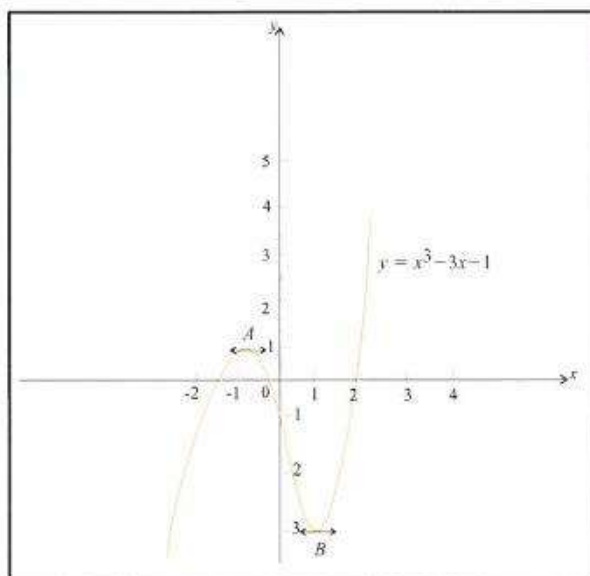
4°)

|         |           |   |      |   |      |   |           |
|---------|-----------|---|------|---|------|---|-----------|
| $x$     | $-\infty$ |   | $-1$ |   | $1$  |   | $+\infty$ |
| $f'(x)$ |           | + | 0    | - | 0    | + |           |
| $f$     | $-\infty$ |   | $1$  |   | $-3$ |   | $+\infty$ |

5°)

|        |       |      |      |      |      |     |      |
|--------|-------|------|------|------|------|-----|------|
| $x$    | $-3$  | $-2$ | $-1$ | $0$  | $1$  | $2$ | $3$  |
| $f(x)$ | $-19$ | $-3$ | $1$  | $-1$ | $-3$ | $1$ | $17$ |

6°)



This graph shows that the curve cuts the  $x$  axis in three different points. Hence there exist 3 distinct values of  $x$  for which  $f(x) = 0$ . An approximate value of each one of these solutions can be read using the graph.

The point  $A(-1, 1)$  is a local maximum for  $C_f$  and the point  $B(1, -3)$  is a local minimum. The tangent lines to  $C_f$  at these points are horizontal.

If  $x + x' = 0$ , then  $x' = -x$  and  $f(x) + f(x') =$   
 $f(x) + f(-x) =$   
 $x^3 - 3x - 1 - x^3 + 3x - 1$   
 $= -2 = 2 \times (-1)$ .  
 this shows that the point  $I(0, -1)$  is the center of symmetry of the curve.

The exact determination of these three solutions is not in our program

## IV

## STUDY OF THE RATIONAL FUNCTION

$$f: x \mapsto \frac{x^2+1}{x-1}.$$

1°) The set of definition of  $f$  is  $] -\infty, 1[ \cup ]1, +\infty[$

$$2^\circ) \quad \diamond \lim_{\substack{x \rightarrow 1 \\ x > 1}} f(x) = +\infty \quad \text{and} \quad \lim_{\substack{x \rightarrow 1 \\ x < 1}} f(x) = -\infty$$

The straight line  $D$  with equation  $x = 1$  is then a vertical asymptote to the curve represented by  $C_f$  of  $f$

$$\diamond \lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \frac{x^2}{x} = \lim_{x \rightarrow +\infty} x = +\infty.$$

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{x^2}{x} = \lim_{x \rightarrow -\infty} x = -\infty.$$

This function does not admit a horizontal asymptote. Furthermore we have, for  $x \neq 1$ ,

$$f(x) = \frac{x^2-1+1+1}{x-1} = \frac{x^2-1}{x-1} + \frac{2}{x-1} = x+1 + \frac{2}{x-1}, \quad \text{and which follows that}$$

$$\lim_{x \rightarrow +\infty} [f(x) - (x+1)] = \lim_{x \rightarrow +\infty} \frac{2}{x-1} = 0 \quad \text{and}$$

$$\lim_{x \rightarrow -\infty} [f(x) - (x+1)] = \lim_{x \rightarrow -\infty} \frac{2}{x-1} = 0.$$

The straight line  $D'$  with equation  $y = x + 1$  is then an oblique asymptote to  $C_f$

$$3^\circ) \text{ For } x \neq 1, \text{ we have } f'(x) = \frac{2x(x-1)-x^2-1}{(x-1)^2} = \frac{2x^2-2x-x^2-1}{(x-1)^2} = \frac{x^2-2x-1}{(x-1)^2}.$$

$$f'(x) = 0 \Leftrightarrow x = 1 + \sqrt{2} \approx 2.414 \text{ ou } x = 1 - \sqrt{2} \approx -0.414$$

$$f'(x) < 0 \Leftrightarrow 1 - \sqrt{2} < x < 1 + \sqrt{2}$$

$$f'(x) > 0 \Leftrightarrow x < 1 - \sqrt{2} \text{ or } x > 1 + \sqrt{2}.$$

4°)

|         |           |               |           |               |           |   |
|---------|-----------|---------------|-----------|---------------|-----------|---|
| $x$     | $-\infty$ | $1-\sqrt{2}$  | $1$       | $1+\sqrt{2}$  | $+\infty$ |   |
| $f'(x)$ | +         | 0             | -         | -             | 0         | + |
| $f$     | $-\infty$ | $2-2\sqrt{2}$ | $-\infty$ | $2+2\sqrt{2}$ | $+\infty$ |   |

$$f(1+\sqrt{2}) = \frac{(1+\sqrt{2})^2+1}{\sqrt{2}}$$

$$= \frac{4+2\sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2}+2}{2}$$

$$= 2+2\sqrt{2} \approx 4.83.$$

$$f(1-\sqrt{2}) = \frac{(1-\sqrt{2})^2+1}{-\sqrt{2}}$$

$$= \frac{4-2\sqrt{2}}{-\sqrt{2}} = \frac{-4\sqrt{2}+4}{2}$$

$$= 2-2\sqrt{2} \approx -0.83.$$

# II

## STUDY OF THE HOMOGRAPHIC FUNCTION

$$f: x \mapsto \frac{2x+1}{x-2}$$

1°) The domain of definition of  $f$  is  $]-\infty, 2[ \cup ]2, +\infty[$ .

$$2^\circ) \lim_{x \rightarrow +\infty} f(x) = 2 \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = 2$$

the straight line of equation  $y = 2$  is then an asymptote to the curve  $C_f$  of  $f$ .

$\lim_{\substack{x \rightarrow 2 \\ x > 2}} f(x) = +\infty$  and  $\lim_{\substack{x \rightarrow 2 \\ x < 2}} f(x) = -\infty$ . The straight line of equation  $x = 2$  is then an asymptote to  $C_f$ .

These two limits are called limit to the right and limit to the left of  $f$  at 2.

3°) for  $x \neq 2$  we have

$$f'(x) = \frac{2(x-2) - (2x+1) \times 1}{(x-2)^2} = \frac{-5}{(x-2)^2} < 0$$

for all  $x \in D_f$  hence  $f$  is decreasing on  $]-\infty, 2[$  and on  $]2, +\infty[$ .

We cannot say that  $f$  is decreasing on  $D_f$  because

$D_f = ]-\infty, 2[ \cup ]2, +\infty[$  is not an interval ...

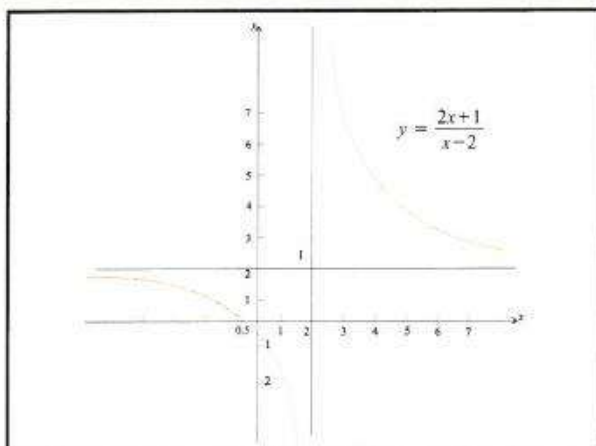
4°)

|         |           |   |           |   |           |
|---------|-----------|---|-----------|---|-----------|
| $x$     | $-\infty$ |   | 2         |   | $+\infty$ |
| $f'(x)$ |           | - |           | - |           |
| $f$     | 2         |   | $+\infty$ |   | 2         |

5°)

|        |               |    |               |               |                |                |    |     |     |   |     |                |   |
|--------|---------------|----|---------------|---------------|----------------|----------------|----|-----|-----|---|-----|----------------|---|
| $x$    | -5            | -3 | -2            | -1            | $-\frac{1}{2}$ | 0              | 1  | 1.5 | 2.5 | 3 | 4   | 5              | 7 |
| $f(x)$ | $\frac{9}{7}$ | 1  | $\frac{3}{4}$ | $\frac{1}{3}$ | 0              | $-\frac{1}{2}$ | -3 | -8  | 12  | 7 | 4.5 | $\frac{11}{3}$ | 3 |

6°)



The point of intersection of the two asymptotes  $I(2, 2)$  is a center of symmetry of the curve. It is then sufficient to represent  $f$  on  $]-\infty, 2[$  and to continue the graph by symmetry. Few values are however useful.

5°) We calculate  $f'(x)$  for  $x \in D_f$ , we determine the intervals on which  $f'$  keeps a constant sign and we deduce the **sense of variation** of  $f$  on these intervals.;

If  $f$  is a rational function, then it is derivable on all the different intervals of  $D_f$ .

6°) We draw the table of variations of  $f$ .

If  $f'(x) > 0$  (resp.  $< 0$ ) for all  $x$  in  $I$ , then  $f$  is increasing (resp. decreasing) on  $I$ .

7°) We draw the table of values of  $f$  and determine if there exist intersection points with the coordinates axes;

If  $f'$  is null at  $x_0$  ( $f'(x_0) = 0$ ) in changing the sign, then  $C_f$  admits in  $M(x_0, f(x_0))$  a local extremum.

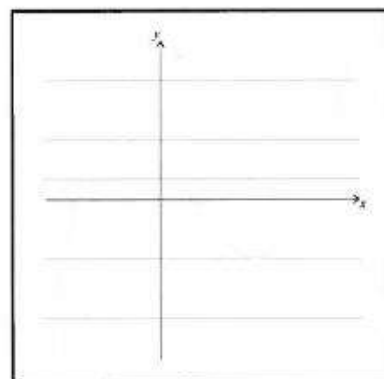
8°) We choose a system of 2 axes and we draw the asymptotes, if they exist, and finally we draw the curve  $C_f$ .

## Graphical resolution of an equation $\frac{P(x)}{Q(x)} = m$

To solve graphically the equation  $\frac{P(x)}{Q(x)} = m$ , we construct in

a system of 2 axes, the curve  $C$  representing the function

defined by  $f(x) = \frac{P(x)}{Q(x)}$  and the straight line  $D$  of equation  $y = m$ .



❖ If  $C$  and  $D$  intersect, then the abscissas of all their points of intersection will be the set of the solution of that equation..

❖ If  $C$  and  $D$  do not intersect, then this equation will not have a solution.

- ❖  $\frac{P(x)}{Q(x)} = m_1$  one solution.
- ❖  $\frac{P(x)}{Q(x)} = m_2$  two solutions.
- ❖  $\frac{P(x)}{Q(x)} = m_3$  three solutions.
- ❖  $\frac{P(x)}{Q(x)} = m_4$  two solutions.
- ❖  $\frac{P(x)}{Q(x)} = m_5$  one solution.

## Study and graphical representation of the rational function

To study and represent graphically a rational function  $f$ , we proceed by the following steps:

1°) we determine the **domain of definition**  $D_f$  of  $f$ ;

2°) we study the parity of  $f$ :

❖ If for all  $x$  in  $D_f$ ,  $(-x) \in D_f$  and  $f(-x) = f(x)$ , then  $f$  is **even**; its representative curve is then symmetric with respect to the ordinate axis;

❖ If for all  $x$  of  $D_f$ ,  $(-x) \in D_f$  and  $f(-x) = -f(x)$ , then  $f$  is **odd**; its graph is then symmetric with respect to the origin;

❖ If  $f$  is even or odd we limit our study of the function  $f$  on  $D_f \cap [0, +\infty[$ ;

3°) We study the limits of  $f$  at the boundaries of the intervals forming its domain of definition and we deduce the asymptotes to the curve represented by  $C_f$ ;

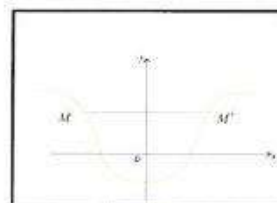
❖ If  $\lim_{x \rightarrow +\infty} f(x) = b$  (or  $\lim_{x \rightarrow -\infty} f(x) = b$ ), then the straight line of equation  $y = b$  is a **horizontal asymptote** at  $+\infty$  (or at  $-\infty$ ) to  $C_f$ ;

❖ If  $\lim_{x \rightarrow a} f(x) = +\infty$  (or  $-\infty$ ), then the straight line of equation  $x = a$  is a **vertical asymptote** to  $C_f$ ;

4°) If it exists  $a \neq 0$  and  $b$  such that

$$\lim_{x \rightarrow +\infty} [f(x) - (ax + b)] = 0$$

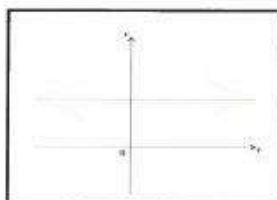
(or  $\lim_{x \rightarrow -\infty} [f(x) - (ax + b)] = 0$ ), then the straight line of equation  $y = ax + b$  is an **oblique asymptote** to  $C_f$ ;



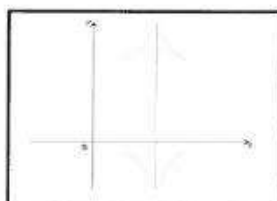
Even function



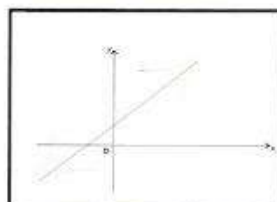
Odd function



Horizontal asymptote



Vertical asymptote



Oblique asymptote

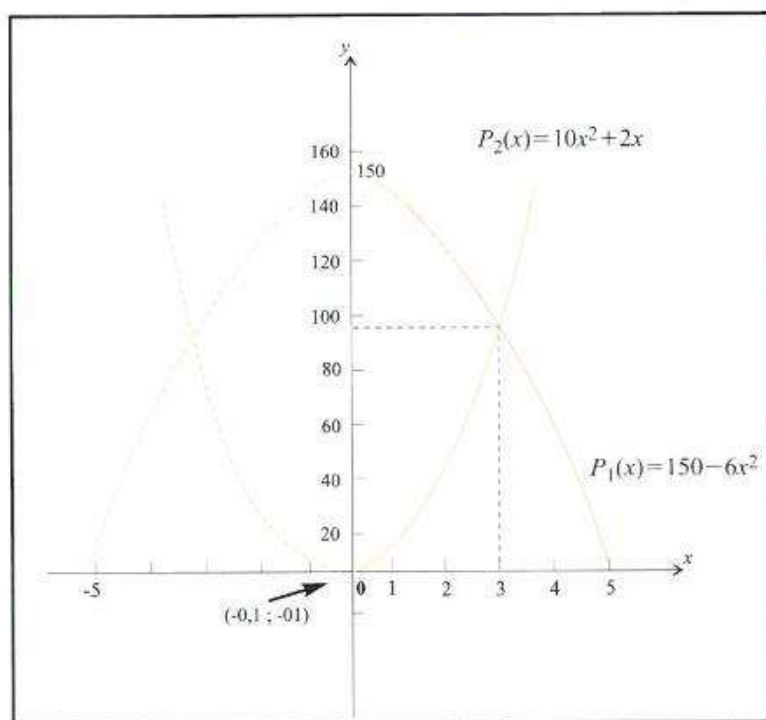
- ❖ The function  $P_2: x \mapsto 10x^2 + 2x$  is defined on  $\mathbb{R}$  and its derivative is defined by  $P'_2(x) = 20x + 2$ . It is null for  $x = -0.1$ , from where the table of variation:

|           |           |        |           |
|-----------|-----------|--------|-----------|
| $x$       | $-\infty$ | $-0.1$ | $+\infty$ |
| $P'_2(x)$ | $+$       | $0$    | $-$       |
| $P_2$     | $+\infty$ | $-0.1$ | $+\infty$ |

and the table of values :

|          |    |    |    |   |      |    |    |
|----------|----|----|----|---|------|----|----|
| $x$      | -3 | -2 | -1 | 0 | -0.1 | 2  | 3  |
| $P_2(x)$ | 84 | 36 | 8  | 0 | -0.1 | 44 | 96 |

Choose a system  $(O; \vec{i}, \vec{j})$  where 1 cm where 1 cm is the unit on the axis of abscissas and 20 units is that of the ordinate axis.



The vertex of a parabola of equation  $y = ax^2 + bx + c$  is the point

$$S\left(-\frac{b}{2a}, \frac{4ac-b^2}{4a}\right).$$

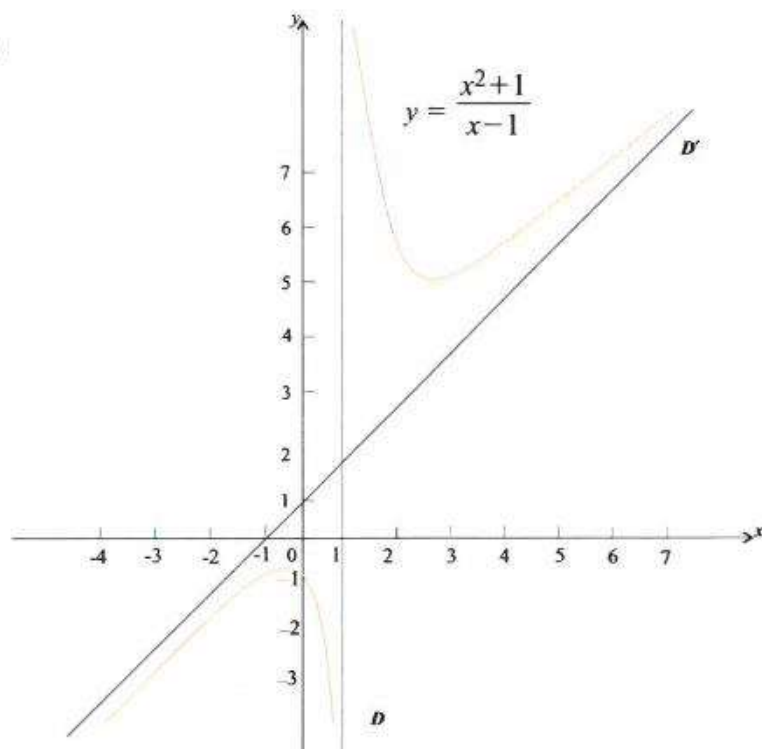
In economics it is enough to represent graphically the economical functions for  $x > 0$ . The production or the price, for example, cannot be negative.

5°)

|        |                 |                |                |    |        |    |      |     |   |       |   |                |
|--------|-----------------|----------------|----------------|----|--------|----|------|-----|---|-------|---|----------------|
| $x$    | -4              | -3             | -2             | -1 | -0.414 | 0  | 0.5  | 1.5 | 2 | 2.414 | 3 | 4              |
| $f(x)$ | $-\frac{17}{5}$ | $-\frac{5}{2}$ | $-\frac{5}{3}$ | -1 | -0.83  | -1 | -2.5 | 6.5 | 5 | 4.83  | 5 | $\frac{17}{3}$ |

The point  $A(-0.414; -0.83)$  is a local maximum and the point  $B(2.414; 4.83)$  is a local minimum.

6°)



The point  $I(1, 2)$  is a center of symmetry.

In fact

$$x + x' = 2 \times 1 \Rightarrow x' = 2 - x \Rightarrow$$

$$f(x') = f(2 - x)$$

$$= \frac{(2-x)^2 + 1}{2-x-1} = \frac{4+x^2-4x+1}{1-x}$$

$$\Rightarrow f(x) + f(x') = \frac{x^2+1}{x-1} + \frac{x^2-4x+5}{1-x}$$

$$= \frac{x^2+1-x^2+4x-5}{x-1} = \frac{4x-4}{x-1}$$

$$= \frac{4(x-1)}{x-1} = 4 = 2 \times 2.$$

## V GRAPHICAL RESOLUTION OF THE EQUATION

$$\frac{x^2+1}{x-1} = m$$

By graphing the function

$x \mapsto f(x) = \frac{x^2+1}{x-1}$  in an orthonormal system (see the above picture) and the constant function  $x \mapsto g(x) = m$  (which is a straight line parallel to the  $x$  axis

of equation  $y = m$ ), we can see that  $\frac{x^2+1}{x-1} = m$  has:

- ❖ Two distinct roots if  $m < 2 - \sqrt{2}$ ,
- ❖ One root only equal to  $1 - \sqrt{2}$  if  $m = 2 - 2\sqrt{2}$ ,
- ❖ No roots if  $2 - 2\sqrt{2} < m < 2 + 2\sqrt{2}$ ,
- ❖ One root only equal to  $1 + \sqrt{2}$  if  $m = 2 + 2\sqrt{2}$ ,
- ❖ Two distinct roots if  $m > 2 + 2\sqrt{2}$ .

The balance of the supply and of the demand of a product is by definition achieved when unit price  $p$  is the same one and consequently  $P_1(x) = P_2(x)$ . The quantity  $x_0$  which carried this equilibrium is then the abscissa of the point of intersection  $M$  of the branches of these two curves corresponding to  $x > 0$ . The point of market equilibrium is thus, as the graph shows,  $M(3, 96)$ . This shows that the equilibrium is realized for  $x = 3$  (means for an equal supply to the demand, equal to 3000 units). The price of equilibrium is thus  $p = 96\ 000$  LL. These results can be verified in writing  $10x^2 + 2x = 150 - 6x^2 \Leftrightarrow 16x^2 + 2x - 150 = 0$ . The solutions of this equation of second degree are 3 and  $-\frac{25}{8}$ . The second solution is excluded and thus  $x_0 = 3$  is the only acceptable solution for which  $p_0 = 96$ .

## FOCUS

To study a function and represent it graphically, we proceed in the following steps:

- ❖ We determine the domain of definition
- ❖ We study the parity of  $f$ ,
- ❖ We study the limits at the boundaries of the intervals forming  $D_f$  and we determine the asymptotes if they exist,
- ❖ We determine the derivative function  $f'$  and we study its sign on the different intervals,
- ❖ We draw the table of variations of  $f$ ,
- ❖ We draw a table of values and we determine the points of intersection with the coordinate axes,
- ❖ We choose judiciously a system of two axes and we construct the asymptotes, if they exist, and the curve  $C_f$ .

# EXERCISES

1 Determine the following limits:

a)  $\lim_{x \rightarrow +\infty} \frac{2x+1}{x-1}$  b)  $\lim_{x \rightarrow -\infty} \frac{x^2+5}{4x-1}$

c)  $\lim_{x \rightarrow -\infty} \frac{x-1}{x^2+4}$  d)  $\lim_{x \rightarrow 1} \frac{3x-1}{x-1}$   
 $x > 1$

e)  $\lim_{\substack{x \rightarrow 2 \\ x > 2}} \frac{2x-5}{x-2}$  f)  $\lim_{x \rightarrow 2} \frac{x-5}{x-2}$   
 $x < 2$

g)  $\lim_{x \rightarrow 3} \frac{x^2-1}{x-3}$  h)  $\lim_{x \rightarrow 3} \frac{x^2-1}{x-1}$

i)  $\lim_{x \rightarrow 1} \frac{x^2+x-2}{x-1}$

2 Study the parity of the following functions:

a)  $x \mapsto f(x) = \frac{1+2x^2}{x}$ ,

b)  $x \mapsto g(x) = \frac{x^2-5}{x^2+1}$ ,

c)  $x \mapsto h(x) = \frac{x}{x-1}$ .

3 Study and represent the following functions graphically:

a)  $x \mapsto f(x) = 2x^2 + 12x + 16$ ,

b)  $x \mapsto g(x) = -x^2 + 4x + 5$ .

4 a) Study and represent the function :  
 $x \mapsto f(x) = x^2 - 2x + 3$  graphically

b) Determine graphically the coordinates of the points of intersection of the curve obtained with the straight line of equation  $y = 6$ .

c) Determine graphically the coordinates of the points of intersection of the curve with the straight line of equation  $y = -x + 5$ .

d) Verify by calculation the results obtained in b) and c).

5 Determine graphically the coordinates of the points of intersection of the two parabolas of equations

$y = x^2 - 2x + 4$  and

$y = 2x^2 + 2x - 1$

respectively and verify the results by calculation.

6 Study and represent graphically the functions in pointing the intersections with the axes :

a)  $x \mapsto f(x) = \frac{x^3}{2} - 4$

b)  $x \mapsto g(x) = x^3 - 3x$

c)  $x \mapsto h(x) = x^3 - 3x^2$ .

7 Study and represent graphically the following functions by giving their intersections with the axes:

a)  $x \mapsto f(x) = \frac{2x-1}{x+1}$ ,

b)  $x \mapsto g(x) = \frac{1-x}{x-2}$ .

## VI APPLICATION TO ECONOMICS

To illustrate the use of the graphical representation of rational functions in economical situations, we will solve the problem of supply and demand according to:

### Problem

Let us suppose that, for a given product, the unit price and the demand are bound by

$$p = 150 - 6x^2$$

where  $p$  is the price in thousands of LL and  $x$  is the number of thousands of units of this product required by the consumers.

Let us suppose in addition that the unit price and the offer are bound by

$$p = 10x^2 + 2x$$

where  $x$  is the number of thousands of units offered on the market by the producer and  $p$  is expressed in thousands of LL.

Represent graphically these two functions

$x \mapsto 150 - 6x^2$  and  $x \mapsto 10x^2 + 2x$  and determine graphically the market point of equilibrium. Then check your result by calculation.

If 2000 units are demanded, then  $x = 2$ .

If 3000 units are offered on the market, then  $x = 3$ .

### Solution

- ❖ The domain of definition of the function

$P_1: x \mapsto 150 - 6x^2$  is  $\mathbb{R}$  and its derivative is defined by

$P_1'(x) = -12x$ , from where the following table of variation:

| $x$       | $-\infty$ | 0   | $+\infty$ |
|-----------|-----------|-----|-----------|
| $P_1'(x)$ | +         | 0   | -         |
| $P_1$     | $-\infty$ | 150 | $-\infty$ |

and the table of values

| $x$      | 0   | 1   | 2   | 3  | 5 |
|----------|-----|-----|-----|----|---|
| $P_1(x)$ | 150 | 146 | 126 | 96 | 0 |

$P_1$  is an even function. We can limit our study on the interval  $[0, +\infty[$ .

- 3 The limit at  $-\infty$  of the function

$$x \mapsto f(x) = \frac{2x+5}{4-x} \text{ is}$$

- a) -1      b) 0      c) -2.

- 4 An asymptote to the represented curve of the function

$$x \mapsto f(x) = \frac{2x+3}{2x-1}$$

is the straight line of equation:

- a)  $x = 1$ ; b)  $x = -1$ ; c)  $x = \frac{1}{2}$ .

- 5 If  $f$  is a function that verify

$$\lim_{x \rightarrow +\infty} (f(x) + 3x - 1) = 0,$$

then an asymptote at  $+\infty$  to the represented curve of  $f$  is the straight line of equation:

- a)  $y = 3x - 1$   
b)  $y = -3x + 1$   
c)  $y = -3x - 1$ .

- 6 Let  $P$  be the parabola of equation  $y = ax^2 + bx + c$  and  $D$  the straight line of equation  $y = mx + n$ . It follows that :

a)  $P$  and  $D$  have always 2 points of intersection,

b)  $P$  and  $D$  can have one and only one point of intersection,

c)  $P$  and  $D$  have always at least one point of intersection.

## SELF-EVALUATION

- 1 a) Represent graphically the parabola with equation  $y = x^2 - 5x + 4$ .

b) Determine graphically the coordinates of the points of intersection of this parabola with the axes and the straight line of equation  $y = -2$ .

c) Verify these results by calculation.

- 2 a) Represent graphically the hyperbola of equation  $y = \frac{x+2}{x-1}$ .

b) Determine graphically the coordinates of the points of intersection of this hyperbola with the axes and with the straight line of equation  $y = 2$ .

c) Verify these results by calculation.

- 8 a) Study and represent graphically the function  $f$  defined by

$$f(x) = \frac{3x+2}{2x+4}.$$

- b) Determine graphically the solutions of the equations

$$\frac{3x+2}{2x+4} = \frac{5}{2} \quad \text{and} \quad \frac{3x+2}{2x+4} = 1.$$

- c) Verify the results by calculation.

- 9 Solve the equations graphically:

$$\frac{2}{1+x} = -2 \quad \text{and} \quad \frac{2}{1+x} = -1.$$

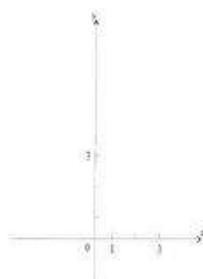
- 10 Determine graphically the coordinates of the points of intersection of the hyperbola of equation

$$y = \frac{x+2}{x-1} \quad \text{and the straight line of}$$

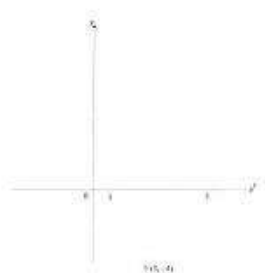
equation  $y = 3x - 2$  and verify these results by calculation.

- 11 Determine the equations of the parabolas represented below by using the coordinates of the points indicated on the graphs.

a)



b)



- 12 Given the function  $f$  defined by

$$f(x) = x - 2 + \frac{1}{x+1}.$$

- a) Show that for  $x \neq -1$ , we have

$$f'(x) = \frac{x(x+2)}{(x+1)^2}.$$

- b) Study the sense of variation of  $f$ .

- c) Study the limits at  $-\infty$ , at  $+\infty$  and at  $-1$  of  $f$ .

Deduce an asymptote  $D$  to the represented curve  $C$  of  $f$ .

- d) Show that the straight line  $D'$  of equation  $y = x - 2$  is an oblique asymptote to  $C$ .

- e) Construct  $D$ ,  $D'$  and  $C$ .

- 13 a) Study and represent graphically the function  $f$  defined by

$$f(x) = \frac{x^2+3}{x-1}.$$

- b) Discuss according to the values of  $m$  the number of solutions of the

$$\text{equation } \frac{x^2+3}{x-1} = m.$$

## M. C. Q.

- 1 The function  $x \mapsto f(x) = \frac{x^2}{x^2-1}$  is:

- a) even  
b) odd  
c) neither even nor odd.

- 2 The function  $x \mapsto f(x) = \frac{x}{x-1}$  is

- a) even  
b) odd  
c) neither even nor odd.

- 7 Suppose that the cost average of production per unit of a certain product is given by

$$C_m(x) = \frac{500}{x+30}.$$

a) Find the average cost for a production per unit when we produce 10 units, 20 units, 50 units, and 100 units.

b) Represent the function  $C_m$  graphically

- 8 To calculate the dose of a medication for a child, the pharmacists can use the formula  $d(x) = \frac{Dx}{x+12}$

where  $x$  is the weight of the child expressed in kg and  $D$  is the dose to the adult.

Suppose that the dose for an adult of a certain medicament is 70 mg.

a) Represent graphically the function  $d$ .

b) What is the weight of the child when we proportionally manage a 50 mg dose?

- 9 The formula which relates the quantity produced of kerosene  $x$  with that of fuel  $y$  from a quantity of crude oil is:  $y = \frac{125000 - 25x}{125 + 2x}$ .

$x$  and  $y$  being expressed in liters.

a) Represent for  $x \geq 0$  and  $y \geq 0$  the function  $x \mapsto \frac{125000 - 25x}{125 + 2x}$ .

b) Find the maximum quantities that can be produced by each of these products?

c) Determine graphically the quantity of kerosene that can be produced by producing 100 liters of fuel.

Verify this results by calculation.

- 10 Suppose that the state revenue, from applying a tax of  $x\%$ , with  $50 \leq x \leq 100$ , in millions of dollars, is given by  $t(x) = \frac{60x - 6000}{x - 120}$ .

a) Find the state income corresponding to a tax of 50%, 60%, 80% and 100%.

b) Represent graphically the function  $t$ .

## JUST FOR FUN

Write the formula defined by  $f(x) = \frac{2x^2 + 3x + 4}{x + 1}$

in the form  $f(x) = ax + b + \frac{c}{x+1}$  and represent it graphically.

# SITUATIONS - PROBLEMS LEADING TO THE SOLUTIONS OF EQUATIONS AND INEQUALITIES

## INTRODUCTION

Often the study of an economical, social or any other situation leads to equations or inequalities of the first degree or second degree in one unknown or to a linear system of equations/inequalities in many unknowns. Those unknowns can be for instance the quantities of several products to be manufactured by a factory knowing that those quantities are submitted to some given constraints, which means to verify certain particular equations or/and inequalities.

The mathematical solution of these equations/inequalities or the systems of equations/inequalities allows us to find possible values of these unknowns. Exploitation of these results makes it possible to make good decisions.

The objectives of this chapter are:

1. To analyze a problem and to express it in terms of equations and/ or inequalities.
2. To solve equations and/or inequalities and to verify the validity of the found solutions.

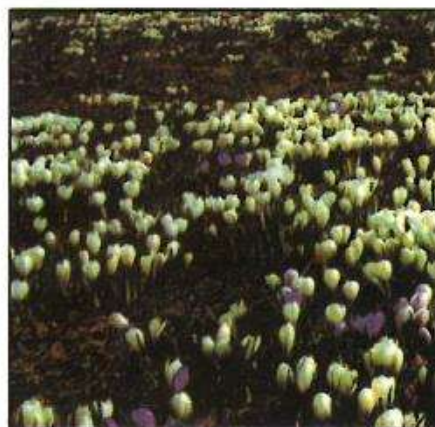
### Prerequisites

- ❖ Equations and inequalities of the first degree in one unknown.
- ❖ System of equations and inequalities of the first degree in two unknowns.
- ❖ Equations and inequalities of the second degree in one unknown.
- ❖ Graphical solution of equations and inequalities.

## ACTIVITIES

### Activity 1. Two pieces of information is good!

The perimeter of a rectangular garden is  $300m$  and its area is  $5000 m^2$ . Determine the dimensions of the garden.



# PROBLEMS

- 1 The profit obtained by the sale of a high technology product is given by the equation  $p(x) = -x^2 + 60x - 400$  where  $x$  is the number of units sold and  $p(x)$  is expressed in thousands of dollars..

a) Represent this function graphically for  $x \geq 0$ .  
 b) Determine the number of units that we must sell to carry out the maximum profit..  
 c) What is then the maximum profit?

- 2 When an object is launched vertically upwards, with an initial velocity of 32 m/s, then the height it reaches, is expressed in meters, after  $t$  seconds is

$$h(t) = 32t - 16t^2.$$

a) Find the maximum height of this object..  
 b) After how many seconds will this object touch again the ground?

- 3 Suppose the unit price  $p$ , expressed in dollars, is dependent on the demand  $x$  (measured in thousands of units) by the formula

$$p = 640 - 5x^2 \text{ and is dependent to the supply by the formula } p = 5x^2$$

a) Represent in a system of 2 axes the two functions

$$x \mapsto P_1(x) = 640 - 5x^2 \text{ et}$$

$$x \mapsto P_2(x) = 5x^2.$$

b) Find the supply that carry out the equilibrium of the market.  
 c) Find the price of equilibrium..

- 4 The functions of the supply and of demand for a given product are

$$q_o = p^2 - 100 \quad \text{and}$$

$$q_d = p^2 - 40p + 400$$

where  $p$  designate the unit price and  $q$  the quantity Determine the price and the quantity, which carry out the equilibrium of the market.

- 5 The total cost function is given by  $C(x) = 100x^2 + 1300x + 1000$  Where  $x$  is the number of thousands of units and  $C(x)$  is expressed in thousands of dollars. The price of selling of one unit of this product is 2000 dollars.

a) Determine the function  $R(x)$  expressing the revenue, where  $x$  and  $R(x)$  being expressed like for the cost function.

b) Determine the necessary level of production to realize no profit.

c) Determine the level of production that realizes the maximum profit.

d) Calculate then this profit..

- 6 The function  $A(x) = -0.1x^3 + x$  give roughly the approximation of the blood alcohol content for a person average  $x$  hours after the absorption of 8 cl of an alcoholic drink.

a) Represent this function graphically for  $0 \leq x \leq 3$ .

b) Estimate graphically the time  $x_0$  that corresponds to the maximum absorption rate of alcohol..

c) Use the graph to estimate the period during which this person is legally considered drunk if the statutory rate of blood content not to be exceeded is 1.

### Choosing the unknown :

The unknown is here, the number of years at the end of which the father's age will be equal to the sum of his sons' ages.

Let  $x$  be that number.

### Forming the equation :

The statement of the problem is translated by the following linear equation in one unknown:

$$3 + x + 5 + x + 8 + x = 36 + x$$

Which can be written as  $16 + 3x = 36 + x$  (E)

A linear equation (resp. inequality) of the first degree is an equation (resp. inequality) of the form

$$ax + b = cx + d$$

(resp.  $ax + b \leq cx + d$  or

$$ax + b < cx + d).$$

### Solving the equation

The equation (E) is written as  $3x - x = 36 - 16$ , therefore  $2x = 20$  and  $x = 10$ .

### Verification of the result

In 10 years, the father's age will be 46 years and his sons ages will be respectively 13, 15 and 18 years and it is clear that:

$$46 = 13 + 15 + 18.$$

### ❖ Example 2

The sum of 3 consecutive positive integers is less than or equal to 13.

What are the possible answers?

### Choosing the unknown

These numbers being consecutive, it is enough to know one of them. Let  $x$  be the smallest number.

## Forming the inequality

The statement of the problem can be translated by the following linear inequality:  $x + x + 1 + x + 2 \leq 13$  which is written as  $3x + 3 \leq 13$  (I).

## Solving the inequality

The inequality (I) is written as  $3x \leq 13 - 3 = 10$  where  $x \leq \frac{10}{3}$ .

## Control and verification of the result

$x$  being a positive integer it can only be 1, 2, or 3. The three possibilities of three positive integers are then: 1, 2 and 3 ; 2, 3 and 4 ; 3, 4 and 5.

It is clear that :

$$1 + 2 + 3 = 6 \leq 13$$

$$2 + 3 + 4 = 9 \leq 13$$

$$3 + 4 + 5 = 12 \leq 13.$$

## II

## SITUATIONS LEADING TO AN EQUATION OR INEQUALITY OF THE SECOND DEGREE IN ONE UNKNOWN

Here are two examples of such situations :

### ❖ Example 1

A household device having an initial price of 800\$ is submitted to two successive price discounts, the second discount percentage is the double of the first.

What is the first discount percentage, knowing that the device is sold for 576\$?

When a price discount is carried out as a percentage of  $x\%$ , it means that if the price was 100\$, the discount will be  $x$  \$ and the new price will be  $(100 - x)$  \$.

## Choosing the unknown

Let  $t\%$  be the percentage of the first discount then the second one is  $2t\%$ .

## Activity 2. Sometimes a third is needed !

**A** - 30 bills, some are of 10,000 LL, the others are of 50 000 LL. make a total sum of 70,000LL.

Find the number of bills of each category.

**B** - 50 bills, some are of 5,000LL., others are of 10,000 LL. and the rest are of 20,000LL., form a sum of 375,000LL. Find the number of bills of each category, knowing that the number of the 5,000LL. bills is equal to the sum of the number of bills of the 2 other categories.

## TEXT

In this chapter, the different applied problems leading to the solution of equations or inequalities are illustrated in some particular examples, their treatment is carried out according to the following general scheme:

1. Choose the unknown or the unknowns.
2. Express the given of the problem, in the form of equation(s) or inequality(ies).

This permits to establish a mathematical model corresponding to the given situation (or constraints)

3. Solve the obtained mathematical model.
4. Control, verify and interpret the obtained results.

The great mathematician "Al' khawarizmi", who wrote "Hissâb al-Jabr wa'l-muqqâbala" contributed in the evolution of algebra, in general, and particularly in solving equations.

## I SITUATIONS LEADING TO AN EQUATION OR INEQUALITY OF THE FIRST DEGREE IN ONE UNKNOWN

Here are two examples of such situations :

### ❖ Example 1

A father is 36 years old and his sons are 3,5 and 8 years old respectively. In how many years will the age of the father be equal to the sum of his sons' ages?

### a) Choosing the unknown

The unknown is the price, which establishes the market balance.

### Forming the equation

The market balance is/are the value(s) of  $p$  which verify(ies) the equation:

$$0.5p^2 - 200 = p^2 - 100p + 2500$$

$$\text{Written as : } 0.5p^2 - 100p + 2700 = 0 \quad (E).$$

### Solving the equation

The equation (E) has two solutions :  $p_1 = 32.1767 \approx 32.18$  and  $p_2 = 167.823$ .

### Control and verification of the results

Only the solution  $p_1$  is in the interval  $[20, 45]$ ; the second solution is rejected and we will have:

$$f(p_1) = g(p_1) \approx 317.67.$$

The market balance price is then 32.18\$, the supply and the demand will be equal to 317.670 units..

### b) Forming the inequality

For the supply to be higher than the demand then the price  $p$  must verify the following inequality :

$$0.5p^2 - 200 > p^2 - 100p + 2500$$

which is written as  $0.5p^2 - 100p + 2700 < 0 \quad (I)$ .

### Solving the inequality

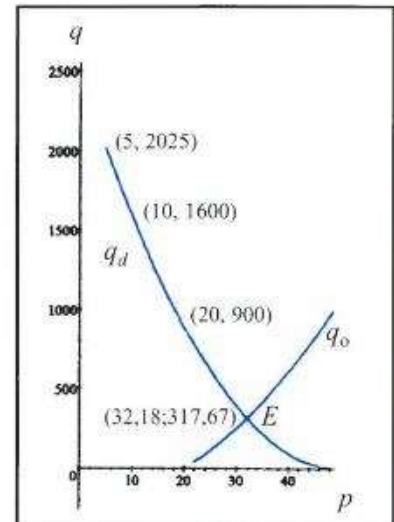
This inequality is verified for the values of  $p$  which verify the following double inequality:

$$32.1767 < p < 167.823$$

### Control and verification of the results

The price interval in which the supply is higher than the demand is  $[32.18; 45]$ .

The market balance is ensured when the supply is equal to the demand



The market balance is the point  $E(32.18 ; 317.67)$  which is the point of intersection of the two parabolas of equations  $q = 0.5p^2 - 200$  and  $q = p^2 - 100p + 2500$  represented graphically in a given coordinate system.

When the trinomial  $ax^2 + bx + c$  has 2 distinct roots ( $\Delta > 0$ ), the trinomial has an opposite sign to  $a$ , between the roots and the same sign as  $a$  at the exterior of the roots.

The trinomial has the same sign as  $a$  in the other cases.

## Forming the equation

At the end of the first discount, the price of the device is:

$$800 - \frac{800 \times t}{100} = (800 - 8t) \$.$$

At the end of the second discount the price of the device is:

$$800 - 8t - \frac{(800-8t) \times 2t}{100} = 800 - 8t - \frac{1600t - 16t^2}{100} =$$

$$800 - 8t - 16t + 0.16t^2 = 0.16t^2 - 24t + 800.$$

the given is expressed by the equation :

$$0.16t^2 - 24t + 800 = 576$$

$$\text{or } 0.16t^2 - 24t + 224 = 0.$$

## Solving the equation

This equation admits two solutions  $t_1 = 10$  and  $t_2 = 140$ .

(You can find here opposite a reminder of the resolution of the second degree and the factorization of trinomials.)

## Control the validity and verification of the results

The discount percentage can not be more than 100 (otherwise we get a negative price). The only acceptable solution is then  $t_1 = 10$ . The first discount percentage is 10% and the second is 20%.

We can verify that  $800 - \frac{800 \times 10}{100} = 720$  and

$$720 - \frac{720 \times 20}{100} = 720 - 144 = 576.$$

Recall that if  $ax^2 + bx + c = 0$  is an equation of the second degree then 3 cases can be present:

**1st case:**  $\Delta = b^2 - 4ac > 0$

The equation has then two solutions:

$$x_1 = \frac{-b - \sqrt{\Delta}}{2a} \quad \text{and}$$

$$x_2 = \frac{-b + \sqrt{\Delta}}{2a}$$

and we will have

$$ax^2 + bx + c = a(x - x_1)(x - x_2).$$

**2nd case:**  $\Delta = 0$

The equation has only one solution

$$x_0 = -\frac{b}{2a} \quad \text{and we have:}$$

$$ax^2 + bx + c = a(x - x_0)^2$$

**3rd case:**  $\Delta < 0$

The equation does not have any solution and the trinomial  $ax^2 + bx + c$  can not be factorized.

## ❖ Example 2

The supply of a particular product is expressed, as a function, in terms of the unit price varying between 20\$ and 45\$ by

$$q_0 = f(p) = 0.5p^2 - 200$$

and the demand function is expressed in terms of the unit price varying between 5\$ and 45\$ by:

$$q_d = g(p) = p^2 - 100p + 2500.$$

In these two expressions  $p$  is expressed in dollars and  $q$  in thousands of units.

a) What is the price, which ensures the market balance?

b) Determine the price interval for which the supply is higher than the demand.

## Forming the equations

The given of this problem is translated by the following linear system of two equations in two unknowns:

$$\begin{cases} 150 = 40m + p & L_1 \\ 300 = 90m + p & L_2 \end{cases}$$

## Solving the system

By subtracting the equation  $L_1$  from  $L_2$ , we obtain  $150 = 50m$  which gives  $m = 3$  and by substituting the value of  $m$  in  $L_1$ , we obtain  $150 = 120 + p$  which gives  $p = 30$ .

$m$  is in fact the slope of the straight line  $(D)$ . Then we could have determined it by the formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{300 - 150}{90 - 40} = \frac{150}{50} = 3$$

So the function  $S$  is in the form of  $S(x) = 3x + p$  and by writing  $S(40) = 120 + p = 150$ , we obtain  $p = 30$ .

## The interpretation of the results

The function  $S$  is defined by  $S(x) = 3x + 30$ , the profit of each sold unit is 3\$ and the weekly salary corresponding to selling 124 units is:

$$S(124) = 3 \times 124 + 30 = 402 \$.$$

### ❖ Example 2

A craftsman manufactures two kinds of furniture, some of type  $A$  and the others of type  $B$ . The manufacturing cost of a furniture of type  $A$  is 100\$ and that of type  $B$  is 300\$, the craftsman can not produce more than 6 pieces per days.

How many units of each type can he produced per day, knowing that the total cost of fabrication must not exceed 900\$ and knowing that he must produce at least 3 units of type  $A$  and one unit of type  $B$

## Choosing the unknowns

The unknowns of this problem are the number of units to be manufactured of each type denoted by  $x$  and  $y$ .

By substituting  $a$  by its value in  $L_1$ , we obtain

$2b = 750,000 - 8 \times 112,500 = -150,000$ , which gives

$b = -75,000$  and by substituting  $a$  and  $b$  by their values in

the equation  $L_1$ , we obtain  $c = 200,000 - 112,500 +$

$75,000$ , which gives  $c = 162,500$ .

then the equation of the parabola is

$$y = 112,500x^2 - 75,000x + 162,500.$$

b) On this graph the year 1995 corresponds to the abscissa  $x = 11$ . The number of subscribers is then:

$$112,500 \times 121 - 75,000 \times 11 + 162,500 = 12,950,000.$$

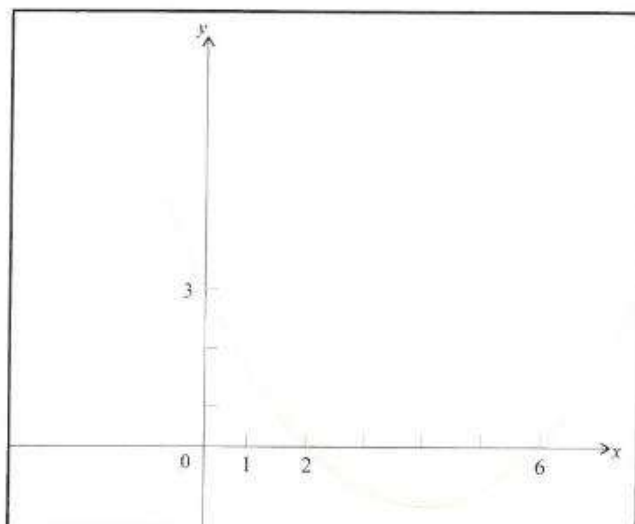
It can be easily verified if  $x$  equals to 1, 3 and 5,  $y$  equals to: 200,000; 950,000 and 2,600,000

## Remark

Sometimes the determination of 3 unknowns in a problem is considerably simplified as is shown in the following example.

### ❖ Example

The temperature readings measured every second permitted to trace the following curve in a parabolic form. Determine the equation of this curve..



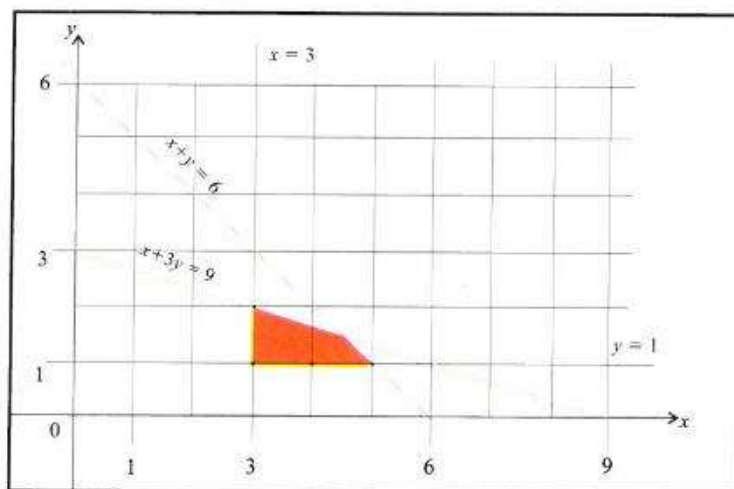
Karl-Freidrich Gauss (1777 - 1855).

## Forming the inequality

The constraints of the problem are translated by the following linear system of inequalities:

$$\begin{cases} x+y \leq 6 \\ x \geq 3 \\ y \geq 1 \\ 100x+300y \leq 900 \end{cases} \quad \text{or} \quad S \quad \begin{cases} x+y \leq 6 \\ x+3y \leq 9 \\ x \geq 3 \\ y \geq 1 \end{cases}$$

## Graphical resolution of the system



In this graph we have represented the straight lines of the equations:  $x + y = 6$ ,  $x + 3y = 9$ ,  $y = 1$  and  $x = 3$

The dashed area is the accepted region associated to the system  $S$ . It is the set of points  $M(x, y)$  such as that real numbers  $x$  and  $y$  verify the system  $S$ .

## Control and verification

An acceptable solution of the problem is the couple of integers  $x$  and  $y$  verifying  $S$ . A graphical reading allows to find 4 acceptable solutions:  $(3, 1)$ ;  $(3, 2)$ ;  $(4, 1)$  and  $(5, 1)$ .

We can easily verify that each one of these pairs verifies the system  $S$ .

## IV

## SITUATIONS LEADING TO A LINEAR SYSTEM OF 3 EQUATIONS IN 3 UNKNOWNNS

The following is an example of a such situation :

### III

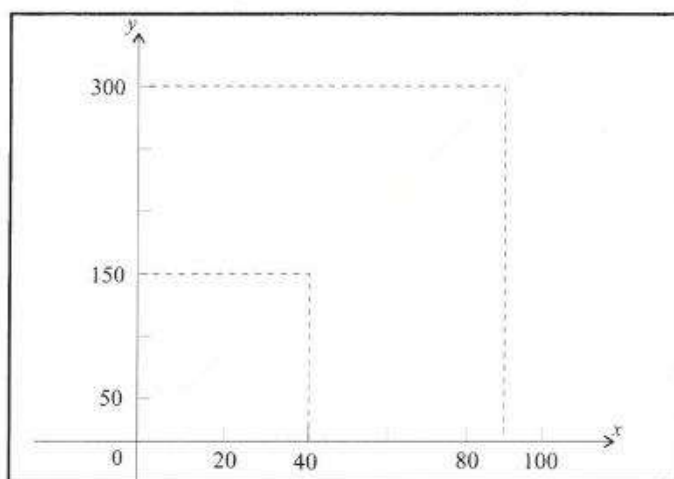
## SITUATIONS LEADING TO A LINEAR SYSTEM OF EQUATIONS OR INEQUALITIES IN TWO UNKNOWNNS

Here are two examples of such situations :

### ❖ Example 1

The weekly salary, in dollars, of a sale agent consists of a fixed sum, to which a sum proportional to the number of sold units is added.

Below is the graphical representation of his weekly salary  $y$  in terms of  $x$ , the number of sold units, on which we can read  $y = 150$  for  $x = 40$  and  $y = 300$  for  $x = 90$ .



How much does he gain for each unit sold and how much is the amount of his weekly salary if he sells 124 units?

### Choosing the unknown

The function that gives, in dollars, the weekly salary, is defined by:

$$y = S(x) = mx + p$$

Where  $x$  is the number of sold units,  $m$ , in dollars, is the amount which is gained by every unit sold and  $p$ , in dollars, is the amount of the fixed weekly salary.

The straight line represented above has the equation

$$y = mx + p.$$

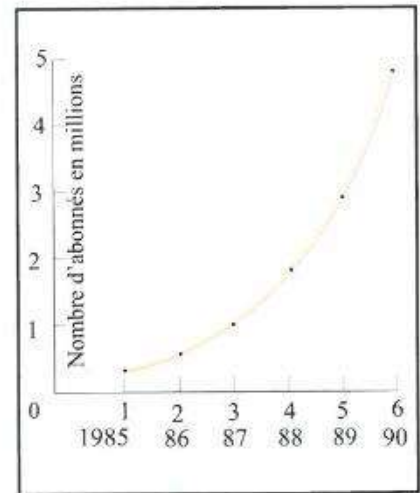
$m$  and  $p$  are the unknowns of the problem.

### ❖ Example

The mobile phone industry in the United States is increasing rapidly since 1985. The statistics done between 1985 and 1990 have permitted the construction of the curve in the diagram, which can be easily found in the form of a parabola.

a) Determine the equation of this parabola knowing that the numbers of subscribers was: 200,000, 950,000 and 2,600,000 in 1985, 1987, 1989 respectively.

b) What is the expected number of subscribers in 1995?



### a) Choosing the unknowns

The equation of this parabola is in the form  $y = ax^2 + bx + c$ , so the unknowns in this problem are  $a$ ,  $b$  and  $c$ .

### Forming the equation

By substituting 1, 3 and 5 for  $x$  in the following equation we must obtain respectively 200,000, 950,000 and 2,600,000 this leads to the following linear system of 3 equations in 3 unknowns.:

$$(S) \quad \begin{cases} 200,000 = a + b + c & L_1 \\ 950,000 = 9a + 3b + c & L_2 \\ 2,600,000 = 25a + 5b + c & L_3 \end{cases}$$

### Solving the system

By subtracting the equation  $L_1$  from  $L_2$ , we obtain

$$8a + 2b = 750,000 \quad L'_1$$

And by subtracting the equation  $L_2$  from  $L_3$ , we obtain

$$16a + 2b = 1,650,000 \quad L'_2$$

Finally by subtracting the equation  $L'_1$  from  $L'_2$ , we obtain

$$8a = 900,000, \text{ which gives}$$

$$a = 112,500.$$

The method used to solve the system is called the method of linear combination.

It can also be solved by the substitution method. These 2 methods are exposed in the opposite paragraph.

2°) Consider the following system:

$$(S) \quad \begin{cases} x+2y-3z=5 & L_1 \\ -2x-y+z=6 & L_2 \\ -2x+2y-4z=22 & L_3 \end{cases}$$

$$\begin{aligned} L_1 + 3L_2 &\rightarrow -5x - y = 23 & L'_1 \\ L_3 + 4L_2 &\rightarrow -10x - 2y = 46 & L'_2 \end{aligned}$$

We notice that  $L'_2 = 2L'_1$ , so  $L'_2$  and  $L'_1$  are equivalent. The previous system of the two equations in two unknowns admits then an infinite number of solutions. The system  $S$  admits also an infinite number of solutions; it is enough to take any  $x$ ,  $y = -5x - 23$  and  $z = 6 + 2x + (-5x - 23) = -3x - 17$  for the triplet  $(x, -5x - 23, -3x - 17)$  to verify the 3 equations of the system  $S$ .

We could notice also that  $L_3 = 2L_1 + 2L_2$ .

## FOCUS

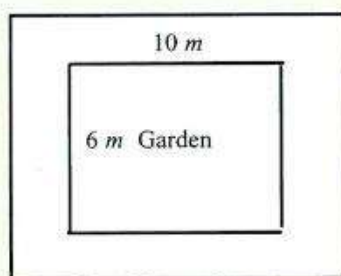
**To solve a given mathematical problem, we must follow the following steps:**

1. read the problem carefully,
2. determine the unknowns of the problem,
3. express every information (or constraints) in the given problem, in an equation (or inequality),

4. solve in an appropriate way the obtained mathematical model,

5. check the validity of the obtained solutions.

- 11 A landscape designer wants to cover with gravel a tape of uniform width around rectangular garden 10m in length and 6m in width. He has just enough gravel to cover  $36\text{m}^2$ . What must be the width of the tape?



- 12 A company manager estimates that total cost of manufacturing  $x$  units of his product is expressed, in dollars, by  $C(x) = 20x + 60$  and the income net of the sale of  $x$  units is expressed, in dollars, by  $R(x) = x^2 - 8x$ . What is the minimum number of units he must sell so that the production is profitable?
- 13 The cost of production of  $x$  units of a product is expressed, in dollars, by  $C(x) = 40x + 150$  and the net selling income of  $x$  units, expressed in dollars, is  $R(x) = 65x - x^2$ . Determine the profit interval of this company.
- 14 Solve by using the substitution and the linear combination methods, the following system:

$$(S) \quad \begin{cases} 2x + y - z = 2 \\ x + 3y + 2z = 1 \\ x + y + z = 2 \end{cases}$$

- 15 Solve the following systems:

$$S_1 \quad \begin{cases} 2x + y - z = 5 \\ 3x + 5y + 2z = 3 \\ 4x + 2y - 2z = 0 \end{cases}$$

$$S_2 \quad \begin{cases} 2x + y - z = 5 \\ 3x + 5y + 2z = 3 \\ 4x + 2y - 2z = 1 \end{cases}$$

- 16 A company director shares an exceptional premium of 7,000\$ among 3 employees, proportionally to their seniority: 15 years, 8 years and 5 years. How much does he give to each one of them?
- 17 Determine the equation of the parabola that passes through the points  $A(2, 15)$ ,  $B(-3, 5)$  and  $C(1, 5)$
- 18 Determine the equation of the parabola that passes through the points  $A(-2, 0)$ ,  $B(3, 0)$  et  $C(0, -6)$ .
- 19 Determine the equation of the parabola that passes through the point  $A(3, 6)$  and having the point  $S(1, 2)$  as a vertex .

By substituting again  $y$  by  $9 - 6z$  in  $L'_1$ , we obtain:

$$9z - 4(9 - 6z) = -3 \text{ or } 9z - 36 + 24z = -3 \text{ which gives}$$

$$33z = 33 \text{ and hence } z = 1 \text{ so}$$

$$y = 9 - 6 = 3 \text{ and } x = 2z = 2.$$

B- the linear combination method consists of replacing any equation of the system by a linear combination of the other two. It is admitted that the new obtained system is equivalent to the initial one.

This operation can be repeated a second time if necessary in order to lead the resolution of the system to a system of two equations in two unknowns.

A linear combination of two equations for examples  $L_1$  and  $L_2$  is systematically represented by:  $\alpha L_1 + \beta L_2$  where  $\alpha$  and  $\beta$  are any two real numbers.

This notation means that the two sides of the equation  $L_1$  are multiplied by  $\alpha$  as well the equation  $L_2$  by  $\beta$  and then we add side by side the 2 obtained equations.

Two systems are equivalent if they have the same solution set.

### ❖ Example

Consider the following system :

$$(S) \quad \begin{cases} x + y + z = 6 & L_1 \\ 2x - y + 3z = 4 & L_2 \\ 4x + 5y - 10z = 13 & L_3 \end{cases}$$

$$L_2 - L_1 \rightarrow -3y + z = -8 \quad L'_1$$

$$L_3 - 4L_1 \rightarrow -14z = -11 \quad L'_2$$

$$L'_1 + 3L'_2 \rightarrow -41z = -41 \Rightarrow z = 1$$

$$y - 14 = -11 \Rightarrow y = 3 \text{ and } x = 6 - 3 - 1 = 2$$

The unique solution of the system is the triplet (2, 3, 1)

### Remark

1°) Consider the following system

$$(S) \quad \begin{cases} -2x + y + 3z = 12 & L_1 \\ x + 2y + 5z = 10 & L_2 \\ 6x - 3y - 9z = 24 & L_3 \end{cases}$$

$$L_1 + 2L_2 \rightarrow 5y + 13z = 32$$

$$L_3 + 3L_1 \rightarrow 0 = 60$$

This last equality being impossible, the proposed system does not admit any solution.

The solution set of the system is the empty set

# EXERCISES

1 A capital is placed at a simple annual interest rate of 10%. After 2 years the value of this capital is 3,360\$. How much was the initial value of this capital?

2 Two amounts of money of 10,000\$ and 8,000\$ are deposited at a simple interest rate of 8% and 12% respectively. After how many years will the values of the two capitals be equal?

3 The difference between two positive integers is 60. If we add 10 to each one, the greater number is 3 times the smaller one. What are the two original integers?

4 Find two numbers having a difference between them of five and the bigger one is strictly bounded between the double and the triple of the smaller number. What are the possibilities?

5 An urn contains a certain number of white balls and 5 black balls. Determine the minimum number of white balls, in this urn, for the probability of drawing a white ball is greater than  $\frac{3}{5}$ .

6 Two amounts of money are deposited at a simple interest for one year: the first at an annual interest rate of 5% and the second at 8%. Their common revenue is 500\$. Determine the amount of these two capitals knowing that if they were deposited for one year, the first at an annual interest rate of 8% and the second at 5% their common revenue will be 410\$.

7 A librarian buys 25 books, some at 20,000LL. each and the others at 30,000 LL. each. How many books of each category did he buy, knowing that the total cost is 650,000 LL?

8 Find two numbers knowing that their sum is : -3 and the difference of their squares is:69

9 Represent graphically the set of solutions of the following linear system of inequalities:

a)

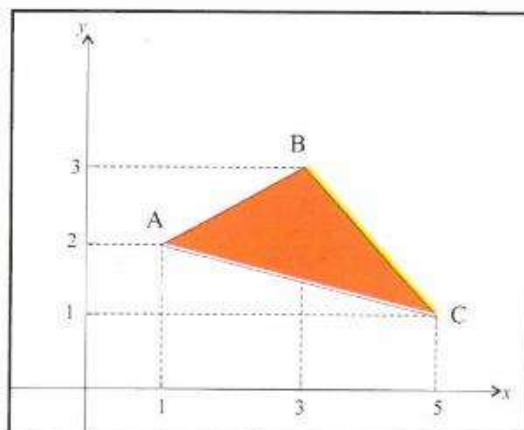
$$(S) \quad \begin{cases} x-y \geq 0 \\ x-3y+5 \leq 0 \\ x+y-6 \leq 0 \end{cases}$$

b)

$$(S') \quad \begin{cases} x+1 \leq 0 \\ 3x-4y+6 \leq 0 \\ 3x+4y \geq -8 \end{cases}$$

10 Consider the coordinate system

$(O; \vec{i}, \vec{j})$ . Characterize by a system of inequalities, the shaded region of the following figure.:



## Solution

The equation of this parabola is of the form:

$y = at^2 + bt + c$ . The unknowns of the problem are the numbers  $a$ ,  $b$  and  $c$ .

However, we know that the number  $c$  is the ordinate of the point of intersection of the curve with the ordinate axis, So we have then  $c = 3$ .

Similarly, by using the properties of the roots of a second order polynomial, we have

$$\frac{c}{a} = t_1 \times t_2 = 2 \times 6 = 12 \text{ which gives } c = 12a \text{ hence}$$

$$a = \frac{c}{12} = \frac{3}{12} = 0.25 \text{ and,}$$

$$-\frac{b}{a} = t_1 + t_2 = 2 + 6 = 8 \text{ which gives } -b = 8a \text{ hence}$$

$$b = -8 \times 0.25 = -2.$$

The equation of the parabola is:

$$y = 0.25t^2 - 2t + 3.$$

To solve this problem we can also use the following system obtained by the 3 points of intersection with the axes:

$$\begin{cases} 3 = 0 + 0 + c \\ 0 = 4a + 2b + c \\ 0 = 36a + 6b + c \end{cases}$$

Which gives by substitution the following system:

$$\begin{cases} -3 = 4a + 2b \\ -3 = 36a + 6b \end{cases}$$

Where its resolution gives  $a = 0.25$  and  $b = -2$ .

## V

## SUBSTITUTION METHOD AND LINEAR COMBINATION METHOD

To solve a linear system of 3 equations in 3 unknowns we can use the substitution method or the linear combination method which lead to the solution of a linear system of 2 equations in 2 unknowns. To solve the obtained system we use again one of the 2 following methods.

A- the substitution method consists of expressing one of three variables in function of the other two using one equation of the system equations and to substitute after that the variable in the two other equations.

### ❖ Example

Consider the system

$$\begin{cases} 2x - 4y + 5z = -3 & L_1 \\ 3x + y = 9 & L_2 \\ x - 2z = 0 & L_3 \end{cases}$$

$L_3$  gives  $x = 2z$

Then  $L_1$  is written  $4z - y + 5z = -3$  ou  $9z - 4y = -3$   $L'_1$

And  $L_2$  is written  $6z + y = 9$   $L'_2$

It is advisable to use the substitution method in the case where one of the 3 variables can be simply expressed in terms of the other two, or in terms of one of them only.

# JUST FOR FUN

We know that:

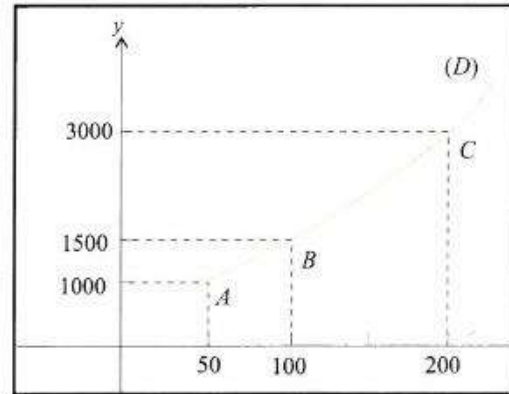
- 1- The sum of ages of: Adib, Rita, Maya and Said is 60 years.
  - 2- The sum of ages of: Adib and Said is equal to 3 times the sum of ages of: Maya and Rita.
  - 3- The sum of ages of: Adib and Rita is equal to 2 times the sum of ages of: Said and Maya.
  - 4- The sum of ages of: Adib and Maya is 10 years more than the sum of ages of: Rita and Said.
- a) Can you determine the age of each one of the 4 persons?
  - b) If we replace the information (4) by the following information:
- 5- When we subtract Maya's age of that of Adib's age, we obtain the sum of ages of Rita and Said,  
What do we get?

# SELF-EVALUATION

1 A large rectangular conference room measures 15 m in length and 12 m in width. We would like to place a fitted carpet there, by leaving a tape of uniform width all around the fitted carpet, without coating. We want to buy  $108 \text{ m}^2$  of carpet only. What must be the width of the tape?

2 60 bills of 5,000LL. ; 10,000LL. and 20,000LL. have a total value of 500,000LL.. Determine the number of bills of each category; knowing that the number of the 5,000LL. bills is equal to the double of the sum of the numbers of 10,000LL. and 20,000LL. bills.

3 The production cost of a product is graphically presented by the part of the parabola as shown in the following graph:  
Determine the equation of the parabola..



# PROBLEMS

1 A type of fertilizer is made of 3 components. It contains 10.8% of the first component and the quantities of the other 2 components are proportional to the 2 numbers 3 and 4. Determine the quantity of each component contained in 750 kg of fertilizers.

2 An animal food is composed of 3 ingredients: wheat, soya and corn. One unit of each ingredient provides the indicated numbers of protein, fats and fibers units as in the following table:

|         | Wheat | Soya | Corn |
|---------|-------|------|------|
| Protein | 2.5   | 4    | 2    |
| Fat     | 4     | 2    | 3    |
| Fibers  | 3     | 2    | 1    |

What is the necessary number of units of each ingredient to obtain a food quantity containing 220 units of protein, 280 units of fats and 180 units of fibers?

3 A dietician must prepare a dish composed of 3 ingredients whose contents in vitamin amount per unit, as well as the global amount of vitamins of the proposed dish are indicated in the following table:

| Ingredient               | Amount of vitamins per unit |           |           |
|--------------------------|-----------------------------|-----------|-----------|
|                          | Vitamin 1                   | Vitamin 2 | Vitamin 3 |
| 1                        | 4 mg                        | 2 mg      | 1 mg      |
| 2                        | 6 mg                        | 8 mg      | 6 mg      |
| 3                        | 3 mg                        | 4 mg      | 2 mg      |
| Total amount of vitamins | 52 mg                       | 56 mg     | 34 mg     |

How many units of each ingredient must she use?

- 4 An enterprise makes 3 types of assorted vegetable sold in 1 kg packing. The oriental assortment is composed of 0.3 kg of beans, 0.3 kg of peas and 0.4 kg of carrots. The French assortment is composed of 0.6 kg of peas, 0.4 kg of carrots. The Italian assortment is composed of 0.2 kg of beans, 0.5 kg of peas and 0.3 kg of carrots.

The enterprise has a stock of 16,200 kg of beans, 41,400 kg of peas and 29,400 kg of carrots.

How many packets of each assortment must the enterprise prepare in order to use all the stock?

- 5 A supermarket cashier has a total of 70 bills of 5,000LL, 10,000 LL, and 20,000LL. The number of bills of 5,000LL, is equal to 3 times the number of bills of 10,000 LL, and the total of all bills is 960,000LL.. Find the number of bills of each category.

- 6 The monthly salaries of 3 representatives  $A$ ,  $B$  and  $C$  are calculated as follows:

$A$  earns a fixed salary of 1,200,000 LL.

$B$  earns only a percentage of his sales, so he has earned 962,000LL. for a 22,640,000LL. of sales.

$C$  earns a fixed salary and a percentage of his sales, so he has earned once 1,128,000LL. for 25,700,000LL. of sales, and another time 1,088,875 LL. for 23,950,000LL. of sales.

a) Determine the basic percentage of the total sale income of the representative  $B$ .

b) Determine the fixed amount and the percentage which determine the salary of  $C$ .

c) We designate by  $f$ ,  $g$  and  $h$  the functions giving respectively the salaries of  $A$ ,  $B$  and  $C$  in function of the total income  $x$  realized.

i) Give the expression of:  $f(x)$ ,  $g(x)$  and  $h(x)$ .

ii) Represent in one coordinate system the 3 functions  $f$ ,  $g$  and  $h$ .

d) Which representative earns the most for 30,000,000LL. of sales?

e) What is the most advantageous remuneration mode?

7

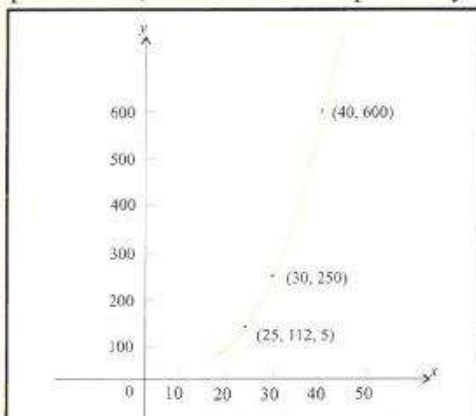
A factory manufactures 3 products. Each one of them must be produced in 3 departments:  $A$ ,  $B$  and  $C$ . the table below, indicates the necessary number of hours to produce one unit of each product, in each department and also the monthly work capacity in the 3 departments.

| Departement | Product |     |   | Available n° of hours per month |
|-------------|---------|-----|---|---------------------------------|
|             | 1       | 2   | 3 |                                 |
| A           | 2       | 3.5 | 3 | 1200                            |
| B           | 3       | 2.5 | 2 | 1150                            |
| C           | 4       | 3   | 2 | 1400                            |

Determine the quantity of each product that must be produced monthly, to use the entire capacity of the departments.

- 8 An investigation is carried out among the providers of a given product indicates that the supply is expressed as a function of the unit price  $p$  by a second degree equation for  $p$  between 20\$ and 50\$

Determine the supply function, knowing that later is 112.5, 250 and 600 (in thousand of units) for the prices 25\$, 30\$ and 40\$ respectively.



- 9 For the same product of the preceding exercise, an investigation among the consumers shows that the demand is expressed as a function of the unit price  $p$  by an equation of the second degree also for  $p$  between 5\$ and 50\$. Determine the demand function, knowing that for a 2025, 1600 and 900 (in thousands of units), the price is 5\$, 10\$ and 20\$ respectively.

- 10 a) For exercises 8 and 9, Determine the unit price and quantity which realize the market balance point (in other terms the supply is equal to the demand).  
b) Represent in a coordinate system, the 2 functions of the exercises 8 and 9 and indicate on the graph the market balance point.

- 11 A leisure club manager disposes of 1,400,000 LL. to buy 60 entry tickets for a show. There are many kinds of tickets: tickets of 25,000LL., 15,000LL., 10,000LL. After this purchase, the 1,400,000LL. are entirely spent. Let  $x$  be the number of purchased tickets of 25,000LL.,  $y$  the number of purchased tickets of 15,000LL. and  $z$  the number of purchased tickets of 10,000LL.

a) Express the given and the constraints in a linear system  $S$  of equations and inequalities.

b) Suppose that the triplet  $(x, y, z)$  is the solution of system  $S$ . prove that:  $3x + y = 160$  and  $x + y \leq 60$ .

c) represent graphically the set of solutions of the system  $S$

❖ Prove that 4 pairs of integers  $(x, y)$  verify the system.

❖ For each pair, calculate the corresponding  $z$  value.

d) Prove that the 4 triplet  $(x, y, z)$  previously found, are solutions of the initial system  $S$ .

- 12 A dietician working in a slimming institute is preparing the dinner menu. Two types of food are used in this meal. The dietician is doubtful of incorporating in the menu, the minimum daily requirement of 2 vitamins. The following table gives the amount of vitamins in 100g of each one of these foods, the daily requirement of the dieting process and the price of 100g of each food:

| Food       | Food ingredient 1 | Food Ingredient 2 | Mimimal daily require-ment |
|------------|-------------------|-------------------|----------------------------|
| Vitamin A  | 2 mg/100g         | 3 mg/100g         | 18 mg                      |
| Vitamin B  | 4 mg/100g         | 2 mg/100g         | 22 mg                      |
| Cost/100 g | 0.12 \$           | 0.15 \$           | --                         |

Formulate a linear mathematical model to determine the quantities of each of the 2 food ingredients which minimize the cost of the meal by ensuring the necessary minimum amount of each vitamin.

- 13 An enterprise manufactures two products. Each one of them must be made in two departments. The product *A* needs 2 hours of work per unit in the department 1, and 4 hours of work per unit in the department 2. The product *B* needs 3 hours of work per unit in the department 1 and two hours per unit in the department 2. The departments 1 and 2 have the work capacity of 60 and 80 hours per week respectively. The profit margin of the two products are 3\$ and 4\$ per unit respectively.

a) Formulate a linear programming problem to determine the quantities of each of these 2 products, which maximize the profit.

b) Suppose that the number of units of the product *B* must be at least equal to the number of units of the product *A*.

Maximize the profit by respecting all the imposed constraints.

- 14 A special food for athletes must be prepared from two different ingredients: ingredient 1 and ingredient 2.

The new obtained food must contain at least: 18 mg of vitamin *A*, 5 mg of vitamin *B* and 1 mg of vitamin *C*. Each kg of ingredient 1 costs 1.5\$ and contains: 1mg of vitamin *A*, 5mg of vitamin *B* and 1mg of vitamin *C*. On the contrary each kg of ingredient 2 costs 2.5\$ and contains: 2mg of vitamin *A*, 3 mg of vitamin *B* and 1 mg of vitamin *C*.

How many kg of each ingredient must be used in a mixture to satisfy at a minimum price, the daily requirement?





2°) What is the probability  $P(B)$  of drawing a white ball from the first draw?

3°) What is the probability  $P(B \cap N)$  of drawing a white ball from the first draw and a black ball from the second?

4°) What is the probability denoted  $P_B(N)$  of getting the black ball in the second draw knowing that a white ball has been drawn in the first draw?

5°) What is the relation between  $P(B \cap N)$ ,  $P_B(N)$  and  $P(B)$ ?

$P_B(N)$  denoted also  $P(N/B)$  is the conditional probability.

## Activity 2.

In a company, the employees are distributed according to the age and the sex as it is indicated in the following table:

| Age<br>Sex | 20 - 30 | 30 - 40 | 40 - 50 | 50 - 60 | Total |
|------------|---------|---------|---------|---------|-------|
| Men        | 80      | 180     | 200     | 100     | 560   |
| Women      | 60      | 150     | 150     | 80      | 440   |
| Total      | 140     | 330     | 350     | 180     | 1000  |

1°) What is the probability for an employee selected at random to be:

- a man?
- a person having an age between 40 and 50 years?
- a man having an age between 40 and 50 years?
- a man knowing that the age is between 40 and 50 years?

2°) What is the relation between the probabilities found in b), c), and d)?

If we denote by  $A$  the event "The selected person is a man" and by  $B$  the event "the selected person has an age between 40 and 50 years", the probabilities of a), b), c), and d) are successively:

$P(A)$ ,  $P(B)$ ,  $P(A \cap B)$  and  $P_B(A)$  or  $P(A/B)$ .

## I CONDITIONAL PROBABILITY

Let  $A$  and  $B$  be two events associated with a random experiment such that  $P(B) \neq 0$ . Let us denote by  $P(A \text{ and } B)$  or  $P(A \cap B)$  the probability of the simultaneous realization of both events  $A$  and  $B$ .

A random experiment can comprise one, two or several stages.  
Ex:- Two successive draws with or without replacement of a ball in a ballot box.  
- The throw of a die and the draw of a card from a deck of 52 cards

### Definition 1

The **conditional probability** of an event  $A$  with respect to an event  $B$  denoted by  $P(A/B)$ , is the real number defined by:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$P(A/B)$  is read probability of  $A$  knowing that  $B$  is carried out or " $P$  of  $A$  knowing  $B$ ", and it is also denoted  $P_B(A)$ .

### ❖ Example

Let us consider 3 boxes each containing 4 balls. The first contains 4 blue balls, the second contains 2 blue and 2 yellow balls and the third contains 4 yellow balls. Let us suppose that we draw at random a ball from one of the boxes and that the result is a yellow ball.

What is the probability that this ball is drawn from the second ballot box? From the third?

The universe  $\Omega$  associated with this experiment is the whole of the 12 balls. Let us note by

$B$  the event "the ball drawn is blue",

$J$  the event "the ball drawn is yellow",

$U_1$  the event "the ball drawn is from the first box",

$U_2$  the event "the ball drawn is from the second box",

$U_3$  the event "the ball drawn is from the third box".

What we need is to calculate the probability of the ball to be drawn from the second box knowing that it is yellow, which means  $P(U_2/J)$ . is the same as  $P(U_3/J)$ .

## INTRODUCTION

Often the realization of an event influences the probability of the occurrence of another event which succeeds it. Thus when one carries out two successive draws without replacement from a deck of 52 cards, the probability of drawing a king, for example, at the second draw is conditioned by the card obtained in the first draw. The universe associated with the second draw (51 cards with 4 kings or 51 cards with 3 kings) is not anymore the same associated with the first (52 cards with 4 kings). We have here an example of the conditional probability. Another example is that of the random selection among the employees of a company, of a qualified person in data processing given that he knows the English language well. The universe  $\Omega_1$  of the employees knowing english is generally different from the universe  $\Omega$  of all the employees of the company.

When the occurrence of an event  $A$  is not influenced by the occurrence of an event  $B$  like the draw of a red card and the output of an odd number at the end of an experiment which consists of drawing a card from a deck of 52 cards and the tossing of a die, we say then that the events  $A$  and  $B$  are independent.

The objectives of this chapter are:

1. To define and calculate the probability of an event  $A$  knowing that an event  $B$  has occurred.
2. To define the concept of two independent events.
3. To recognize the formula of the total probability.

### Prerequisites

- ❖ Set notions
- ❖ The Language of Events
- ❖ Fundamental laws of Probability
- ❖ Probability formulas:  
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$   
 $P(\bar{A}) = 1 - P(A).$

## ACTIVITIES

### Activity 1. with or without replacement ?

A ballot box contains 2 white balls denoted  $B_1$  and  $B_2$  and a black ball denoted by  $N$ .

A- In this part, the experiment consists in carrying out two successive draws with replacement of the first.

1°) Complete the tree below and the two result columns by indicating the missing probability on the branches and in the last column.

$$P(B) = \frac{4+2}{4+4+4} = \frac{6}{12} = \frac{1}{2}$$

$$P(J) = \frac{2+4}{12} = \frac{6}{12} = \frac{1}{2}$$

$$P(U_2 \cap J) = \frac{2}{12} = \frac{1}{6}$$

$$P(U_2/J) = \frac{P(U_2 \cap J)}{P(J)} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

$$P(U_3 \cap J) = \frac{4}{12} = \frac{1}{3}$$

$$P(U_3/J) = \frac{P(U_3 \cap J)}{P(J)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

Let us notice that  $U_1 \cap J = \emptyset$  and consequently  $P(U_1 \cap J) = 0$ , from where  $P(U_1/J) = 0$ ; a result which one already knows since the yellow ball cannot be drawn from the first box.

## Remark

the preceding formula is often used to determine  $P(A \cap B)$  knowing  $P(A/B)$  and  $P(B)$ . This formula is written indeed:

$$P(A \cap B) = P(A/B) \times P(B)$$

## ❖ Example

Determine the probability to draw 2 red cards by carrying out two successive draws without replacement from a deck of 52 cards.

Let  $A$ : "the first card drawn is red"

Let  $B$ : "the second card drawn is red".

$$P(A) = \frac{26}{52} = \frac{1}{2}, \quad P(B/A) = \frac{25}{51}$$

$$P(A \cap B) = P(B/A) \times P(A) = \frac{1}{2} \times \frac{25}{51} = \frac{25}{102}$$



## Use of the tree

In certain cases the use of the tree diagram and the formula of the conditional probability facilitates the calculation of the different probabilities.



Pierre de Fermat (1601 - 1665)

### ❖ Example

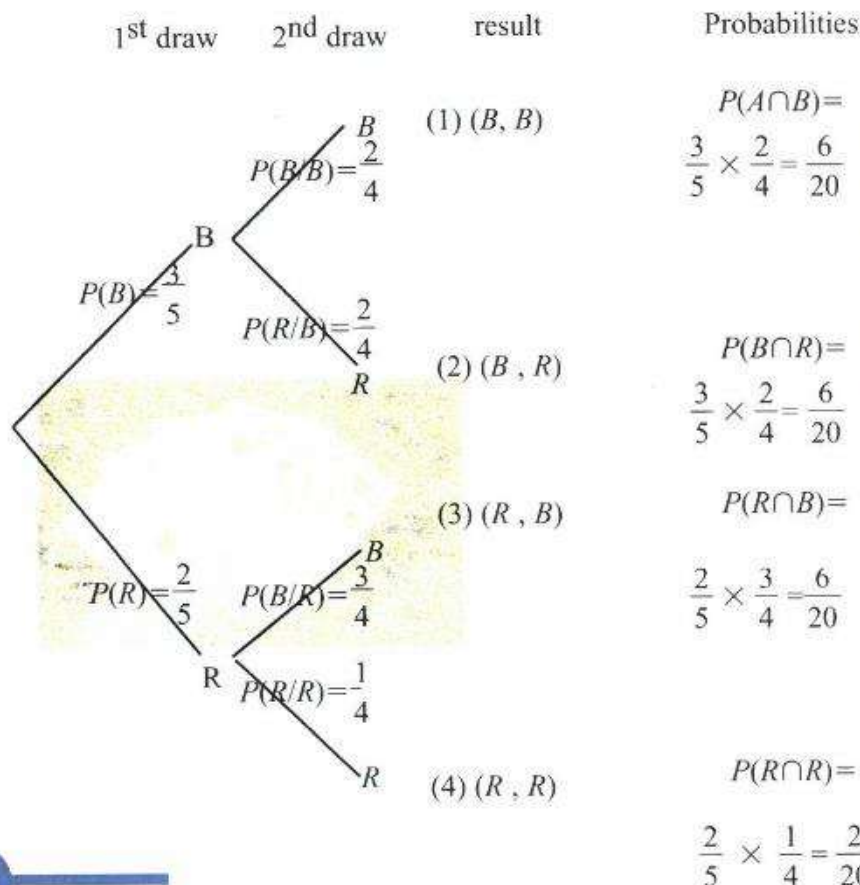
Suppose that we carry out 2 successive draws without replacement in a ballot box containing 3 white balls and 2 red balls and we are interested in determining the probabilities of the following events:

$E_1$ : "the two balls drawn are white"

$E_2$ : "one of the two balls, is white while the other is red"

$E_3$ : "the two balls drawn are red".

We can supplement the tree below:



We can get these results by naming the balls  $B_1, B_2, B_3, R_1, R_2$  and completing the tree table indicating all the events:

$(B_1, B_2) (B_1, B_3) (B_1, R_1)$   
 $(B_1, R_2), (B_2, B_1) \dots\dots\dots$

On the last column of this tree we can read:

$$P(E_1) = P(B \cap B) = \frac{6}{20} = 0.3.$$

$$P(E_2) = \frac{6}{20} + \frac{6}{20} = \frac{12}{20} = \frac{3}{5} = 0.6.$$

$$P(E_3) = P(R \cap R) = \frac{2}{20} = \frac{1}{10} = 0.1.$$

**Let us notice that:**

- ❖ the probability in the last column of the event is the product of the probabilities indicated on the branches leading to this event;
- ❖ The sum of the probabilities on the branches of a vertex is 1.
- ❖ The sum of the probabilities in the last column is 1.

Event  $E_2$  is the union of the two following events:

$E'_2$ : "the first ball drawn is white and the second is red" and

$E''_2$ : "the first drawn ball is red and the second is white".

## II INDEPENDENT EVENTS

### Definition 2

Two events  $A$  and  $B$  are said to be **independent** if the occurrence of one is not affected by the occurrence of the other.

Two events  $A$  and  $B$  such that  $P(A) \neq 0$  and  $P(B) \neq 0$  are independent if, and only if, we have one of the following properties:

$$P(A/B) = P(A), \quad P(B/A) = P(B),$$

$$P(A \cap B) = P(A) \times P(B).$$

### ❖ Example

In drawing a card from a deck of 52 cards, find the probability of drawing a heart followed by a club if the drawing is done with replacement of the first card.

If  $A$  is the event "the first drawn card is a heart" and  $B$  "the second drawn card is a club" then:

$$P(A) = \frac{13}{52} = \frac{1}{4}, \quad P(B) = \frac{13}{52} = \frac{1}{4}.$$

$A$  and  $B$  being independent we have:

$$P(A \cap B) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}.$$

If the drawing is without replacement then we would have

$$P(A) = \frac{13}{52} = \frac{1}{4}$$

$$P(B/A) = \frac{13}{51}$$

$$P(A \cap B) = \frac{13}{51} \times \frac{1}{4} = \frac{13}{204}.$$

### III

## INDEPENDENT AND INCOMPATIBLE EVENTS

### Definition (recall)

Two events  $A$  and  $B$  of the same universe  $\Omega$  (associated with a random experiment) are said to be **incompatible** if  $A \cap B = \emptyset$ , which means that they cannot occur simultaneously.

We can also say that the occurrence of one event excludes the occurrence of the other.

### Example

if the experiment consists of throwing a dice and the outcome is the number that shows, then the events:  
"getting a number  $\leq 2$ " and "getting a number  $\geq 4$ " are incompatible.

If two events  $A$  and  $B$  are independent, they can occur simultaneously, but we must have

$$P(A \cap B) = P(A) \times P(B)$$

Often the independence of two events is obvious like the output of a number 6 during a throwing of a die followed by the output of a spade drawn from a deck of 52 cards. But sometimes, in order to show the independence of two events we have to use the formula of the conditional probability.

The events  $C$ : "getting an even number" and  $D$ : "getting an odd number" are also incompatible. The events are in fact complementary.

### Example

if the experiment consists in simultaneously drawing 2 balls from a box containing 4 balls numbered from 1 to 4, let us denote by  $A$  the event "the sum of the chosen balls is 5" and by  $B$  the event "the two chosen numbers are consecutive".

Show that  $A$  and  $B$  are independent..

The universe  $\Omega$  contains 6 outcomes

$$\Omega = \{(1, 2); (1, 3); (1, 4); (2, 3); (2, 4); (3, 4)\}$$

$A$  contains two outcomes  $A = \{(1, 4); (2, 3)\}$ , hence

$$P(A) = \frac{2}{6} = \frac{1}{3}$$

$B$  contains 3 outcomes  $B = \{(1, 2); (2, 3); (3, 4)\}$ , hence

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

However  $A \cap B = \{(2, 3)\}$ , then  $P(A \cap B) = \frac{1}{6} = P(A) \times P(B)$ ,

Hence  $A$  and  $B$  are independent.

We can verify that :

$$P(A/B) = \frac{1}{3} = P(A) \text{ et } P(B/A) = \frac{1}{2} = P(B)$$

In the example mentioned opposite, the experiment consists of throwing a die and drawing a card. The universe  $\Omega$  is a set product:

$$\Omega = \Omega_1 \times \Omega_2 \text{ where}$$

$\Omega_1 = \{1, 2, \dots, 6\}$  and  $\Omega_2$  is the set of a deck of 52 cards.

If  $S$  is the set of 13 cards of spade and if  $A = \{6\} \times \Omega_2$  and  $B = \Omega_1 \times S$ , then the mentioned event is

$$A \cap B = (\{6\} \times \Omega_2) \cap$$

$$(\Omega_1 \times S) = \{6\} \times S.$$

Its cardinality is 13. The probability of this event is :

$$\frac{13}{6 \times 52} = \frac{1}{6} \times \frac{13}{52}$$

# FOCUS

The steps to be followed in resolving a problem in probability are;

- 1°) To read carefully the statement of the problem and understand the data;
- 2°) To identify the universe associated with the random experiment and the given events;
- 3°) To determine if the situation is that of a uniform probability and to determine, if necessary, the cardinality of the universe and of the events.

- 4°) To see whether the considered events are incompatible, independent, and if they form a partition of the universe.;
- 5°) If it is the case of a conditional situation of probability, we must use the formula

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \quad \text{which is written}$$

$$P(A \cap B) = P(A/B) \times P(B);$$

❖ If the events are independent, use the formula  $P(A \cap B) = P(A) \times P(B)$ .

# EXERCISES

- 1 We throw a die perfectly balanced and we designate by  $A$  the event "getting an even number", by  $B$  the event "getting a number greater or equal to 4", and by  $C$  the event "getting a multiple of 3".  
a) Are  $A$  and  $B$  independent?  
b) Same question for  $A$  and  $C$ .

- 2 Let  $A$  and  $B$  be two events such that  $P(A) = 0.3$  and  $P(B) = 0.6$ . Find  $P(A \cup B)$  when these two events are:  
a) Incompatible ;  
b) Independent.

- 3 Suppose that  $E$  and  $F$  are two events verifying  $P(E) = 0.3$ ,  $P(F) = 0.7$  and  $P(E \cup F) = 0.79$   
a) Are  $E$  and  $F$  incompatible?  
b) Are  $E$  and  $F$  independent?

- 4 A Survey about household equipment of families in a village showed that:  
80% of the families have a household machine,

75% of the families have a refrigerator,  
10% of the families have neither a washing machine nor a refrigerator.

a) Complete the following table:

|                                    | Families having a washing machine | Families not having a washing machine | Total |
|------------------------------------|-----------------------------------|---------------------------------------|-------|
| Families having a refrigerator     |                                   |                                       | 75    |
| Families not having a refrigerator |                                   | 10                                    |       |
| Total                              | 80                                |                                       | 100   |

- b) A family is selected at random from this village and we denote by  $M$  the event "the family has a washing machine" and  $R$  the event "the family has a refrigerator". Calculate the probabilities  $P(M/R)$  and  $P(R/M)$ .

# JUST FOR FUN

## Follow the needle....

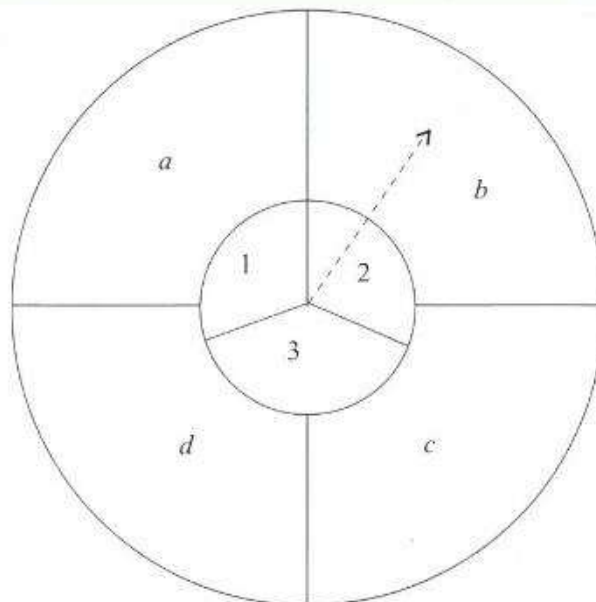
In the opposite figure, we represented a needle, which turns and which can stop randomly at any position. The big circle is decomposed into four equal regions denoted  $a$ ,  $b$ ,  $c$ , and  $d$ . The small circle is decomposed into three equal regions numbered 1, 2, 3 as the figure indicates. The used notations down being easily identified, find the following probabilities:

$P(a)$ ,  $P(b)$ ,  $P(c)$ ,  $P(1)$ ,  $P(3)$ ,  $P(1, d)$

$P(2, b)$ ,  $P(a/1)$ ,  $P(b/2)$ ,  $P(d/3)$ ,  $P(c/2)$

$P(1/a \text{ or } d)$ ,  $P(b/2 \text{ or } 1)$ ,  $P(1 \text{ or } 2/b)$

$P(b/1 \text{ or } 2 \text{ or } 3)$  et  $P(1 \text{ or } 2/a \text{ or } b \text{ or } d)$ .



- 11 The probabilities of the events  $A_1$ ,  $A_2$ , and  $A_3$  pairwise incompatible are  $P(A_1) = 0.2$ ,  $P(A_2) = 0.5$  and  $P(A_3) = 0.3$ . The conditional probabilities of the event  $B$  with respect to  $A_1$ ,  $A_2$  et  $A_3$  are  $P(B/A_1)=0.5$ ,  $P(B/A_2)=0.4$  and  $P(B/A_3)=0.3$ .

- Calculate  $P(B \cap A_1)$ ;  $P(B \cap A_2)$  and  $P(B \cap A_3)$ .
- Deduce  $P(B)$ .

## TRUE OR FALSE

Are the following assertions true or false?

- If  $A$  and  $B$  are two events then  $P(A) = P(A \cap B) + P(A \cap \bar{B})$ .
- If  $A$  and  $B$  are two independent

events then  $P(A \cup B) =$

$$P(A) + P(B) - P(A) \times P(B).$$

- If  $A$  and  $B$  are two incompatible events with  $P(A) \neq 0$  and  $P(B) \neq 0$ , then  $A$  and  $B$  are independents.

d) If we suppose that  $P(A) = 0.6$ ,  $P(B) = 0.4$  and  $P(A \cap B) = 0.2$  alors  $P(A/B) = \frac{1}{2}$ .

- When we draw successively two cards with replacement from a deck of 52 cards we are in a conditional probability situation.

f) from a box having a number of balls, we draw successively two balls without replacement. This situation represents a conditional probability.

## SELF-EVALUATION

- 1 The probabilities that a student passes his math exam, his history exam, and the 2 exams at the same time, are respectively:

$$P(M) = 0.6; P(H) = 0.8$$

$$P(M \cap H) = 0.5.$$

Are the two events  $M$  and  $H$  independent?

- 2 The 32 students in a class study English or French. 18 students study English while 24 French.

a) How many students are bilingual (study English and French at the same time)?

b) We choose a student at random. Calculate:

1) The probability that he studies English.

2) The probability that he is bilingual.

3) The probability that he studies French knowing that he studies English

- 4) The probability that he studies English knowing that he studies French..

- 3 60% of the students of a certain college are girls. 40% of the boys and 25% of the girls play basketball. Find the probability of choosing at random:

- a boy that plays basket-ball,
- a girl that does not play basket-ball.

- 4 In a given village 60% of the car drivers are less than 35 year old. Among these 7%, receive fines for violation of the highway code during the year. Among the drivers, 35 year old or more, 4% only will receive fines during the same period. What is the probability to select at random a driver from this village who will receive fines during this year

- 5 The probability that a student studies is 0.75, the probability that a student who studied passes his examination is 0.8 and the probability that the student who didn't study pass his exam is 0.25. Find the probability that a student

a) studies and passes the examination  
b) does not study and pass the examination  
c) does not pass the examination

- 6 The probability that a bus making the trip Beirut-Tripoli leave on time is 0.9 and the probability that it leaves on time and arrives on schedule is 0.8.

a) What is the probability that a bus leaving on time arrives on schedule?  
b) What is the probability for a bus leaving on time does not arrive on schedule?

- 7 40% of the employees of a certain company have a university diploma. Meanwhile 25% of the employees have a university diploma and are qualified in data processing. Find the probability of selecting randomly a qualified employee in data processing knowing that he has a university diploma

- 8 What is the probability that the third child of a couple be:

a) a girl, knowing that the first two children are boys?  
b) A boy knowing that the first two are boys?

- 9 A new drug is tested on a given population having a rate of glycemia superior to the average. 75% of the individuals take the

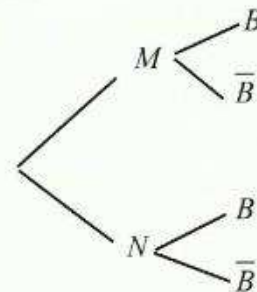
correct drug, the others receive a placebo in place. The result is rather convincing since one notes a fall of the rate at 90% of the individuals who took the drug and only at 10% of the others.

Let us denote by  $M$  the event " the person took the medication"

$N$  the event " the person took the placebo"

$B$  the event " the person has a drop in the rate of glycemia"

a) Indicate the probabilities on the following table tree::



b) Calculate  $P(B \cap M); P(B \cap N)$  and deduce  $P(B)$

- 10 On a road, two cross roads are provided with two tricolor traffic signals  $A$  and  $B$ . One supposes that these two traffic signals are not synchronized and that for a motorist moving on this road, the appearance of a given color is random

Assume that the probability that the light  $A$  is green is  $\frac{3}{4}$ , and that

the light  $B$  is green is  $\frac{1}{2}$ .

A motorist passes two cross roads successively.

a) Calculate the probability that he will meet two green lights  
b) Calculate the probability that he will meet at least one green light

b) If the hostesses go on strike, what is the probability that the pilots follow them in solidarity?

c) What is the probability that at least one of the two groups go on strike?

- 5 A company has purchased a raw material from two suppliers  $A$  and  $B$ . If none of the requests arrive in the following week, the product manufacturing will stop. The probability that the supplier  $A$  delivers the material in time is 0.7. The probability that the supplier  $B$  delivers the material on time is 0.5.
- a) What is the probability that the two providers  $A$  and  $B$  deliver the material on time?
- b) What is the probability that at least one of them delivers the material on time?
- c) What is the probability that the manufacturing stops in a week?

- 6 A company manufacturing a certain detergent launches a promotion on television. On the basis of a survey carried out among a given population one finds that the probability of the following events:
- $A$ : "The individual bought this product"
- $T$ : "The individual saw that product on the television",
- are  $P(A) = 0.15$ ,  $P(T) = 0.3$  and that  $P(A \cap T) = 0.08$ .

- a) What is the probability that an individual has bought this product knowing that he saw the promotion on television?
- b) What is the probability that an individual has bought the product or saw the promotion on television?

- 7 An epidemiological study concerning a certain disease was carried out in families having 2 children less than 10 years, a girl and a boy. We assume that the disease affected 20% of the girls and 50% of the boys. In addition, in the families where the girl is infected with the disease, the boy also has the disease in 70% of the cases. We select at random one of the families in the study.

Calculate the probability of the following events:

$A$ : "the two children are affected by the disease";

$B$ : "at least one of the two children has the disease";

$C$ : "None of the two has the disease"

$D$ : "Knowing that the boy is affected, the girl is also affected";

$E$ : "Knowing that the boy does not have the disease, the girl has it".

Denote by :

$F$ : the event "the girl of the family has the disease".

$G$ : the event "the boy of the family has disease".

(French Bac)

- 8 A new vaccine has been tested on a given population. Half of the population was vaccinated, and we note that during an epidemic, of 10 patients two people were vaccinated. We designate: by  $M$  the event "a person is sick" and by  $V$  the event "a person was vaccinated"

a) Compare  $P(M|V)$  to  $P(M)$  and deduce that the vaccine is efficient.

b) By supposing that of 100 vaccinated people, 10 people are sick, determine the proportion of sick people in the population.

# PROBLEMS

- 1 The following table gives information about a population of smokers and those of which who have lung disease.

|             | Men     |      | Men and Women |      |
|-------------|---------|------|---------------|------|
|             | smokers | sick | smokers       | sick |
| 20-40 years | 2500    | 30   | 4500          | 50   |
| 40-60 years | 3000    | 50   | 5000          | 75   |
| 60-80 years | 2000    | 80   | 4000          | 140  |
| Total       | 7500    | 160  | 13500         | 265  |

An individual is selected at random from this population, find the probabilities of the following events:

- being sick;
- being sick knowing that he is between 60 and 80 years old;
- being sick and between 20 and 40 years old;
- being sick knowing that he is a man;
- being sick knowing that he is a man between 40 and 60 years old;
- being sick knowing that she is a woman between 40 and 60 years old.

- 2 A school establishment of 930 students includes two sections: one academic and one technical.
- ❖ 30% of the students are enrolled in the technical section;
  - ❖ 40% of the students are boys;
  - ❖ 25% of the boys are enrolled in the technical section.
- a) Show that the number of the boys in the technical section is 93.
  - b) Calculate the number of girls in the academic section.

2") We choose randomly a student from the establishment. What are the probabilities of the following events:  
*A*: "is a boy in the technical section".  
*B*: "knowing that he is a boy, he is a student of the technical section";  
*C*: "knowing that he is a student of the technical section, he is a boy".

- 3 The department of the consumer protection carries out two independent inspections in all supermarkets of the city. The two inspectors must grant or not to each supermarket a certificate of conformity in obeying the rules. 2.5% of the cases of violations of the rules escape the attention of the inspector *A*. The inspector *B*, less experimented lets pass 5% of the violating cases. Determine the probability that:
- The inspector *A* grants the certificate to a supermarket knowing that the inspector *B* has found a violation,
  - The two inspectors grant the certificate to a supermarket having a violation.

(French Bac).

- 4 A director of an airline company is interested of the possibility of a strike by his employees. He was informed that the probability that his pilots go on strike is 0.65, and his airhostesses go on strike is 0.75. Moreover, he knows that if the pilots go on strike there is 90% possibility that the hostesses will go on strike in solidarity.
- What is the probability for the two groups go on strike simultaneously?

### ❖ Example

1) The following data set gives the marks of 7 students of a class of English language:

17 18 17 12 13 18 10

$\bar{x} = 15$ ; the variance of this data set is:

$$V = \frac{1}{n} \sum_{i=1}^k n_i x_i^2 - \bar{x}^2 = \frac{1}{7} \times 1639 - (15)^2 = 9.143$$

2) Take again the example of the distribution in classes of the height of the students.

| Note $x_i$ | Freq.   | $n_i x_i$ | $n_i x_i^2$ |
|------------|---------|-----------|-------------|
| 10         | 1       | 10        | 100         |
| 12         | 1       | 12        | 144         |
| 13         | 1       | 13        | 169         |
| 17         | 2       | 34        | 578         |
| 18         | 2       | 36        | 648         |
| Total      | $n = 7$ | 105       | 1639        |

| Height        | Frequency<br>$n_i$ | Class<br>center $x_i$ | $n_i x_i$ | $n_i x_i^2$ |
|---------------|--------------------|-----------------------|-----------|-------------|
| [1.50 ; 1.60[ | 6                  | 1.55                  | 9.3       | 14.415      |
| [1.60 ; 1.70[ | 7                  | 1.65                  | 11.55     | 19.057      |
| [1.70 ; 1.80[ | 8                  | 1.75                  | 14        | 24.500      |
| [1.80 ; 1.90[ | 2                  | 1.85                  | 3.7       | 6.845       |
| [1.90 ; 2.00] | 1                  | 1.95                  | 1.95      | 3.802       |
| Total         | 24                 |                       | 40.5      | 68.620      |

$$V = \frac{1}{24} (68.62) - (1.69)^2 = 0.003$$

### Definition 4

**The standard deviation** is the square root of the variance. It is generally denoted :

$$\sigma = \sqrt{V}$$

### ❖ Example

The standard deviation for the previous example is :

$$\sigma = \sqrt{0.003} = 0.0547$$

### ❖ Example

We have  $\bar{x} = \frac{105}{7} = 15$  and the absolute mean deviation is :

$$\frac{1}{n} \sum_{i=1}^k n_i |x_i - \bar{x}| = \frac{20}{7} = 2,857.$$

2. Observations grouped in classes.

We consider the distribution of the height of 24 students as follows :

| Height        | Frequency<br>$n_i$ | Class<br>center $x_i$ | $n_i x_i$ | $n_i  x_i - \bar{x} $ |
|---------------|--------------------|-----------------------|-----------|-----------------------|
| [1.50 ; 1.60[ | 6                  | 1.55                  | 9.3       | 0.84                  |
| [1.60 ; 1.70[ | 7                  | 1.65                  | 11.55     | 0.28                  |
| [1.70 ; 1.80[ | 8                  | 1.75                  | 14        | 0.48                  |
| [1.80 ; 1.90[ | 2                  | 1.85                  | 3.7       | 0.32                  |
| [1.90 ; 2.00] | 1                  | 1.95                  | 1.95      | 0.26                  |
| Total         | 24                 |                       | 40.5      | 2.18                  |

$\bar{x} = \frac{40.5}{24} = 1.69$  ; the absolute deviation from the mean is:  $\frac{2.18}{24} = 0.09$

## III VARIANCE AND STANDARD DEVIATION

### Definition 3

**The variance** of a data set is the average of the square of the deviation from the mean of the values.

Generally we denote it by  $V$ , and we have:

$$V = \frac{1}{n} \sum_{i=1}^k n_i (x_i - \bar{x})^2$$

or

$$V = \frac{1}{n} \sum_{i=1}^k n_i x_i^2 - \bar{x}^2$$

### ❖ Example

#### a) Discrete variable

Consider the data : 3 3 5 7 7 49.

The range is of:  $49 - 3 = 46$ .

#### b) Classes repartition. Consider the distribution:

| Class                     | [2 ; 4[ | [4 ; 6[ | [6 ; 8[ | [8 ; 10] |
|---------------------------|---------|---------|---------|----------|
| Frequency                 | 2       | 5       | 3       | 8        |
| The range is $10 - 2 = 8$ |         |         |         |          |

## II THE ABSOLUTE DEVIATION FROM THE MEAN

In all the following,  $n$  is the total number of observations,  $x_i$  is a value of the variable or a class center,  $n_i$  is the relative frequency of  $x_i$ ,  $k$  is the number of modalities or of classes.

### Definition 2

We call a **deviation from the mean of  $x_i$**  the value  $x_i - \bar{x}$ .

So the sum of the corresponding deviations corresponding to  $x_i$ , is :  $n_i(x_i - \bar{x})$

### Properties

The sum of deviations from the mean is equal to zero

$$\sum_{i=1}^k n_i(x_i - \bar{x}) = 0.$$

### Definition 3

**The absolute mean deviation** of a statistical data is the average of the absolute values of the deviations from the mean.

The absolute mean deviation is given by one of these two formulas

$$\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}| \quad \text{or}$$

$$\frac{1}{n} \sum_{i=1}^k n_i |x_i - \bar{x}|.$$

| Note  | Eff.    | $n_i x_i$              | $n_i  x_i - \bar{x} $             |
|-------|---------|------------------------|-----------------------------------|
| 10    | 1       | 10                     | 5                                 |
| 12    | 1       | 12                     | 3                                 |
| 13    | 1       | 13                     | 2                                 |
| 17    | 2       | 34                     | 4                                 |
| 18    | 2       | 36                     | 6                                 |
| Total | $n = 7$ | $\sum_i n_i x_i = 105$ | $\sum_i n_i  x_i - \bar{x}  = 20$ |

### ❖ Example

- The following data gives the marks of 7 students in an English language class: 17 18 17 12 13 18 10. The opposite table gives the distribution of the frequency of this data.

## INTRODUCTION

The measures or the indicators of position summarize and synthesize the values of statistical data. But these measures do not give any indication about the repartition of these values. We can then have two groups of values having the same average but with different range. Of this comes the necessity of introducing new parameters, which will characterize this range and will allow us to know the extent to which measure of the position is representative of the data. These parameters are called "measures or indicators of the dispersion" or variability. These measures have been studied in the case of a discrete variable. The aim of this chapter is to review these notions, to develop them and to extend them to the case where the variable values are regrouped by classes, both in the case that the variable is discrete or continuous.

The objectives of this chapter are:

1) To define and apply in the case of a continuous variable, the notions of :

- ◆ the range of a statistical data;
- ◆ absolute deviation from the mean;
- ◆ variance;
- ◆ standard deviation

2) To compare and interpret two statistical data sets of the same mean average and of different standard deviations.

### Prerequisites

- ❖ Average of statistical data.
- ❖ Dispersion measures in the case of discrete variable.

## ACTIVITIES

### Activity 1.

The following values represent the salaries of 24 employees where 12 of them work in the factory *A* and 12 in the factory *B*.

|                    |     |     |     |     |     |     |     |     |     |     |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Factory <i>A</i> : | 100 | 100 | 100 | 300 | 300 | 400 | 400 | 500 | 600 | 600 |
|                    | 700 | 700 |     |     |     |     |     |     |     |     |
| Factory <i>B</i> : | 100 | 200 | 300 | 300 | 400 | 400 | 400 | 400 | 500 | 500 |
|                    | 600 | 700 |     |     |     |     |     |     |     |     |

- 1- Calculate the average, the variance and the standard deviation of these two given sets.
- 2- Compare the obtained results. Which one of the two sets has its values the nearest to the average?

## Activity 2.

The weights (in kg) of 100 new-born babies are given in the following table:

1. Calculate the range of this distribution.
2. We consider the obtained data by replacing every class by its class center.  
By supposing that the values of the observations that belong to every class are equal to their class center, find the average, the variance and the standard deviation of this new data.

| Weight      | Frequency |
|-------------|-----------|
| [2 ; 2.3[   | 4         |
| [2.3 ; 2.6[ | 12        |
| [2.6 ; 2.9[ | 20        |
| [2.9 ; 3.2[ | 26        |
| [3.2 ; 3.5[ | 20        |
| [3.5 ; 3.8[ | 20        |
| [3.8 ; 4.1[ | 16        |
| [4.1 ; 4.4[ | 10        |
| [4.4 ; 4.7] | 6         |

## TEXT

In the study of measures of position, we have already seen that some definitions and some formulas are the same if the values of the variable are not regrouped or grouped in classes and represented by their class centers.

We shall see that the same is applicable in the case of the treatment of dispersion measures..

In this chapter like in chapter 3 , we consider only classes of the same range.

### I THE RANGE

#### Definition 1

- ♦ **The range** of statistical data is the difference between the biggest and the smallest value of the data.
- ♦ In a **class repartition**, it is the difference between the upper boundary of the last class and the lower boundary of the first class.

The classes being sorted in increasing order.

### Remarque importante

The standard deviation  $\sigma$  point out the size of the dispersion of the values  $x_i$  about their mean  $\bar{x}$ . The interval  $[\bar{x}-\sigma, \bar{x}+\sigma]$  contains, in fact, a large number of values  $x_i$ . The smaller is the standard deviation  $\sigma$ , the most regrouped about the mean  $\bar{x}$  are the values  $x_i$  and the most representative of the population is the mean.

#### IV

### COMPARISON BETWEEN THE DISPERSION MEASURES

| Measure                                                                      | Advantage                                                                                                                                                                                                                                                                                                                                                                                                                | Disadvantage                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Range</b>                                                                 | <ol style="list-style-type: none"> <li>1. Easy to solve and to understand.</li> <li>2. Wildly used in the quality control.</li> </ol>                                                                                                                                                                                                                                                                                    | <ol style="list-style-type: none"> <li>1. It takes only in consideration the extreme values, and it is sensible to it.</li> <li>2. It cannot be defined for the open distributions.</li> <li>3. Its use is very limited.</li> </ol> |
| <b>Absolute mean deviation</b><br>$\frac{1}{n} \sum_{i=1}^n  x_i - \bar{x} $ | <p>It is the minimum absolute deviation with respect to any other value of statistical data. :</p> $\frac{1}{n} \sum_{i=1}^k  x_i - \bar{x}  \leq \frac{1}{n} \sum_{i=1}^n  x_i - \text{median} ,$ $\frac{1}{n} \sum_{i=1}^k  x_i - \bar{x}  \leq \frac{1}{n} \sum_{i=1}^n  x_i - \text{mode} ,$ $\frac{1}{n} \sum_{i=1}^k  x_i - \bar{x}  \leq \frac{1}{n} \sum_{i=1}^n  x_i - x_j  \text{ for each } j = 1, \dots, n.$ | Hard to calculate                                                                                                                                                                                                                   |
| <b>Variance and standard deviation</b>                                       | The standard deviation is the dispersion measure the most commonly used. It is expressed in the same unit as the variable.                                                                                                                                                                                                                                                                                               | We cannot compare the standard deviations of two data sets if they do not have the same unit..                                                                                                                                      |

- 2 Consider the following data:  
100, 90, 90, 85, 80, 75, 75, 75, 70, 70, 65, 65, 60, 40, 40, 40.

- a) Find the variance of this data.  
b) Find the standard deviation of this data.

- 3 Consider the following two data sets:  
 $S_1$ : 4, 6, 8, 10, 12, 14, 16

$S_2$ : 4, 7, 9, 10, 11, 13, 16

- a) Calculate the standard deviation of each set.  
b) Which one of the two sets has values closer to their average?

## PROBLEMS

- 1 We give the class centers and the relative frequencies of two different distributions in the following tables:

| Data A       |           | Data B       |           |
|--------------|-----------|--------------|-----------|
| Class center | Frequency | Class center | Frequency |
| 18           | 43        | 2            | 1         |
| 23           | 36        | 5            | 6         |
| 28           | 25        | 8            | 11        |
| 33           | 14        | 11           | 23        |
| 38           | 8         | 14           | 16        |
| 43           | 16        | 17           | 7         |
| 48           | 29        | 20           | 4         |
| 53           | 33        | 23           | 1         |
| 58           | 41        |              |           |

Calculate the average, the variance and the standard deviation of each of the two distributions.

## FOCUS

In all the following,  $n$  is the total number of observations,  $x_i$  is the value of the variable or the class center,  $n_i$  is the relative frequency of  $x_i$ ,  $k$  is the number of modalities or classes.

### The range

❖ Non-grouped values: is the difference between the biggest and the smallest value of a statistical data.

❖ Distribution in classes: is the difference between the lower boundary of the first class and the upper boundary of the last class.

### Absolute mean deviation

$$\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}| \quad \text{or} \quad \frac{1}{n} \sum_{i=1}^k n_i |x_i - \bar{x}|.$$

### Variance

$$V = \frac{1}{n} \sum_{i=1}^k n_i (x_i - \bar{x})^2$$

or also

$$V = \frac{1}{n} \sum_{i=1}^k n_i x_i^2 - \bar{x}^2.$$

The standard deviation  $\sigma = \sqrt{V}$ .

## EXERCISES

- 1 8 students obtained in an exam the following grades:  
37, 65, 32, 49, 50, 69, 70, 62  
Put these grades in increasing order and find the range of the data.
- 2 Fouad has paid, during the last six months, the following telephone bills:  
35 500 LL, 33 600 LL, 55 000 LL, 29 300 LL, 45 300 LL  
Find the range of this data..
- 3 The minimum temperatures registered during 8 consecutive days were:  
20, 19, 18, 19, 20, 21, 22, 21. Find:  
a) the sum of the deviations from the mean.  
b) the absolute deviation from the mean .
- 4 The following given represents the waiting time (in minutes) of 10 clients in a bank:  
10, 8, 7, 15, 14, 5, 10, 11, 9, and 11. Find:  
a) the range;  
b) the absolute deviation from the mean ;  
c) the variance;  
d) the standard deviation;
- 5 Calculate the mean and the standard deviation of the following sequence:  
36, 36, 36, 36, 36.  
What conclusion you can make about the standard deviation of a sequence where all the modalities are equal? Justify.

- 6 By considering the following distribution find:  
 a) the range and absolute deviation from the mean;  
 b) the variance and the standard deviation..

| Class         | Frequency |
|---------------|-----------|
| [40.5 ; 50.5[ | 10        |
| [50.5 ; 60.5[ | 20        |
| [60.5 ; 70.5[ | 30        |
| [70.5 ; 80.5[ | 20        |
| [80.5 ; 90.5] | 10        |

- 7 By considering the following distributions:  
 Find for each distribution:  
 a) The class centers;  
 b) the mean;  
 c) the variance and the standard deviation;

| Distribution A |           | Distribution B |           |
|----------------|-----------|----------------|-----------|
| Class          | Frequency | Class          | Frequency |
| [50 ; 55[      | 6         | [6 ; 8[        | 4         |
| [55 ; 60[      | 15        | [8 ; 10[       | 6         |
| [60 ; 65[      | 24        | [10 ; 12[      | 7         |
| [65 ; 70[      | 33        | [12 ; 14[      | 4         |
| [70 ; 75]      | 22        | [14 ; 16]      | 3         |

- 8 A gas station has registered the number of liters (within an error of one liter) of gas bought, by car, during a given day. We have obtained the following distribution.  
 Find the average and the standard deviation of the distribution.

| Class     | Frequency |
|-----------|-----------|
| [0 ; 10[  | 37        |
| [10 ; 20[ | 96        |
| [20 ; 30[ | 140       |
| [30 ; 40[ | 52        |
| [40 ; 50] | 24        |
| [50 ; 60] | 3         |

- 9 To estimate the dosage (by tablets) of a medicine needed for healing some patients from a certain sickness, a medical agency took a sample of 12 patients and administered a test on them. Its result was as follows:  
 20, 22, 23, 19, 18, 17, 22, 21, 23, 21, 17, 17.  
 a) Find the average, the variance and the standard deviation of this sample.  
 b) To prevent the return of the sickness, the agency has decided to give to the similar cases, 2 extra tablets to each patient. Calculate the new number of tablets. Recalculate the average, the variance and the standard deviation of this new sequence.  
 c) Compare the obtained results in the parts a) and b). What do you conclude?

- 10 Consider the following sequence:  
19,18,16,16,17,18,20,19,18,18,17,17,19,20,18.  
a) Find the standard deviation of this sequence.  
b) Suppose that we have doubled each value of this sequence. Find the new obtained data, calculate its standard deviation.  
c) Compare the obtained results in the parts a) and b). What do you conclude?
- 11 Find the standard deviation of a set of values such that:  
 $n=20$ ,  $\sum x_i = 200$  and  $\sum x_i^2 = 3600$ .  $x_i$  represents the modalities and  $n$  the total number of the observations.
- 12 A group of values satisfy the following:  
 $\sum x_i^2 = 40$  and  $\sum x_i = 10$ .  
How many values are in this group if the standard deviation is equal to 2?
- 13 In five days, we measured the time in minutes taken by a person to reach his work. We obtained the following results: 37 35 41 35 32.  
a) Find the standard deviation of this sequence.  
b) If we subtract 30 of each value of the sequence, what will be then the standard deviation of this new sequence?  
c) Compare the results of the parts a) and b). Can you find from that solution a conclusion that allows simplifying the calculation of the standard deviation?

## SELF-EVALUATION

- 1 Answer the following propositions by TRUE or FALSE:

|                                                                                                                              | TRUE | FALSE |
|------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 1. The range of a statistical data is the difference between the standard deviation and the average.                         |      |       |
| 2. Two different data sets can have the same average.                                                                        |      |       |
| 3. Small standard deviation indicates that the values are widely dispersed.                                                  |      |       |
| 4. The sum of the deviations from the mean is null.                                                                          |      |       |
| 5. We can calculate the absolute deviation from the mean, the variance and the standard deviation without using the average. |      |       |
| 6. A statistical data of equal modalities has a zero standard deviation.                                                     |      |       |

When the representative curve  $C_f$  of a function  $f$  is traced in a given system of axis, we are able to draw a great number of information on this function.

## I DOMAIN OF DEFINITION AND CONTINUITY

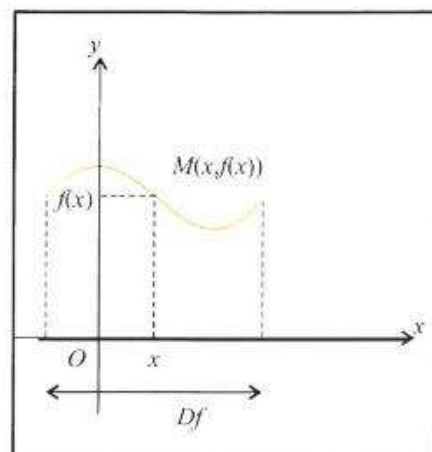
### Definition

In a given system of axis, the representative curve  $C_f$  of a function  $f$  is the set of the points  $M(x, f(x))$  where  $x$  belongs to the domain of definition of  $f$ .

### Property 1

A point  $M_0$  belongs to a curve representing, in a given system of axes, a function  $f$  if, and only if, its coordinates  $x_0$  and  $y_0$  verify  $y_0 = f(x_0)$ .

Consequently, if the curve  $C_f$  of  $f$  is drawn in a given system of axes, the domain of definition of  $f$  is the same as the set of abscissas of all points  $M$  belonging to  $C_f$ .

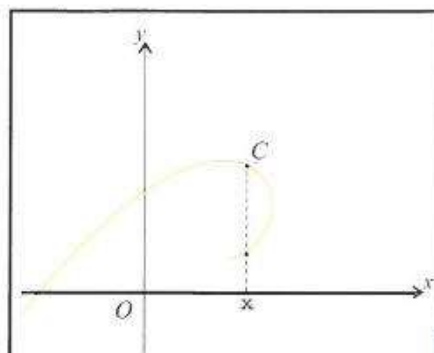


To the real number  $x$  of  $D_f$  corresponds one and only one point  $M(x, f(x))$  of  $C_f$ , but to a real number  $m$  it can correspond many real  $x$  of  $D_f$  such that  $f(x) = m$ .

We suppose here that for all  $x_0 \in I$ , it exists one and only one point of intersection of  $C$  with the straight line of equation  $y = x_0$ ; Otherwise  $C$  is not a representative curve of a function  $f$ .

### Property 2

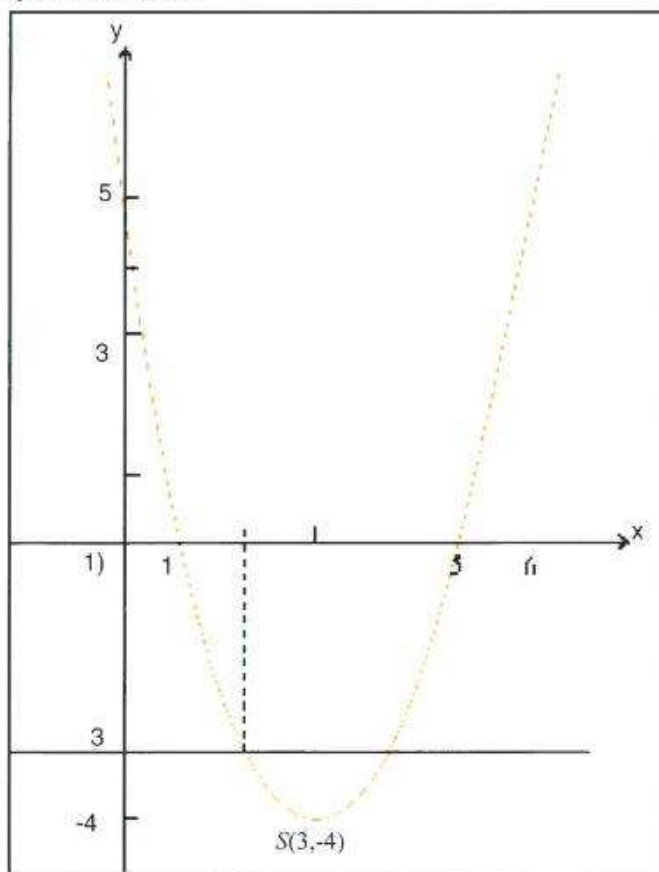
A curve  $C$  is the graphical representation of a continuous function  $f$  on a given interval  $I$  if, and only if, there is no cut or jump in the curve..



# ACTIVITIES

A good graph makes it clear..

The following is the representation of a function  $f$  in an orthogonal system of axes:



The curve representing a function  $f$  is the set of points  $M$  of the plane of abscissa  $x$  and ordinate  $f(x)$  where  $x$  varies according to the set of definition of  $f$ .

$M(x_0, y_0) \in C_f$  if, and only if,  $x_0 \in D_f$  and  $y_0 = f(x_0)$ .

$x_0$  and  $C_f$  being given, we can determine  $f(x_0)$  by finding the point  $M$ , the intersection of the straight line  $x = x_0$  with  $C_f$ , and by reading the ordinate  $y_0 = f(x_0)$  of this point.

- 1°) What is the domain of definition of  $f$ ?
- 2°) Is  $f$  continuous? Is it differentiable in the domain of definition?
- 3°) Determine graphically  $f(0)$  and  $f(6)$ .
- 4°) Solve graphically the equation  $f(x) = 0$  and the inequalities  $f(x) \leq 0$  and  $f(x) \geq 0$ .
- 5°) Solve graphically the equation  $f(x) = -3$  and the inequalities  $f(x) < -3$  and  $f(x) > -3$ .
- 6°) Study the variation of  $f$ .
- 7°) What are the extreme values of  $f$  on its domain of definition?
- 8°) Knowing that this curve is a parabola with equation  $y = ax^2 + bx + c$ , determine  $a$ ,  $b$  and  $c$ .

The extreme values of  $f$  are the maximum or minimum values, if they exist, of  $f$  on its domain of definition.

- 5 According to the "fortune directory" (15 June 1967), the following data is the rate of profit of 50 big commercial banks in the United State.

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 12.2 | 9.9  | 11.2 | 12.5 | 9.8  | 11.5 | 11.8 | 11.1 | 12.3 | 10.1 |
| 11.4 | 9.2  | 12.8 | 9.8  | 12.6 | 9.9  | 10.2 | 12.6 | 14.4 | 10.9 |
| 10.2 | 10.3 | 11.6 | 10.2 | 13.1 | 10.4 | 10.9 | 8.4  | 14.6 | 13.4 |
| 12.3 | 11.4 | 9.2  | 12.8 | 11   | 11.2 | 10.9 | 10.1 | 10.9 | 12.9 |
| 11.2 | 13.2 | 10.2 | 16   | 13.6 | 10.9 | 11.4 | 11.6 | 11.7 | 13   |

- Put the data by range equal to 1.
- Calculate the centers of the obtained classes.
- Find the average, the variance and the standard deviation of this data.

## JUST FOR FUN

Two teachers of a class of 20 students, exchange their impressions after an exam. The French teacher expresses his satisfaction. He says: My students did very well in their French exam because the class average is 12/20 and the success rate is 85%. The math teacher expresses his worries. He says, most of my students have failed their math exam. The average is 12/20 but the success rate is only 45%. Knowing that the passing mark in each subject is any mark  $\geq 10/20$ , give two sequences of grades showing these two situations. Justify your answer by using the statistical techniques.

- 2 In a study on 230 students needing a treatment, their pulse rate was taken and is given in the following table.  
Find the variance and standard deviation of this distribution.

| pulse rate  | Frequency |
|-------------|-----------|
| [55 ; 60[   | 1         |
| [60 ; 65[   | 4         |
| [65 ; 70[   | 23        |
| [70 ; 75[   | 56        |
| [75 ; 80[   | 76        |
| [80 ; 85[   | 34        |
| [85 ; 90[   | 17        |
| [90 ; 95[   | 9         |
| [95 ; 100[  | 7         |
| [100 ; 105] | 3         |

- 3 The following values represent the numbers of tapes sold in 56 shops during the feast period of a given year.

|    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|
| 69 | 41 | 20 | 86 | 79 | 92 | 90 | 22 | 26 | 37 | 26 | 57 |
| 88 | 83 | 9  | 18 | 83 | 71 | 76 | 3  | 25 | 29 | 44 | 22 |
| 11 | 92 | 8  | 17 | 95 | 41 | 6  | 77 | 57 | 1  | 76 | 13 |
| 75 | 75 | 45 | 52 | 92 | 17 | 16 | 1  | 75 | 83 | 28 | 8  |
| 74 | 46 | 22 | 40 | 84 | 23 | 18 | 85 |    |    |    |    |

- Find the range of this statistical data.
- 1) Put this statistical data in classes equal to 10.  
2) Calculate the variance and the standard deviation of this distribution.
- 1) Regroup this statistical data in classes equal to 20.  
2) Calculate the variance and the standard deviation of this new distribution.
- Compare the obtained results in b) and in c). What do you conclude?

- 4 A company has 10 employees, their monthly salaries are as follows:  
600\$; 700\$; 800\$; 800\$; 900\$; 1000\$; 1000\$; 1000\$; 1100\$; 1100\$.

- Find the average and the standard deviation of this sequence.
- Knowing that each employee has received a bonus of 200\$ by the end of the month of December, find the average and the standard deviation for this month.
- What is the relation between the obtained averages in a) and in b)? What is the relation between the standard deviation obtained in a) and in b)? Justify.

## INTRODUCTION

Often the analysis of a situation in physics, economics, sociology or other fields, permits, starting from certain number of data, to specify the relation of dependence of a variable  $y$  as a function of an independent variable  $x$  on the form  $y = f(x)$  where  $f$  is a known function. It is the case, for example, when the total cost of production, expressed in dollars, is written under the form of  $C(x) = 5x + 10000$  where  $x$  is the number of produced units and 10000 is the amount of the fixed costs, in dollars. The graphical representation, in an orthonormal system, of  $f$  as a curve gives a visual representation of the behavior of  $f$ .

In other cases, statistical or empirical data lead to curves, built point by point, which cannot be characterized by an analytical equation; in other words they are not the graphical representation of known functions.

In this case, a good graphical interpretation provides very useful information whose exploitation makes it possible to make adequate decisions.

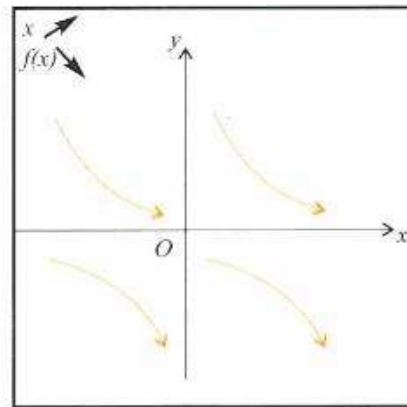
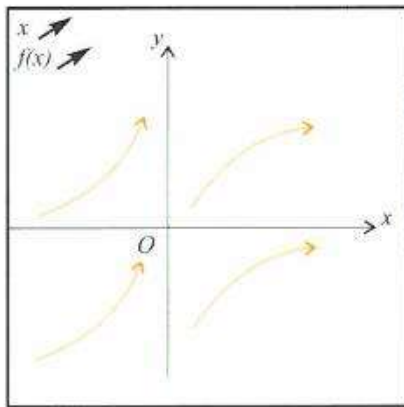
The objectives of this chapter are:

1. To interpret a graph and draw the essential information;
2. To use the representative curve of a function  $f$  to
  - Study the sense of variation of  $f$  and to determine the extreme values;
  - Determine the asymptotes if they exist
  - Solve the equations ( resp. the inequalities) of the form  $f(x) = m$  (resp.  $f(x) \leq m$  or  $f(x) \geq m$ ).

### Prerequisites

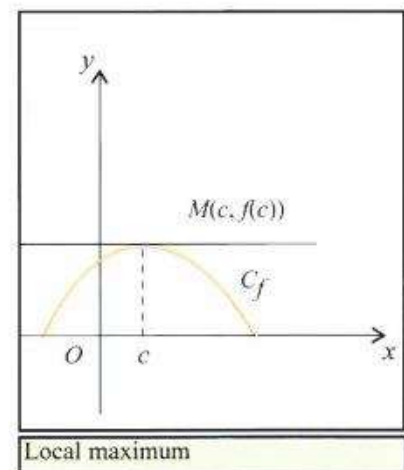
❖ **General information about functions: limits, continuity, differentiability, asymptotes and graphical representation.**

❖ **Notions on equations and inequalities**



### Property 1

A function  $f$  differentiable in an interval  $I$  is increasing (resp. decreasing) on  $I$  if, and only if,  $f'(x) \geq 0$  (resp.  $f'(x) \leq 0$ ) for all  $x$  of  $I$ .



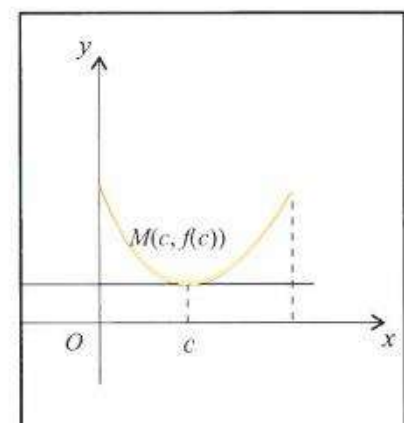
Local maximum

### Property 2

If a function  $f$  is differentiable on an interval  $]a, b[$  and is increasing on  $]a, c[$  and decreasing on  $]c, b[$  for  $c \in ]a, b[$ , then  $f'(c) = 0$  and  $f$  admits a local maximum at  $c$ .

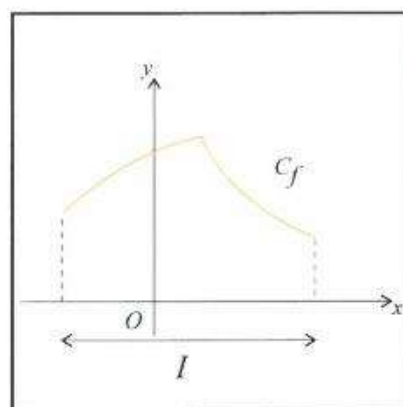
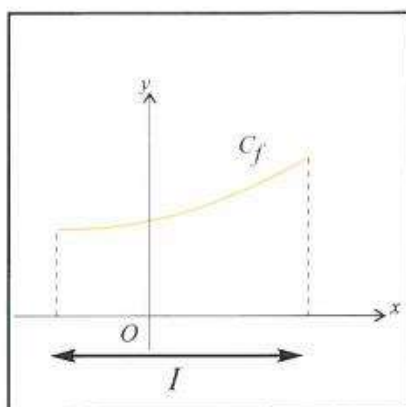
If it is decreasing on  $]a, c[$  and increasing on  $]c, b[$ , then  $f'(c) = 0$  and  $f$  admits a local minimum at  $c$ .

In these two cases, the tangent line at  $M(c, f(c))$  to  $C_f$  is horizontal.



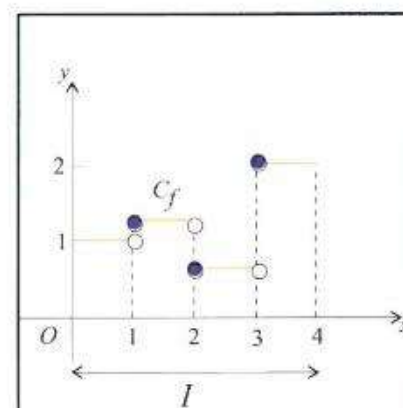
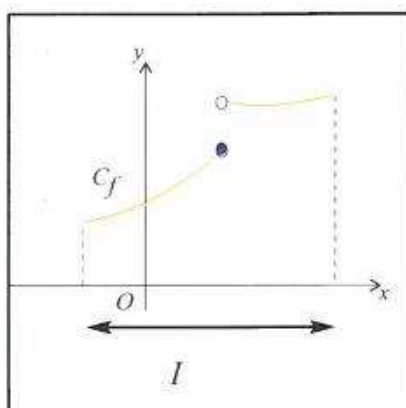
Local minimum

## Examples of graphs of continuous functions



Continuous functions on  $I$

## Examples of graphs of non continuous functions



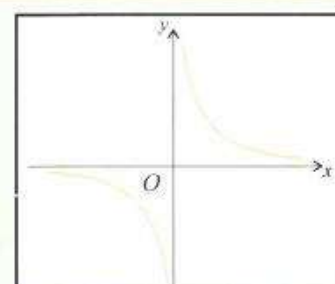
Non continuous functions on  $I$

The curve on the right is the graphical representation of a constant function by intervals (**or step function**).

It is the function defined by.

$$\begin{aligned} f(x) &= 1 & \text{for} & & 0 \leq x < 1 \\ f(x) &= 1.5 & \text{for} & & 1 \leq x < 2 \\ f(x) &= 0.5 & \text{for} & & 2 \leq x < 3 \\ f(x) &= 2 & \text{for} & & 3 \leq x \leq 4. \end{aligned}$$

Such curves occur in statistical data.

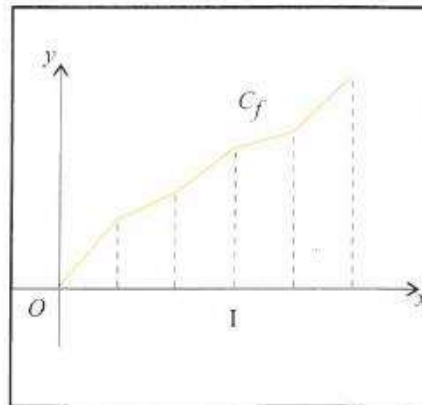
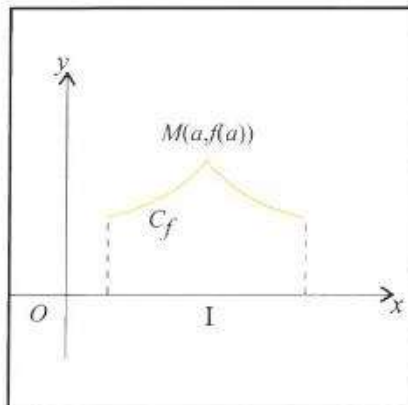


The hyperbola drawn above is the graphical representation of the function

$$x \mapsto f(x) = \frac{1}{x}.$$

This function is continuous on the set of definition  $D_f = ]-\infty, 0[ \cup ]0, +\infty[$  which is not an interval.  $f$  is clearly continuous on each interval in  $D_f$ .

## Examples of functions which are not differentiable



At  $M(a, f(a))$ , the left curve does not admit a tangent. It admits in fact two different tangents, one to the right and one to the left.  $M$  is an angular point.

These two curves are continuous on the interval  $I$ , but not everywhere differentiable on this interval.

The second graph is that of a linear function by intervals. It is the juxtaposition of line segments.

In statistics, the polygons of frequencies and the cumulative frequencies are examples of such curves.

## IV SENSE OF VARIATION AND EXTREMA

### Property

When the point  $M(x, f(x))$  of  $C_f$  moves from the left to the right (in the increasing sense of  $x$ ), two cases are possible:

1. The point  $M$  moves upwards (means in the sense of increasing  $y$ ) and the function  $f$  is increasing;
2. The point  $M$  moves downwards (means in the sense of decreasing  $y$ ) and the function  $f$  is decreasing.

The different figure cases are illustrated below:

$f$  is increasing when  $x \geq x' \Rightarrow f(x) \geq f(x')$  and decreasing when  $x \geq x' \Rightarrow f(x) \leq f(x')$ .

We can define the strictly increasing and decreasing functions by replacing the inequalities above by strict inequalities.

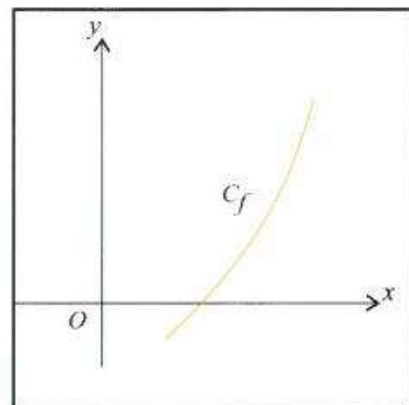
## Remark

In a great number of cases, the point  $M$  moves indefinitely on the curve  $C_f$  without verifying one or the other of the three preceding properties. The curve does not admit in this case an asymptote.

It is the case, for example, when

$$\lim_{x \rightarrow +\infty} f(x) = +\infty.$$

without the existence of the two point  $a$  and  $b$  verifying the condition of definition 3.



## III DIFFERENTIABILITY AND THE TANGENT LINE

### Property

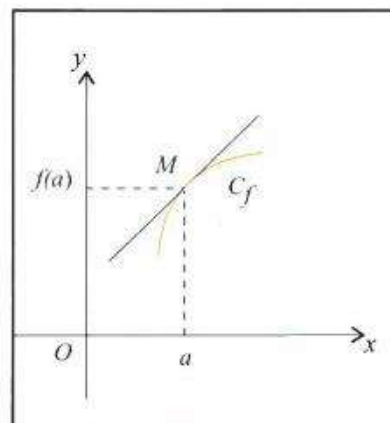
The function  $f$  is **differentiable** at  $a$  if, and only if, the curve  $C_f$  admits at a point  $M(a, f(a))$  a non vertical tangent.  $f'(a)$  is then the slope of the tangent line at point  $M$ .

An equation of this tangent line is :

$$y - f(a) = f'(a)(x - a) \quad \text{or}$$

$$y = f'(a)x + f(a) - af'(a)$$

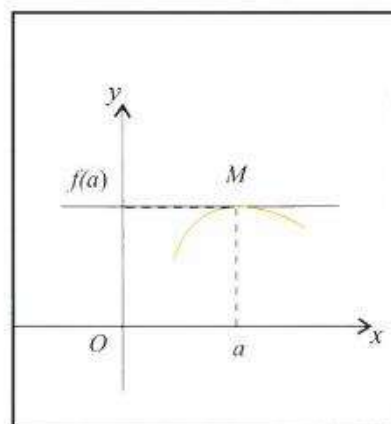
Which shows that the slope is  $f'(a)$  and that for  $x = a$ , we have :  $y = f(a)$ .



Tangent line at point  $M(a, f(a))$  to the curve  $C_f$ .

### Remark

If  $f'(a) = 0$ , then the equation of the tangent is  $y = f(a)$  and the tangent is the horizontal straight line passing by  $M$ .



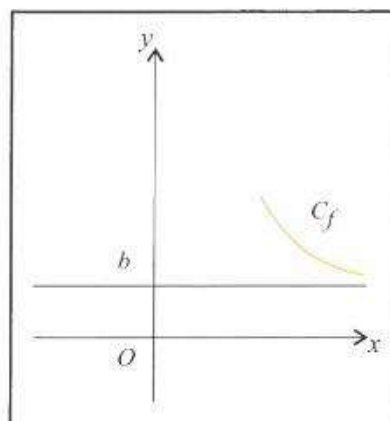
Horizontal tangent

## II LIMITS AND ASYMPTOTES

### Definition 1

When the point  $M$  moves indefinitely on the curve  $C_f$  and approaches more and more the horizontal line of equation  $y = b$ , this straight line is called a horizontal asymptote to the curve.

It is the case, for example, when  $\lim_{x \rightarrow +\infty} f(x) = b$ .



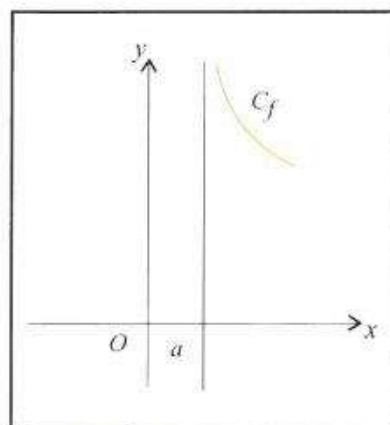
Horizontal asymptote

### Definition 2

When the point  $M$  moves indefinitely on the curve  $C_f$  and approaches more and more the vertical straight line of equation  $x = a$ , this straight line is called a vertical asymptote to the curve.

It is the case, for example, when

$$\lim_{\substack{x \rightarrow a \\ x > a}} f(x) = +\infty.$$



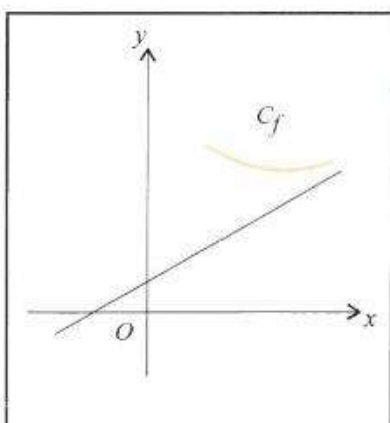
Vertical asymptote

### Definition 3

When the point  $M$  moves indefinitely on the curve  $C_f$  and approaches more and more a straight line of equation  $y = ax + b$  with  $a \neq 0$ , this straight line is called an oblique asymptote to the curve.

This is the case, for example, when

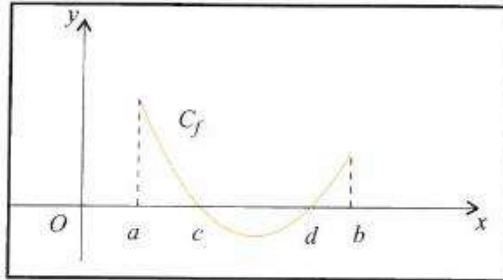
$$\lim_{x \rightarrow +\infty} [f(x) - (ax + b)] = 0.$$



Oblique asymptote

## Particular case

If  $C_f$  is representing  $f$ , then we can determine graphically the values of  $x$  such that  $f(x) = 0$ ,  $f(x) < 0$  and  $f(x) > 0$ .



In other words, we can solve graphically the equation  $f(x) = 0$  and the inequalities  $f(x) < 0$  and  $f(x) > 0$ .

|            |     |                              |
|------------|-----|------------------------------|
| $f(x) = 0$ | for | $x = c$ or $x = d$ .         |
| $f(x) < 0$ | for | $x \in ]c, d[$ .             |
| $f(x) > 0$ | for | $x \in [a, c[ \cup ]d, b]$ . |

## Plane region (decomposition)

### Property

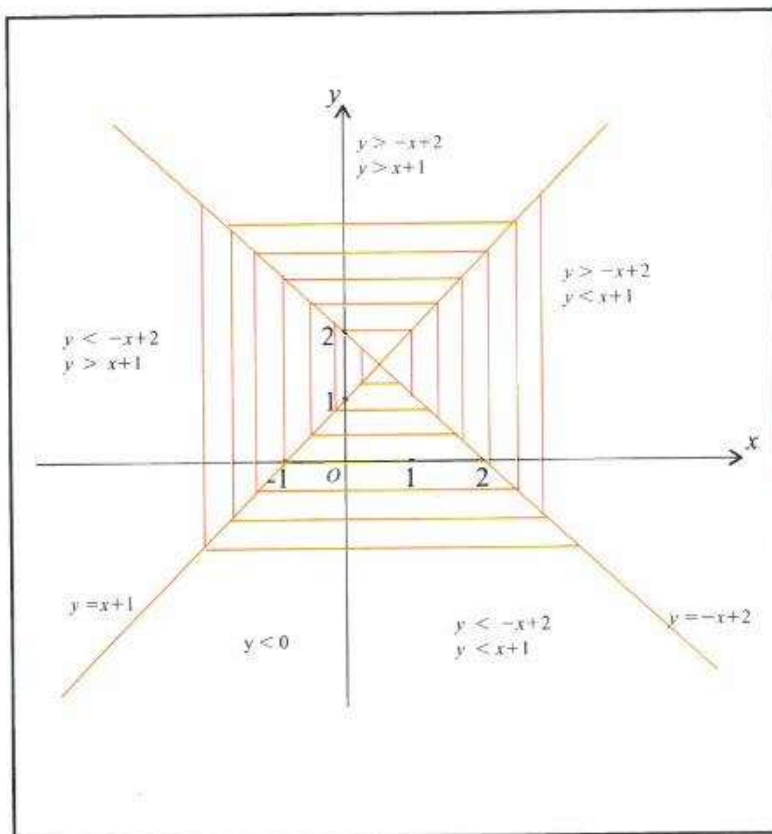
A straight line with equation  $y = ax + b$  being given, we can determine graphically the two regions of the plane corresponding to the points  $M(x, y)$  such that  $y > ax + b$  and  $y < ax + b$ , consequently we can solve graphically a system of linear inequalities.



René Descartes (1596 - 1650). French philosopher and mathematician.

### ❖ Example

On the graph below, we have characterized the different regions of the plane



The region below the two straight lines is the set of points  $M(x, y)$  whose coordinates verify the system of inequalities

$$\begin{aligned} y - x - 1 &< 0 \\ y + x - 2 &< 0. \end{aligned}$$

If we limit to the part of the region situated below the  $x$  axis, then  $M(x, y)$  belongs at this region if, and only if,  $x$  and  $y$  verify the following system

$$\begin{aligned} y - x - 1 &< 0 \\ y + x - 2 &< 0 \\ y &< 0. \end{aligned}$$

## VI GRAPHICAL DETERMINATION OF THE EQUATION OF A STRAIGHT LINE

The graphical reading of the coordinates of two distinct points belonging to a given straight line  $D$  permits us to determine its equation. If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are two points of a non vertical straight line  $D$ , then the equation of  $D$  is:

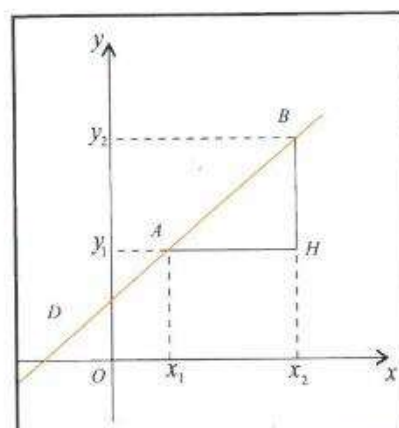
$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

The ratio  $\frac{y_2 - y_1}{x_2 - x_1} = m$  is the slope of  $D$ .

In particular, if  $A(a, 0)$  and  $B(0, b)$  are the points of intersection of the straight line  $D$  with the axes of coordinates, then the equation of the straight line is:

$$\frac{y}{x - a} = \frac{b}{-a} \quad \text{or} \quad y = -\frac{b}{a}(x - a) \quad \text{or}$$

$$y = -\frac{b}{a}x + b$$



$$m = \frac{\overline{HB}}{\overline{AH}} = \frac{y_2 - y_1}{x_2 - x_1}$$

This relation shows that by writing the equation of a straight line  $D$  by  $y = mx + p$ , the slope  $m$  is equal to  $-\frac{b}{a}$  and  $p = b$ .

### Remarks

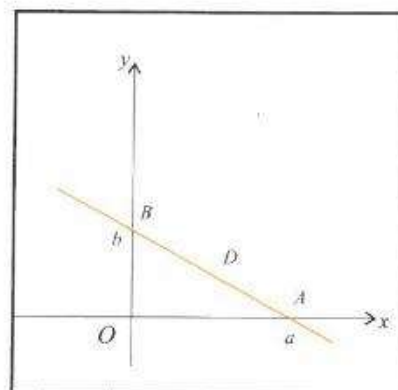
1. When we know the slope  $m$  of a straight line  $D$  and the coordinates  $x_1$  and  $y_1$  of any point  $A$  of  $D$ , the equation of  $D$  is

$$\frac{y - y_1}{x - x_1} = m \quad \text{or}$$

$$y - y_1 = m(x - x_1)$$

2. When a point  $M(x_0, y_0)$  of a vertical straight line is known, the equation of this straight line is

$$x = x_0$$



$$m = \frac{\overline{OB}}{\overline{AO}} = -\frac{b}{a}$$

## FOCUS

The curve  $C_f$  of a function  $f$  being traced in a system  $(O; \vec{i}, \vec{j})$ , we can determine graphically:

- ❖ The domain of definition  $D_f$  of  $f$ ,
- ❖ The points of discontinuity of  $f$  and the intervals on which  $f$  is continuous,
- ❖ The limits at the boundaries of these intervals as well as the asymptotes if they exist,
- ❖ The intervals where  $f$  is differentiable, and the intervals where  $f'$  is positive ( $f$  is increasing) and the intervals where  $f'$  is negative ( $f$  is decreasing).
- ❖ The local and absolute extrema, if they exist, of  $f$  as well as the values of  $x$  for which these extrema are realized.
- ❖ The solutions, if they exist, of the equation  $f(x) = m$  and the inequalities  $f(x) < m$  and  $f(x) > m$  and in particular the roots of  $f$  and the sign of  $f(x)$  for  $x \in D_f$ .
- ❖ We can also, starting from the graphical representations of two functions  $f$  and  $g$ , compare  $f$  and  $g$  and solve graphically the equation  $f(x) = g(x)$  where  $x$  belongs to the set of definition of  $f$  and  $g$  at the same time.

# V

## COMPARISON OF TWO FUNCTIONS GRAPHICAL SOLUTION OF EQUATIONS AND INEQUALITIES

### Property 1

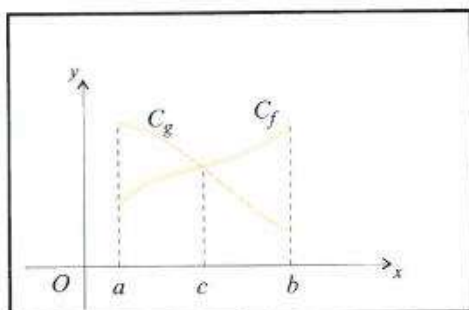
Let  $C_f$  and  $C_g$  the curves representing two functions  $f$  and  $g$  on an interval  $I$ .

$C_g$  is above  $C_f$  if and only if,  $f \leq g$  on  $I$ .

If  $C_f$  and  $C_g$  have points of intersection, then we can determine graphically the coordinates of these points as well as the different intervals such that  $f \leq g$  or  $f \geq g$ .

$f \leq g$  on  $I$  means  $f(x) \leq g(x)$  for all  $x \in I$ .

The abscissas of the points of intersection of  $C_f$  and  $C_g$  are the solutions of the equation  $f(x) = g(x)$  for  $x \in I$ .



From this graph, we can see:

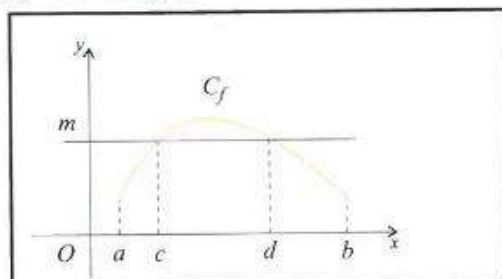
$f < g$  on  $[a, c[$ ,

$f > g$  on  $]c, b]$ ,

$f(c) = g(c)$  and  $c$  is the only solution of the equation  $f(x) = g(x)$  on  $I$ .

### Property 2

If  $C_f$  is representing  $f$  and  $D$  is the straight line with equation  $y = m$ , then we can determine graphically the values of  $x$  such that  $f(x) = m$ ,  $f(x) < m$  and  $f(x) > m$ .



In other words, we can solve graphically the equation  $f(x) = m$  and the inequalities  $f(x) < m$  and  $f(x) > m$ .

From this graph we can read:

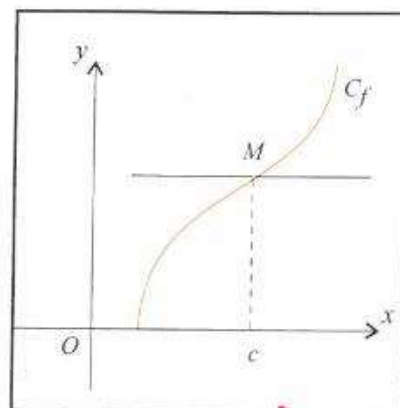
$f(x) = m$  for  $x = c$  or  $x = d$ .

$f(x) < m$  for  $x \in [a, c[$  or  $x \in ]d, b]$ .

$f(x) > m$  for  $x \in ]c, d[$ .

## Remark

If  $C_f$  has a horizontal tangent at a point  $M(c, f(c))$  or, if  $f'(c) = 0$ , then  $f$  has a local extremum at  $c$  if  $f$  changes its sense of variation around  $c$  (or when  $f'(x)$  changes its sign around  $c$ ) and it does not have an extremum if  $f$  keeps the same sense of variation around  $c$  (or when  $f'(x)$  keeps the same sign around  $c$ ).



Curve representing a function strictly increasing  $f$ , admitting at  $M(c, f(c))$  a horizontal tangent ( $f'(c) = 0$ ).

## Absolute extremum

### Definition

An absolute maximum (resp. minimum) of a function  $f$  defined and continuous on a closed interval  $I = [a, b]$  is the greatest value (resp. smallest value) of  $f$  on  $I$ .

Let  $C_f$  be the curve of a continuous function  $f$  on  $[a, b]$ , the absolute maximum (resp. minimum) of  $f$  is the ordinate of the point(s) having the greatest (resp. the smallest) ordinate. To determine the absolute extrema of  $f$  on  $[a, b]$ , we determine:

1°) the local extrema of  $f$ ,

2°) The values of  $f$  at the points where  $f$  is not differentiable,

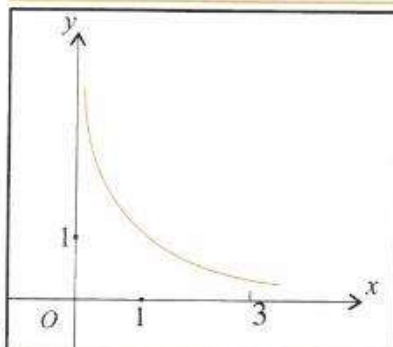
3°)  $f(a)$  and  $f(b)$ .

The absolute maximum (resp. minimum) of  $f$  on  $[a, b]$  is the greatest (resp. the smallest) of all these values.

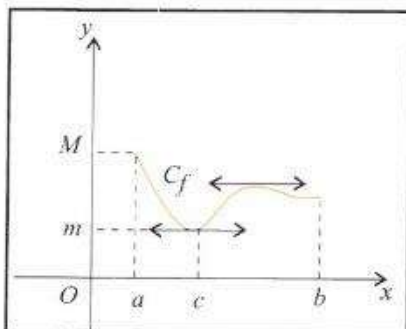
$M = f(c)$  is the absolute maximum value of  $f$  on  $I$  if, for all  $x$  in  $I$ , we have  $f(x) \leq M$ .

$m = f(c)$  is the absolute minimum value of  $f$  on  $I$  if, for all  $x$  in  $I$ , we have  $f(x) \geq m$ .

If  $I$  is not closed,  $f$  **may not be bounded** finite on  $I$  and the absolute extrema **may not exist**.

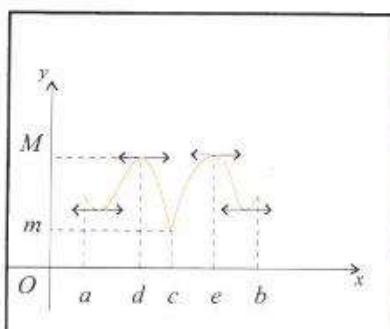


The function  $x \mapsto f(x) = \frac{1}{x}$  is continuous on  $]0, 3]$ . Its absolute minimum on  $]0, 3]$  is  $f(3) = \frac{1}{3}$ . But  $f$  does not admit an absolute maximum on this interval



$M = f(a)$  is the absolute maximum.

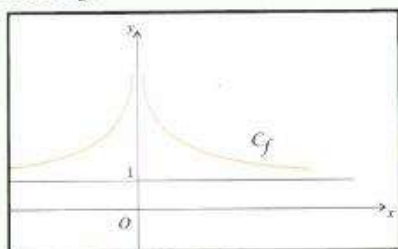
$m = f(c)$  is the absolute minimum.



$M = f(d) = f(e)$  is the absolute maximum

$m = f(c)$  is the absolute minimum.

- 7 Here is the curve representing a function  $f$ :

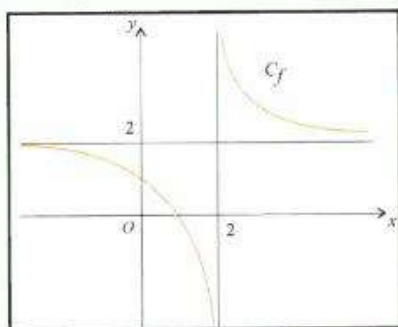


Determine  $\lim_{x \rightarrow 0} f(x)$ ,  $\lim_{x \rightarrow 0} f(x)$ ,  
 $\lim_{x > 0}$   $\lim_{x < 0}$

$\lim_{x \rightarrow +\infty} f(x)$ ,  $\lim_{x \rightarrow -\infty} f(x)$

and deduce the asymptotes to this curve.

- 8 Here is the curve representing a function  $f$ :



Determine  $\lim_{x \rightarrow 2} f(x)$ ,  $\lim_{x \rightarrow 2} f(x)$ ,  
 $\lim_{x > 2}$   $\lim_{x < 2}$

$\lim_{x \rightarrow +\infty} f(x)$ ,  $\lim_{x \rightarrow -\infty} f(x)$

and deduce the asymptotes to this curve.

- 9 Determine the region of the plane formed by the points  $M(x, y)$  whose coordinates in a given system verify:

a) 
$$\begin{cases} x + y \geq 1 \\ x - y \geq 0 \\ x \leq 2 \end{cases}$$

b) 
$$\begin{cases} 2x - y \geq 2 \\ y \geq 0 \\ x \leq 3 \end{cases}$$

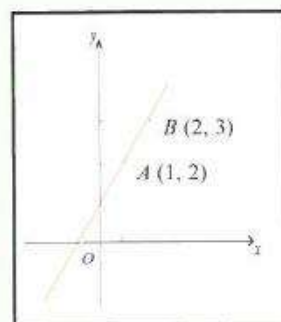
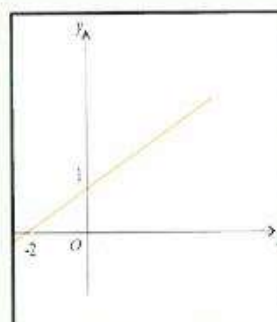
c) 
$$\begin{cases} x + y \geq 4 \\ 7 - x \geq 1 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

d) 
$$\begin{cases} y - 2x \leq 0 \\ x + y \leq 4 \\ 2y - x \leq 2 \end{cases}$$

- 10 Determine the equations of the following straight lines:

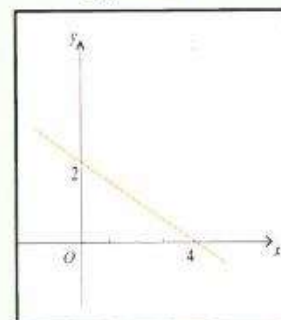
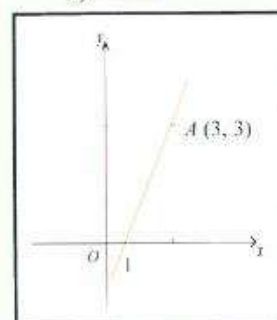
a)

b)



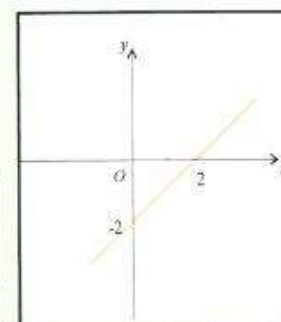
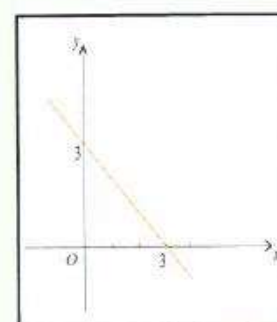
c)

d)



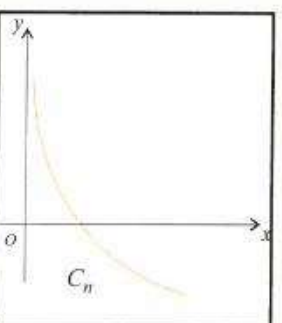
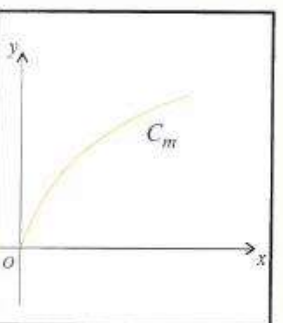
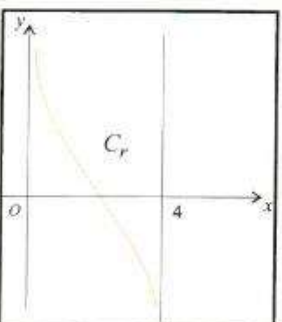
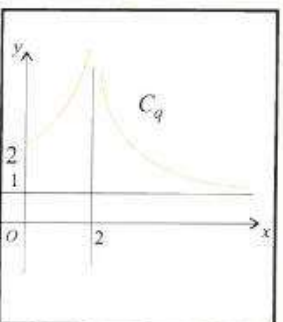
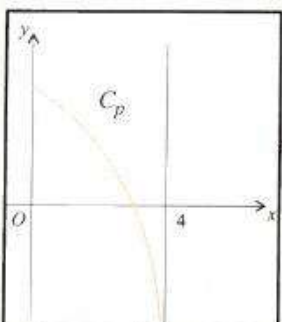
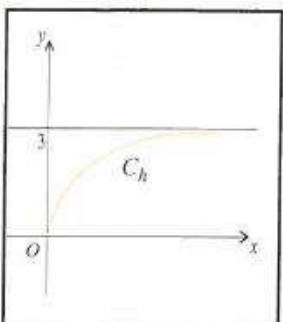
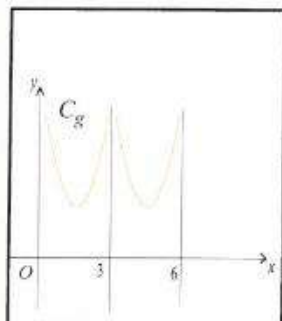
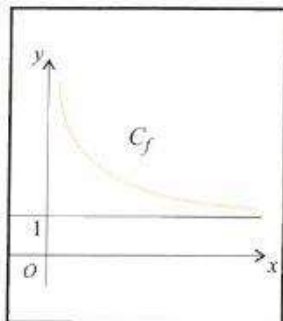
e)

f)



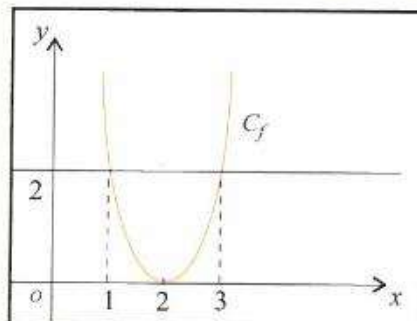
- 11 Determine the equations of the parabolas whose graphs are drawn below:

- 3 Here are the graphical representations of the functions  $f, g, h, p, q, r, m$  and  $n$ :

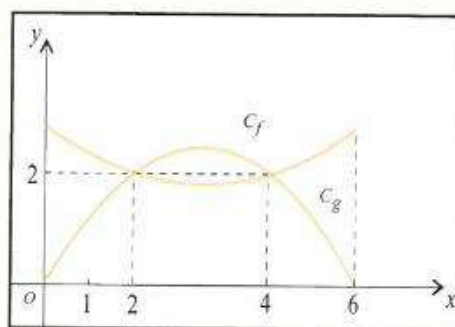


- Determine their sets of definition
- Determine the limits at the boundaries of the intervals forming their domains of definition
- Determine the asymptotes of these curves

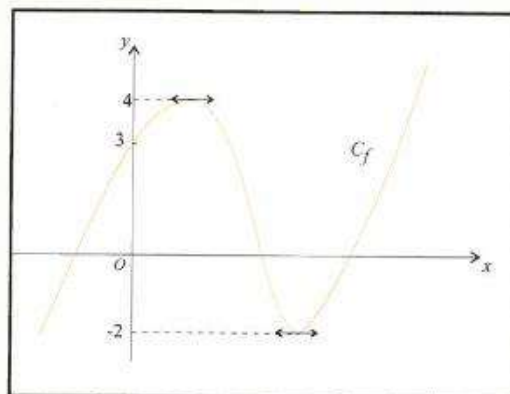
- 4 Solve graphically the equations:  $f(x) = 0$  and  $f(x) = 2$  and the inequalities  $f(x) < 0$  and  $f(x) > 2$ .

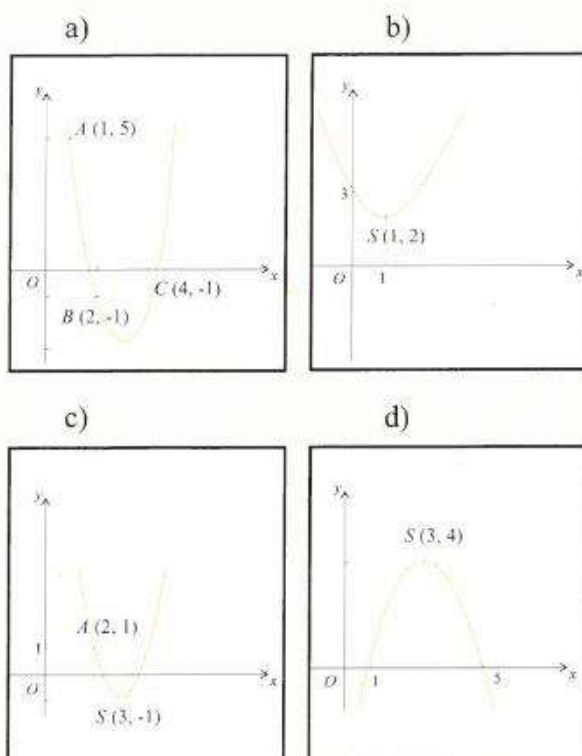


- 5 Solve graphically the equation  $f(x) = g(x)$  and compare  $f$  and  $g$ .

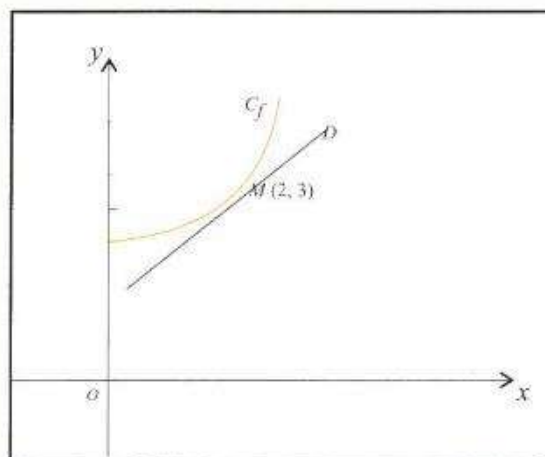


- 6 By using the graphical representation of the function  $f$  below, discuss according to the values of  $m$  the number of solutions of the equation  $f(x) = m$  and determine in each case their signs.



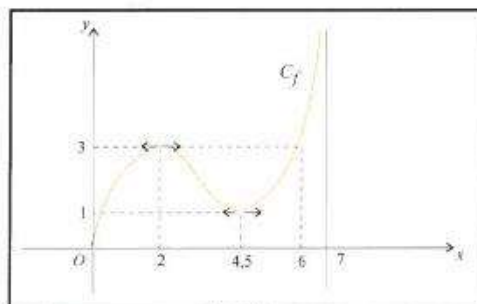


- 12 Determine the equation of the tangent  $D$  to the graph of the function  $f$  at the point  $M(2, 3)$  such that  $f'(2) = 1$



## SELF-EVALUATION

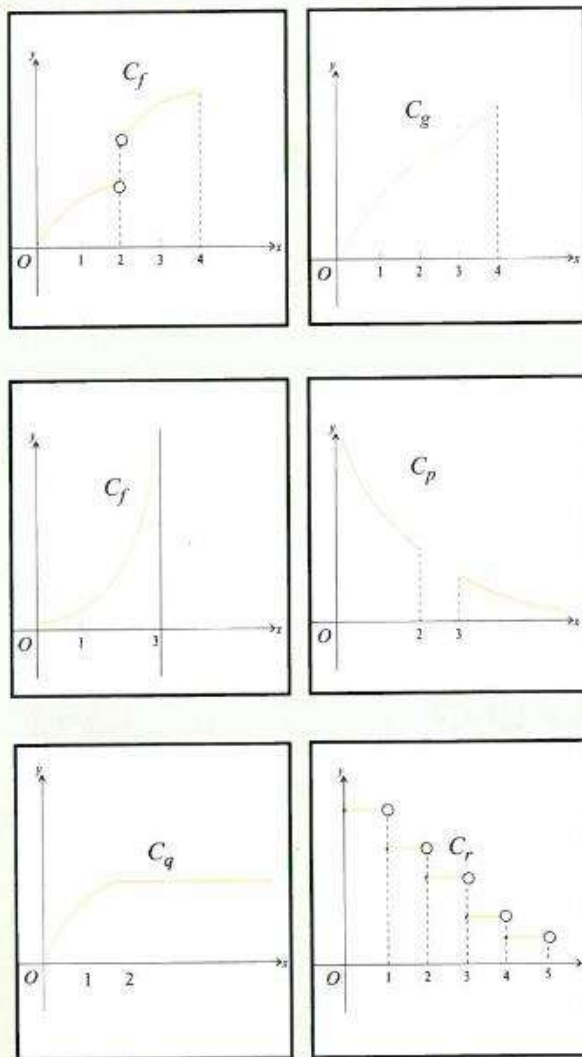
Given below the curve  $C_f$  representing in an orthonormal system the function  $f$ :



- 1°) Determine the domain of definition  $D_f$  of  $f$ .
- 2°) Study the continuity and the differentiability of  $f$  on  $D_f$ .
- 3°) Study the sense of variation of  $f$  and determine its local extrema. Do we have absolute extrema?
- 4°) Study the limits of  $f$  at the boundaries of  $D_f$  and determine the asymptote.
- 5°) Discuss according to the values of  $m$  the number of solutions of the equation  $f(x) = m$ .
- 6°) Solve the inequality  $f(x) < 3$ .

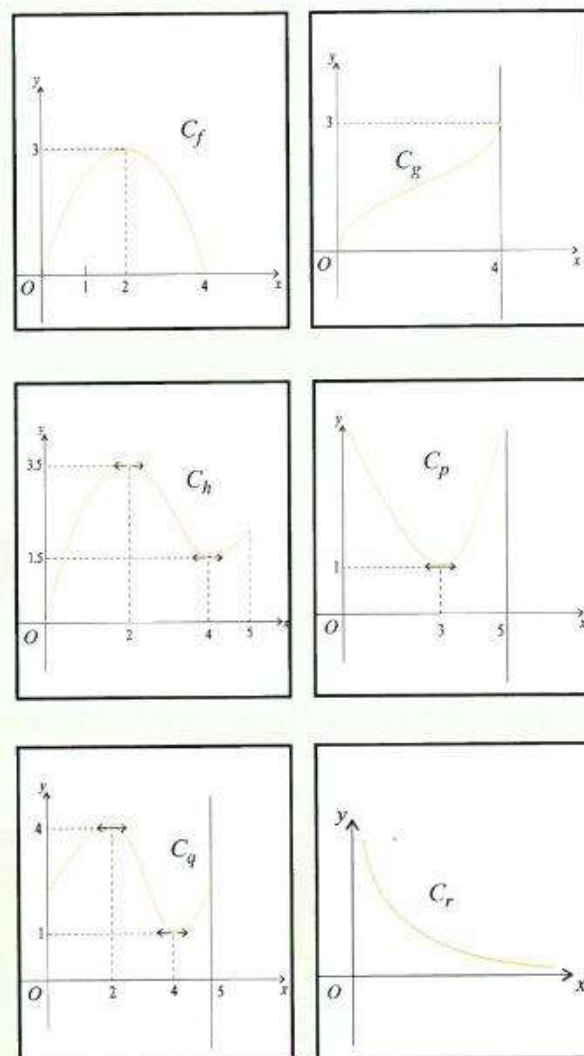
# EXERCISES

- 1 Given the following representative graphs of the functions  $f, g, h, p, q$  and  $r$ :



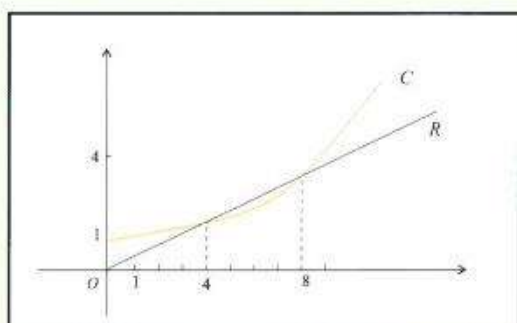
- Determine their domains of definition.
- Determine the intervals on which they are continuous
- Determine the open intervals on which they are differentiable.

- 2 Determine the intervals of definition, the sense of variation and the local and absolute extrema of the functions  $f, g, h, p, q$  and  $r$  whose graphical representations are the following:



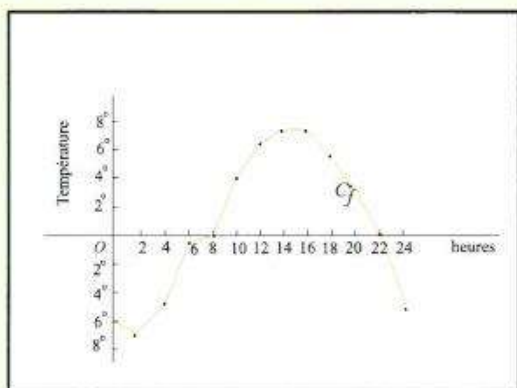
# PROBLEMS

- 1 A company manufactures a quantity  $q$  of a certain product. The total cost of production is given by the function  $C(q)$  and the revenues by the function  $R(q)$ . On the figure below, we have the graphical representations of these two functions (on the abscissa axis, 1cm represents 50 units and on the ordinate axis, 1cm represents 1000000 LL)



- a) Determine the interval for which this company makes a profit.  
b) Give an estimate of the quantity that must be produced to maximize the profit (recall that the profit is  $B(q) = R(q) - C(q)$ ).

- 2 The temperature reading given by a thermometer during one day allowed us to draw the following curve:



- a) Determine the intervals on which  $f$  is increasing,  $f$  is decreasing and  $f$  is constant.  
b) Determine the intervals on which  $f$  is positive and  $f$  is negative.  
c) At what time is the temperature minimum? What is the interval of time where the temperature is maximum?

- 3 a) Determine graphically the regions where the following systems of linear inequalities are satisfied:

$$S_1 \quad \begin{cases} 2x + y \leq 30 \\ 3x + 5y \leq 75 \\ x \geq 0, y \geq 0 \end{cases}$$

$$S_2 \quad \begin{cases} 3x + 2y \leq 120 \\ 4x + 6y \leq 260 \\ x \geq 0, y \geq 0 \end{cases}$$

$$S_3 \quad \begin{cases} x + y \geq 2 \\ 0 \leq x \leq 4 \\ 0 \leq y \leq 4 \end{cases}$$

- b) In each case, maximize the function  $z = 2x + 5y$ .

- 4 The depreciation of a machine, measured in thousands of dollars, with time  $t$ , measures per year, is described by the following function:  $V(t) = 20 - 4t$ .

Giving the graphical representation of the function, in the indicated figure.

- a) Determine the interval of definition of  $V$  (the time interval on which the value of the machine is not zero).  
b) Determine the image by  $V$  of this interval.

In this chapter,  $a$  designates a positive real number. For any real  $x$ , we can define a unique positive real number denoted  $a^x$  (read as  **$a$  to the power of  $x$  or  $a$  exponential  $x$** ). An exact or approximate value of this number is obtained by using the key  $x^y$  of a calculator.

If  $a \leq 0$ , then we can define  $a^{\frac{p}{q}}$  for certain numbers, but we can not define  $a^x$  for any irrational number.

## Properties

If  $a$  and  $b$  are two real positive numbers and if  $x$  and  $y$  are any two real numbers, then we have:

$$a^x \times a^y = a^{x+y} \quad (a^x)^y = a^{xy} \quad a^x \times b^x = (ab)^x$$

### ❖ Example

$$3^{0,1} \times 3^{1,9} = 3^2 = 9.$$

$$(4^{-\frac{1}{3}})^{\frac{3}{2}} = 4^{-\frac{1}{2}} = \left(\frac{1}{4}\right)^{\frac{1}{2}} = \frac{1}{\sqrt{4}} = \frac{1}{2} = 0,5.$$

$$(3^{\sqrt{2}})^{\sqrt{8}} = 3^{\sqrt{16}} = 3^4 = 81.$$

$$(5^{\frac{1}{2}})^{\sqrt{3}} = 5^{\frac{\sqrt{3}}{2}} \approx 4,030192405.$$

$$2^{\sqrt{2}} \times 3^{\sqrt{2}} = 6^{\sqrt{2}} \approx 12,60294532.$$

$$[(-8)^{-\frac{2}{3}}]^{\frac{1}{2}} = (-8)^{-\frac{1}{3}} = \frac{1}{(-8)^{\frac{1}{3}}} = \frac{1}{-2} = -0,5.$$

## INTRODUCTION

If  $n$  is a natural integer, the definition of the  $n^{\text{th}}$  power  $a^n$  of any real number  $a$  doesn't cause any problem. It is equal to 1 if  $n = 0$  and it is the product of  $a$  by itself  $n$  times if  $n \geq 1$ .

If  $p = -n$  is a negative integer, then we can define  $a^p$  for all  $a \neq 0$  being the inverse of  $a^n$  ( $a^p = \frac{1}{a^n}$ ).

For the rational powers in the form of  $\frac{p}{q}$ , where  $p$  and  $q$  are irreducible positive integers, we can define  $a^{\frac{p}{q}}$  as soon as we can define the  $q^{\text{th}}$  root of  $a$ . It is the case of any real number  $a$  if  $q$  is odd, and all positive or zero real if  $q$  is even. If  $a \neq 0$  and if  $a^{\frac{p}{q}}$  is defined, we can define  $a^{-\frac{p}{q}}$ .

As for the irrational real powers  $x$ , we can define  $a^x$  for a positive number  $a$ . The properties of the operations for these real positive numbers  $a^x$  are the same as the operations about the integers or rational powers.

For the positive  $a$  numbers we obtain then the functions in the form  $x \mapsto a^x$ , defined in  $\mathbb{R}$  and called **exponential functions**. These functions are frequently used in problems of the current life and particularly in sociology, economy and biology for problems where certain quantities increase or decrease in a particular way called exponential.

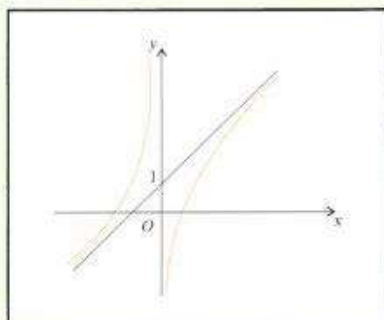
**Prerequisites**

- ❖ The integer powers of a non zero real number.
- ❖ Rational powers of a positive real number.

This chapter has for objectives:

- 1° To calculate  $a^x$  for all the real numbers  $x$  in the two cases:  
 $a > 1$  and  $0 < a < 1$ ;
- 2° To know and use the properties  $a^x \cdot a^y = a^{x+y}$ ,  $(a^x)^y = a^{xy}$ ;
- 3° To represent graphically, point by point, the functions  $x \mapsto a^x$  where  $a$  is any real positive number.

c)



Without studying the variations of these functions, associate each one to its representative graph. justify your answer.

8 Given below the table of variation of the function  $f$  defined on  $[-4, 3]$ :

|         |    |               |                |               |               |   |
|---------|----|---------------|----------------|---------------|---------------|---|
| $x$     | -4 | -3            | 0              | 1             | 3             |   |
| $f'(x)$ | -  | -             | 0              | +             | 0             | - |
| $f$     | 3  |               |                | $\frac{3}{2}$ |               |   |
|         |    | $\frac{3}{2}$ | $-\frac{3}{2}$ |               | $\frac{1}{2}$ |   |

- Draw the graph of  $f$  on  $[-4, 3]$ .
- Solve the inequality  $f(x) < \frac{3}{2}$ .
- For which values of  $m$  does the equation  $f(x) = m$  admit 2 positive solutions? One solution?

9 Given below the following table of variations of a function  $f$ :

|         |           |    |       |           |
|---------|-----------|----|-------|-----------|
| $x$     | $-\infty$ | -3 | 4     | $+\infty$ |
| $f'(x)$ | -         | -  | 0 + 0 | -         |
| $f$     | 5         |    | 8     | 5         |

- Determine the asymptotes of  $f$ .
- Draw the curve.
- Solve the inequality  $f(x) < 8$ .

10 Given below the table of variations of the function  $f$ :

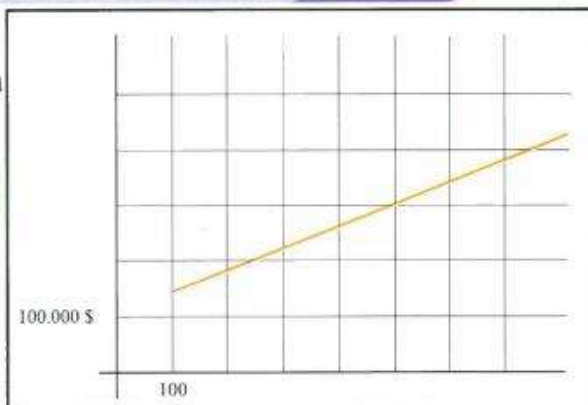
|         |           |    |      |    |   |           |           |   |
|---------|-----------|----|------|----|---|-----------|-----------|---|
| $x$     | $-\infty$ | -3 | -1.5 | -1 | 0 | 1         | $+\infty$ |   |
| $f'(x)$ | +         | 0  | -    | -  | + | -         | 0         | + |
| $f$     |           | 5  |      | 3  |   | $+\infty$ | 3         |   |

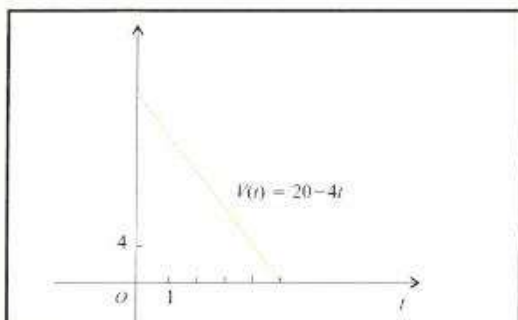
- Use this table to draw the curve representing  $f$ .
- Solve graphically the equation  $f(x) = 3$ .
- Solve graphically the inequality  $f(x) > 0$  (Let  $\alpha$  be the solution of the equation  $f(x) = 0$ ).

## JUST FOR FUN

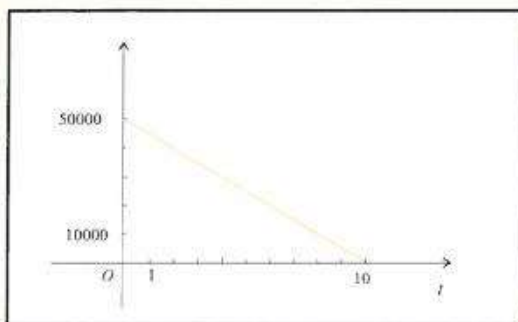
This is the curve that represents the total cost of a given function:

How many units can we produce by \$1 900 000?





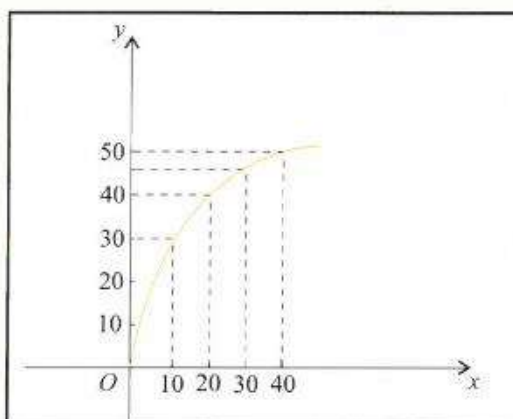
- 5 We suppose that the demand for a product decreases with time, according to the linear function whose graphical representation is given below by:



Where one unit on the abscissa axis represents 1 year and one unit on the ordinate axis represents 10000 articles.

Determine from this graph, the function  $d(t)$  describing the demand in function of time.

- 6 The graph below indicates the total sales  $y$ , in thousands of dollars, as a function of the distribution of  $x$  thousands of pamphlets.



Find the average rate of increase of sales resulting from the increase of pamphlets' distributions:

- From 10 000 to 20 000
- From 10 000 to 40 000
- From 20 000 to 30 000
- From 30 000 to 40 000.

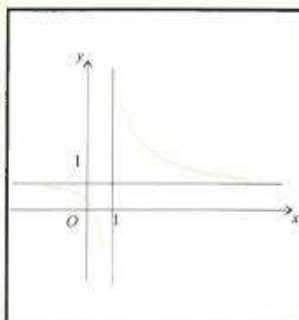
- 7 Given the graphical representations of the following three functions:

$$u: x \mapsto \frac{x+1}{x-1}$$

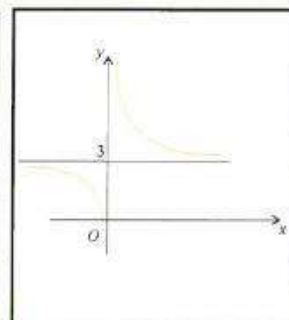
$$v: x \mapsto 3 + \frac{1}{x}$$

$$w: x \mapsto x+1 - \frac{1}{x}$$

a)



b)



# ACTIVITIES

## Unique activity. The powers ... it is a whole program

A- 1°) a) Calculate without using a calculator  $2^0$ ,  $2^1$ ,  $2^2$ ,  $2^3$ ,  $2^4$ ,  $2^{-1}$ ,  $2^{-2}$  and  $2^{-3}$ .

b) Calculate by using the button of the calculator  $x^y$  :  $2^{50}$  and  $2^{-50}$ .

c) Determine  $\lim_{n \rightarrow +\infty} 2^n$  and  $\lim_{n \rightarrow -\infty} 2^n$ .

2°) a) Calculate without calculator  $(0.1)^0$ ,  $(0.1)^1$ ,  $(0.1)^2$ ,  $(0.1)^3$ ,  $(0.1)^{-1}$ ,  $(0.1)^{-2}$  and  $(0.1)^{-3}$ .

b) Calculate  $(0.1)^8$  and  $(0.1)^{-8}$ .

c) Determine  $\lim_{n \rightarrow +\infty} (0.1)^n$  and  $\lim_{n \rightarrow -\infty} (0.1)^n$ .

B- 1°) Calculate without using the calculator  $(-2)^2$ ,  $(-2)^3$  and  $(-2)^4$ , then by using the calculator  $(-2)^{20}$  et  $(-2)^{21}$ .

2°) Calculate without calculator  $(-8)^{\frac{1}{3}}$ ,  $(-8)^{\frac{2}{3}}$ ,  $(-8)^{\frac{4}{3}}$ ,

$(-27)^{\frac{1}{3}}$  and  $(-27)^{\frac{2}{3}}$ , then try to find these numbers by using the calculator.

3°) Can we calculate  $(-9)^{\frac{1}{2}}$  and  $(-9)^{\frac{1}{4}}$ ?

What does the calculator show for  $(-9)^{\frac{3}{2}}$ ,  $(-9)^{\frac{3}{4}}$  and  $(-9)^{\frac{5}{6}}$ ?

4°) Calculate by using the calculator  $5^{\sqrt{2}}$ ,  $5^{\sqrt{3}}$ ,  $5^\pi$ ,  $5^{0.1}$ ,  $5^{0.2}$ ,  $5^{0.3}$ ,  $5^{0.4}$ .

5°) What does the calculator show for  $(-5)^{\sqrt{2}}$ ,  $(-5)^{\sqrt{3}}$ ,  $(-5)^\pi$ ,  $(-5)^{0.1}$ ,  $(-5)^{0.3}$ ,  $(-5)^{0.2}$ ,  $(-5)^{0.4}$ ?



Some old calculators show E for  $(-2)^n$ . So we must calculate  $2^n$  then add the sign + if  $n$  is even and - if  $n$  is odd.

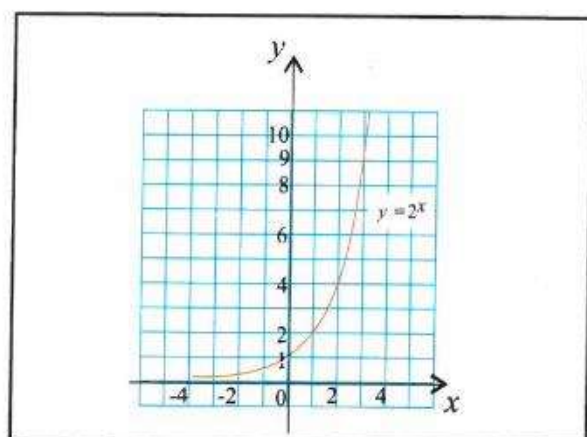
Also some calculators not of a good quality show E for  $(-8)^{\frac{2}{3}}$ , for example, despite that

$$(-8)^{\frac{2}{3}} = [(-8)^{\frac{1}{3}}]^2 = (-2)^2 = 4.$$

But a good calculator must show E for the 5 first cases and a solution for the last two cases because  $0.2 = \frac{1}{5}$  and  $0.4 = \frac{2}{5}$  et  $\sqrt[5]{-5}$  is well defined.

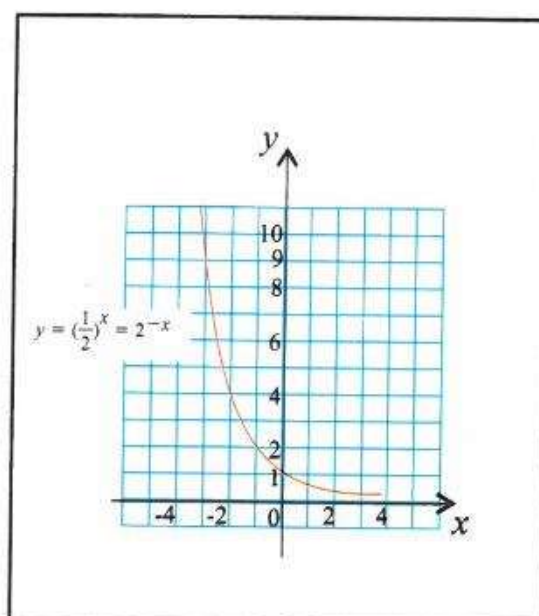
|       |               |               |               |   |   |   |   |    |
|-------|---------------|---------------|---------------|---|---|---|---|----|
| $x$   | -3            | -2            | -1            | 0 | 1 | 2 | 3 | 4  |
| $2^x$ | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | 1 | 2 | 4 | 8 | 16 |

|                   |    |    |    |    |   |               |               |               |
|-------------------|----|----|----|----|---|---------------|---------------|---------------|
| $x$               | -4 | -3 | -2 | -1 | 0 | 1             | 2             | 3             |
| $(\frac{1}{2})^x$ | 16 | 8  | 4  | 2  | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |



Note that the  $x$  axis is a horizontal asymptote for these half-curves, for  $-\infty$  for the first and for  $+\infty$  for the second.

Note also that if we represent these two functions in the same coordinate system, we obtain two symmetric curves with respect to the  $y$ -axis.



## Definition

The exponential function of base  $a$  is a function defined in  $\mathbb{R}$  and having a value in  $\mathbb{R}_+^*$ , defined by  $x \mapsto a^x$  where  $a > 0$  and  $a \neq 1$ .

If  $a = 1$ , then  $1^x = 1$  for any real  $x$ .

## Graphical representation

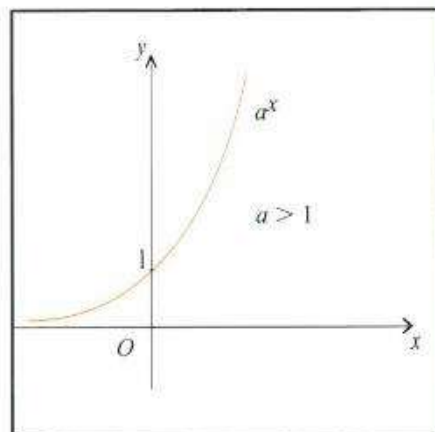
The shape of the representative curve of an exponential function  $x \mapsto a^x$  depends entirely on the value  $a$ .

Two cases to study:

1)  $a > 1$ , the function  $x \mapsto a^x$  is then increasing and we get:

$$a^x > 1 \text{ for } x > 0 \quad a^x < 1 \text{ for } x < 0$$

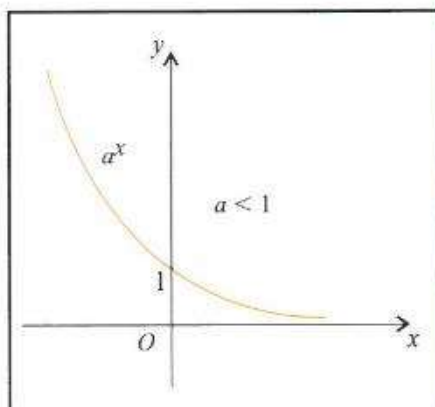
$$\lim_{x \rightarrow +\infty} a^x = +\infty \quad \lim_{x \rightarrow -\infty} a^x = 0$$



2)  $0 < a < 1$ , the function  $x \mapsto a^x$  is then decreasing and we have:

$$a^x < 1 \text{ for } x > 0 \quad a^x > 1 \text{ for } x < 0$$

$$\lim_{x \rightarrow +\infty} a^x = 0 \quad \lim_{x \rightarrow -\infty} a^x = +\infty$$



Notice that in the two cases we have:  
 $a^x > 0$  for any real  $x$  and  $a^0 = 1$ .

## ❖ Example

By taking a sufficient number of points, we can draw the curves of the two functions  $f: x \mapsto 2^x$  et  $g: x \mapsto \left(\frac{1}{2}\right)^x$ .

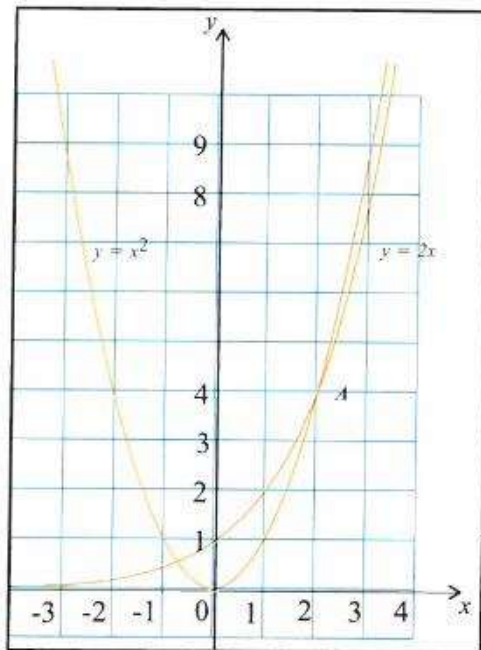
## Comparison of power functions of $x$ and exponential functions

As an example, we will graphically represent the two functions  $x \mapsto x^2$  and  $x \mapsto 2^x$  in the same orthonormal system:

For that let's refer to the following table:

| $x$   | $-\infty$ | -2  | -1  | 0 | 1 | 2 | 3 | 4  | 5  | 6  | $+\infty$ |
|-------|-----------|-----|-----|---|---|---|---|----|----|----|-----------|
| $x^2$ | $+\infty$ | 4   | 1   | 0 | 1 | 4 | 9 | 16 | 25 | 36 | $+\infty$ |
| $2^x$ | 0         | 1/4 | 1/2 | 1 | 2 | 4 | 8 | 16 | 32 | 64 | $+\infty$ |

Where the limits at  $+\infty$  and  $-\infty$  are marked for these two functions.



These two curves have two points of intersection for  $x > 0$ :  $A(2, 4)$  and  $B(4, 16)$ .  
The point  $B$  is not marked on the graph.

Note that as  $x$  is getting bigger ( $x > 4$ ) we have  $x^2 < 2^x$

### Property

If  $a > 1$  and if  $n$  is a positive integer, then we have  $a^x > x^n$  for all the real numbers  $x$  sufficiently large..

This fact is expressed by saying that the increase of any exponential function is more rapid than the increase of any power function.

## Comparison of the exponential functions

### Property

If  $0 < a < b$ , then

$$\diamond a^x < b^x \quad \text{for } x > 0$$

$$\diamond a^x > b^x \quad \text{for } x < 0.$$

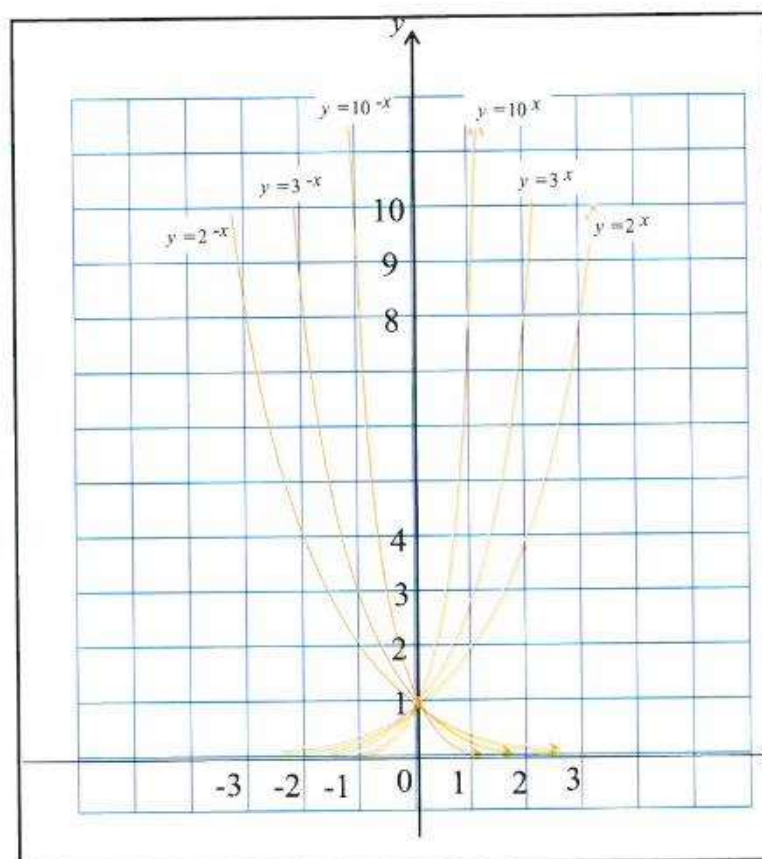
$\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$ .  $\frac{a}{b}$  being  $< 1$ , then  
 $\left(\frac{a}{b}\right)^x < 1$  for  $x > 0$  and  $\left(\frac{a}{b}\right)^x > 1$  for  
 $x < 0$ .

### Property

- ❖ If  $a > 1$ , then the bigger the base  $a$  is the faster the increase of  $a^x$  is.
- ❖ If  $a < 1$ , then the smaller the base  $a$  is the faster the decrease of  $a^x$  is.

### Graphical comparison

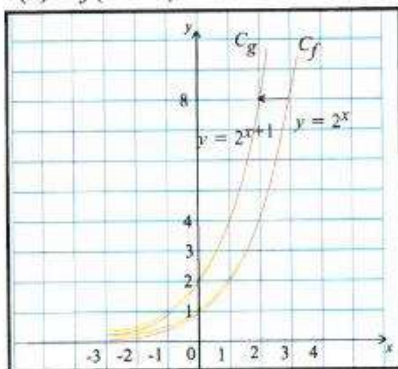
Here is, in one orthonormal system, the graphical representations of the exponential functions of base  $\frac{1}{10}$ ,  $\frac{1}{3}$ ,  $\frac{1}{2}$ ,  $2$ ,  $3$ ,  $10$ .



Note that the points with abscissa 2 of the last 3 curves are respectively:  $2^2 = 4$ ;  $3^2 = 9$ ;  $10^2 = 100$  and the points with abscissa 1 of the first three are  $\frac{1}{2}$ ,  $\frac{1}{3}$ ,  $\frac{1}{10}$ .

## Remark

From the representative curve of the function  $f: x \mapsto 2^x$ , we can obtain by translation the graphical representation of the function  $h: x \mapsto 2^{x+1}$  because for any  $x$  of  $\mathbb{R}$  we have  $h(x) = f(x+1)$ .



The points  $(-1, 1)$ ,  $(0, 2)$  and  $(2, 8)$  of  $C_g$  are obtained from the points  $(0, 1)$ ,  $(1, 2)$  and  $(3, 8)$  by translation.

$C_g$  is also obtained from  $C_f$  by taking instead of each point  $M(x, y)$  of  $C_f$ , the point  $M'(x, 2y)$  which belongs to  $C_g$ .

In fact we have

$$g(x) = 2^{x+1} = 2 \times 2^x = 2f(x).$$

$C_g$  can also be constructed point by point.

## Solution of equations

For  $a \neq 1$ , the function  $f: x \mapsto a^x$  is strictly increasing for  $a > 1$ , and is strictly decreasing for  $0 < a < 1$ ; it is a bijection from  $]-\infty, +\infty[$  to  $]0, +\infty[$ .

It follows that, if  $a^x = a^y$  then  $x = y$ .

### ❖ Example

Determine  $x$  knowing that  $4^x = 32$ .

From one side we have  $4^x = (2^2)^x = 2^{2x}$  and from the other side we have  $32 = 2^5$ , Therefore  $2^{2x} = 2^5$  then  $2x = 5$ , and  $x = \frac{5}{2}$ .

## Exponential function of base $e$

From the entire positive numbers, a particular irrational number denoted  $e$  plays an important role in practice.

An approximate value of this number given by a calculator is, within an error of  $10^{-5}$  approximately,

$$e = 2.71828.$$



John Napier (1550-1617)

Let  $a > 0$ ; for any real  $x$  we can define a real  $a^x > 0$  which is the same as the  $n^{\text{th}}$  power of  $a$  if  $x = n \in \mathbb{N}$ .

- The function  $x \mapsto a^x$  of  $\mathbb{R}$  in  $\mathbb{R}_+^*$  as defined is called the exponential function of base  $a$ .

❖ If  $a > 1$ , this function is strictly increasing and we have  $\lim_{x \rightarrow +\infty} a^x = +\infty$  and  $\lim_{x \rightarrow -\infty} a^x = 0$ .

❖ If  $a < 1$ , this function is strictly decreasing and we have  $\lim_{x \rightarrow +\infty} a^x = 0$  and  $\lim_{x \rightarrow -\infty} a^x = +\infty$ .

- If  $a$  and  $b$  are 2 positive real numbers. We have:

$$\text{❖ } a^x \times a^y = a^{x+y} \quad (a^x)^y = a^{xy}$$

$$\text{❖ } a^x = a^y \Leftrightarrow x = y.$$

$$\text{❖ } a^x \times b^x = (ab)^x.$$

❖ If  $0 < a < b$ , then  $a^x < b^x$  for  $x > 0$  and  $a^x > b^x$  for  $x < 0$ .

- For any real  $a > 1$  and any integer  $n > 0$ , we have  $a^x > x^n$  for all the real numbers  $x$  greater than a certain given number.

- The functions  $x \mapsto Ca^x$  and  $x \mapsto Ce^{kx}$ , where  $e$  is an irrational number equal, to  $10^{-5}$  approximately, to 2.71828, and  $k$  is a given constant, are very useful to describe some phenomena of growth in a large number of domains.

|     |   |   |   |   |   |      |   |   |   |   |     |
|-----|---|---|---|---|---|------|---|---|---|---|-----|
| $t$ | 0 | 1 | 2 | 3 | 4 | 5    | 6 | 7 | 8 | 9 | 10  |
| $y$ |   |   |   |   |   | 0.77 |   |   |   |   | 0.6 |

b) Represent graphically in a suitable chosen coordinate system, the function defined by

$$y = f(t) = (0.95)^t.$$

- 6 Suppose that the evolution of the population of a certain city is described by the formula

$$P(t) = 100\,000 \cdot 2^{0.05t}$$

where  $t$  is the time measured by years.

a) Calculate  $P(0)$ ,  $P(2.5)$ ,  $P(5)$  and  $P(10)$ .

b) Represent, in a suitable chosen coordinate system, the function  $t \mapsto P(t)$ .

- 7 Suppose that the quantity of a certain detected radioactive substance in a time  $t$  is given by

$$y = Q(t) = 100 e^{-0.1t}$$

where  $t$  is measured in days and  $y$  in grams.

Find the quantity of this detected substance in the times  $t = 0$ ,  $t = 4$ ,  $t = 10$  and  $t = 20$ .

- 8 Suppose that the population of a city is given by

$$P(t) = 100\,000 e^{0.02t}.$$

Find  $P(0)$ ,  $P(3)$ ,  $P(5)$  and  $P(10)$ .

## TRUE OR FALSE

Answer by True or False :

Do the following expressions give real numbers?

a)  $(-8)^{\frac{1}{3}}$

b)  $(-4)^{\frac{1}{2}}$

c)  $(-8)^{\sqrt{2}}$

d)  $(-27)^{\frac{2}{3}}$

e)  $(-5)^{0.2}$

f)  $(-5)^{0.1}$

g)  $(-5)^{0.4}$

h)  $(3)^{-\sqrt{2}}$

i)  $3^e$

j)  $(-2)^e$

k)  $5^{-e}$

l)  $2^\pi$

m)  $\pi^{-\frac{1}{2}}$

n)  $(-\pi)^{\frac{1}{4}}$

o)  $(-e)^{0.3}$

# EXERCICES

1 Represent in an orthonormal system, the following exponential functions:

a)  $x \mapsto 2^{-\frac{x}{2}}$

b)  $x \mapsto 4^{\frac{x}{2}}$

c)  $x \mapsto 3^{0.25x}$

d)  $x \mapsto 4^{\frac{x}{8}}$

e)  $x \mapsto 3^{x-2}$

2 Represent in an orthonormal system, the following functions:

a)  $x \mapsto 5^{1-x}$

b)  $x \mapsto 2^{-2x}$

c)  $x \mapsto 2^{-x^2}$

3 Solve in  $\mathbb{R}$  the following equations:

a)  $5^x = 25$

b)  $2^x = \frac{1}{8}$

c)  $3^x = \frac{1}{9}$

d)  $9^x = 27$

e)  $4^x = 64$

f)  $(\frac{1}{5})^x = 5$

g)  $a^x = \frac{1}{a^3} (a > 0)$

h)  $(\frac{3}{4})^x = \frac{16}{9}$

i)  $5^{-2x} = \frac{1}{25}$

j)  $3^{x-1} = 9$

k)  $16^{-x+1} = 8$

l)  $16^{x+2} = 64 \cdot 2^{x-1}$

m)  $9^x = 3^{x^2}$

n)  $8^{x^2} = 2^{5x}$

o)  $3^{x^2+x} = 1$

4 If you deposit 1\$ in an account that pays a rate of 8% compounded annually, your account will contain after  $t$  years the sum of

$$y = (1 + 0.08)^t = (1.08)^t \$.$$

a) Complete the following table, to two decimal places.

| $t$ | 0 | 1 | 2 | 3 | 4 | 5    | 6 | 7 | 8 | 9 | 10   |
|-----|---|---|---|---|---|------|---|---|---|---|------|
| $y$ |   |   |   |   |   | 1.47 |   |   |   |   | 2.16 |

b) Represent graphically in a suitable chosen coordinate system, the function defined by

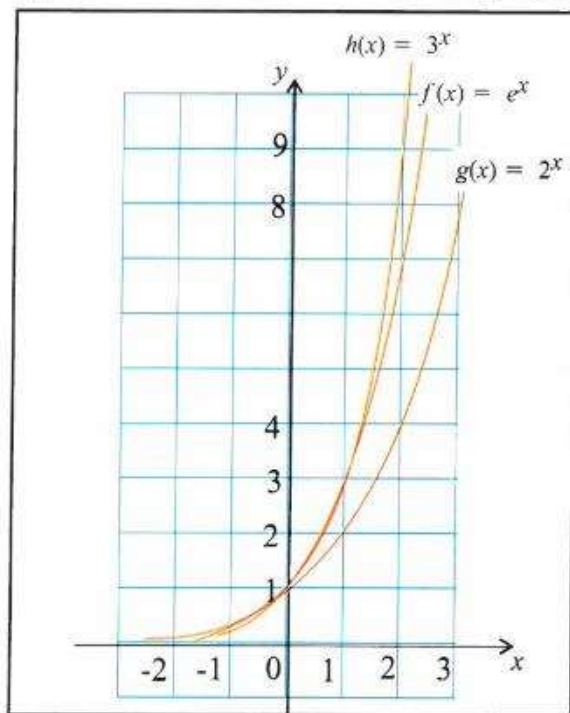
$$y = f(t) = (1.08)^t$$

5 If the money loses annually 5% of its buying power, then after  $t$  years, the value of 1\$ is given by

$$y = (1 - 0.05)^t = (0.95)^t$$

a) Complete the following table to two decimal places.

The graphical representation of the function  $f: x \mapsto e^x$  is drawn here below in the same coordinate system as the functions  $g: x \mapsto 2^x$  and  $h: x \mapsto 3^x$  for comparison purposes.



The representative curve of  $f$  is situated between  $g$  and  $h$  since we have  $2 < e < 3$ .

## Application

In sociology, economy and biology, some values increase or decrease exponentially in functions of the time  $t$ . A general formula giving such values in the function of time  $t$ , is in the form  $Q(t) = Q_0 e^{kt}$  where  $Q_0$  is the measure of the value at the initial instant  $t = 0$ .

$$Q(0) = Q_0 e^{k \times 0} = Q_0 e^0 = Q_0 \times 1 = Q_0$$

### ❖ Example

The evolution of the population in a certain city, for  $t \geq 0$ , is given by

$$P(t) = 100\,000 e^{0.02t},$$

Where  $t$  is expressed in years and  $t = 0$  corresponds to the actual hour. What is the effective population in 10 years?

Remark first that  $P(0) = 100\,000 \times e^0 = 100\,000 \times 1 = 100\,000$ . So there is in fact 100,000 habitants in this city.

In 10 years there will be

$$P(10) = 100\,000 \times e^{0.02 \times 10} = 100\,000 \times e^{0.2} \approx 122\,140 \text{ habitants.}$$

# SELF-EVALUATION

- 1 Represent in an orthonormal system the exponential function defined by  

$$f(x) = 2^{\frac{x}{3}}$$
- 2 Solve in  $\mathbb{R}$  the two following equations:  
 a)  $16^x = 64$                       b)  $81^{-2x} = 3^{x-1}$
- 3 The evolution of a population in a certain city, since 1<sup>st</sup> January 1990 is described by the formula  $P(t) = 200\,000 e^{0.03t}$  where  $t$  where  $t$  is the time measured in years since that date. What was the city population in the first of January 1990? In the first of January 2010?

# PROBLEMS

- 1 Suppose that the quantity in grams of a certain detected radioactive substance at time  $t$  is  

$$Q(t) = 1000 \times 3^{-0.5t}$$
 where  $t$  is measured in months.  
 a) Calculate  $Q(0)$ ,  $Q(1)$ ,  $Q(2)$ ,  $Q(4)$  and  $Q(10)$ .  
 b) Represent graphically in a suitable chosen coordinate system the function  $t \mapsto Q(t)$ .
- 2 Suppose that the price of a car is given in function of the used time  $t$ , expressed in years, by the formula  

$$P(t) = 20\,000 (1 - 0.12)^t$$

$$= 20\,000 \times (0.88)^t \text{ \$}$$
 (It expresses the fact that the value of the car decreases by 12% every year.)  
 a) Calculate  $P(0)$ ,  $P(1)$ ,  $P(2)$ ,  $P(5)$  and  $P(10)$ .  
 b) Represent graphically the function  $t \mapsto P(t)$ .
- 3 A person, who is learning typing, starts by learning at a certain rhythm, which decreases with time. Suppose that the number of characters typed per minute a learning person is capable to type, is given by :  

$$C(t) = 200 - 100 \times 3^{-0.5t}$$
 where  $t$  is the duration expressed in months since the beginning of learning.  
 a) Calculate the number of characters the person is capable to type per minute after:  
 1 month, 2 months, 4 months and 10 months of learning.  
 b) Represent graphically the function  $t \mapsto C(t)$ .

## Remark

When the time is given indirectly as the time between dates, it is customary to count the exact number of days including either the first or the last day, but not both. When time is computed in this manner, it is called the exact time.

### ❖ Example

Find the simple interest on 10,000\$ at 10% from January 15, 1999 to June 12, 1999.

The exact number of days is calculated in the margin.

And the interest is  $I = 10000 \times 0,1 \times \frac{148}{365} \approx 405.48$  \$.

|          |               |
|----------|---------------|
| January  | 31-15=16 days |
| February | 28 days       |
| March    | 31 days       |
| April    | 30 days       |
| May      | 31 days       |
| June     | 12 days       |
| Total    | 148 days      |

## Remark

The formula of the simple interest utilizes four values  $I$ ,  $P$ ,  $r$ , and  $t$ . The knowledge of any three among them makes it possible to calculate the fourth one.

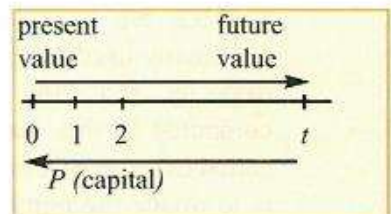
## Present value

The value of an amount  $S$  on some date prior to its due date is called the present value of the amount on that (prior) date.

From  $S = P(1 + rt)$  we have

$$P = S / (1 + rt)$$

$P$  is called the present value at  $r$  simple interest of  $S$  due in  $t$  years.



- 4 The sales of a new product increase rapidly at the beginning and tend to stabilize after a certain time. Suppose the sales of a new product are given by the formula:

$$S(t) = 1000 - 900 e^{-t} \text{ units}$$

where  $t$  represents the time measured in years since the launching time of this product.

a) Calculate  $S(0)$ ,  $S(1)$ ,  $S(2)$ ,  $S(3)$  and  $S(4)$ .

b) Represent graphically the function  $t \mapsto S(t)$  after finding the horizontal asymptote

- 5 Suppose the number of bacteria,  $B(t)$ , found in an organism is given by

$$B(t) = 50\,000 e^{0.2t}$$

where  $t$  is measured in hours and where  $t = 0$  refers to noon.

Calculate the number of bacteria found at: 12:00, 13:00, 14:00, 16:00 and 18:00.

- 6 An anti bacterial product is put in a bacterial plantation. The number of bacteria is given by

$$B(t) = 100\,000 e^{-0.1t}$$

where  $t$  is the time measured in hours and where  $t = 0$  corresponds to 9:00 am.

Calculate the number of bacteria present at: 9:00h, 10:00h, 12:00h, 14:00h, 17:00h and 20:00h

- 7 Suppose that a new employee working in a factory produces  $P(t) = 100 - 100 e^{-t}$  items per day, where  $t$  is the time measured in days since his recruitment

a) Calculate  $P(0)$ ,  $P(1)$ ,  $P(2)$ ,  $P(5)$  and  $P(10)$ .

b) Represent graphically the function  $t \mapsto P(t)$  after determining the horizontal asymptote.

## JUST FOR FUN

On your calculator there is a key (ln) and a key (log).

a) For which values of  $x$  the two keys give real numbers?

b) What are the relations between these two new functions and the functions  $x \mapsto e^x$  and  $x \mapsto 10^x$ ?

# SIMPLE INTEREST - COMPOUND INTEREST - ANNUITIES

## INTRODUCTION

The concept of interest plays an important role in the business world of finance. In some words, the interest can be defined as being the income from invested capital or the fee paid for the use of money. If you deposit the money in a bank, the bank will pay you interest for the use of your money. In this case, the bank is the borrower and you are the lender. At the same time, if you borrow the money from the bank you will pay interest for the use of the money of the bank. In this case, the bank is the lender and you are the borrower.

The sum of money loaned is the principal. An agreement is usually made concerning the period of time for which the money is loaned. The sum of the interest and the principal, due at the end of this time period, is called the amount. The interest rate per period is the ratio of the interest for the period to the principal. This rate is always expressed as a percentage or the equivalent decimal.

There are two kinds of interest, the simple interest and the compound interest. In the first case, the interest is calculated only at the end of the entire life of the transaction. In the second case, the interest is periodically added to the principal and this new sum is used as principal for the following time period.

The objectives of this chapter are:

1. To know how to calculate the simple interest and the compound interest.
2. To find an element among the four elements concerned by the calculation of interest (principal, rate, interest, period) knowing the three others.
3. To find the amount knowing the present value and conversely, the period and the interest rate being given
4. To know and use the annuities' formulas.

### ***Prerequisites***

- ❖ **Arithmetic operations**
- ❖ **Percentages.**
- ❖ **Notions on numerical sequences.**

## Definition

When only the principal earns interest for the entire life of the transaction, the interest due at the end of the time is called simple interest. The simple interest on a principal  $P$  for  $t$  years at the rate  $r$  is given by

$$I = Prt$$

And the simple amount is given by

$$S = P + I = P + Prt = P(1+rt)$$

### ❖ Example

Find the simple interest on 24,000,000 L.L. at 12% for two years. What is the amount ?  
Here  $P = 24,000,000$ ,  $r = 0.12$ , and  $t = 2$ . Then

$$\begin{aligned} I &= Prt = 24,000,000 \times 0.12 \times 2 \\ &= 5,760,000 \text{ L.L.} \end{aligned}$$

$$\text{and } S = P + I = 24,000,000 + 5,760,000 = 29,760,000 \text{ LL}$$

## Simple interest calculation

In general, the simple interest is used for short-term (less than one year). The time period is expressed in months or in days.

When the time is given in months,  $t$  is simply the number of months divided by 12.:

When the time is given in days, two different methods are in use for computing  $t$ . The most commonly used method is to divide the number of days by 365 (366 in leap years). When  $t$  is computed in this manner, the interest obtained is called exact simple interest. A second method in use is to divide the number of days by 360. When  $t$  is computed in this manner, the interest obtained is called ordinary simple interest.

**Example** Find the exact and ordinary simple interest on 2,000,000 L.L. for 50 days at 5%.

Exact simple interest:

$$\text{Using 365-day year, } t = 50/365 = 10/73$$

$$\text{And } I = Prt = 2000000(0.05)(10/73) = 13698.63 \text{ L.L.}$$

Ordinary simple interest:

$$\text{Using a 360-day year, } t = 50/360 = 5/36$$

$$\text{And } I = Prt = 2000000(0.05)(5/36) = 13888.89 \text{ L.L.}$$

The interest can be paid at the end of the borrowed date, or at the beginning of this date, as well as the interest can be paid monthly, or quarterly, or annually ...etc

The use of a 360-day year increases the interest collected by the lender of money.

# ACTIVITIES

## Single activity

Suppose that you want to deposit in your savings account a sum of \$ 10,000 for five years and that you have the choice between the two following formulas:

A- The annual interest rate is 10% of the principal (\$10,000).

B- The annual interest is added at the end of the year to produce with this principal an interest during the following year. The annual interest rate is 8% of the capital accumulated at the beginning of the year.:

1- Calculate according to formula A, the annual interest and deduce the total interest for a 5 years duration.

2- By supposing that you have chosen the formula B, fill the following table:

| End of the year | Interest | Capital + interest |
|-----------------|----------|--------------------|
| 1               | 800      | 10 800             |
| 2               | 864      | 11 664             |
| 3               | .....    | .....              |
| 4               | .....    | .....              |
| 5               | .....    | .....              |

3°) For the five years interval, which is the more advantageous formula?

4°) Is this report true for a period exceeding six years?

This type of interest is known as the simple interest and the percentage of 10% is the simple interest rate.

This type of interest is known as the compound interest, and percentage of 8% is the compounded annual rate

# TEXT

## I SIMPLE INTEREST

When money of value  $P$  on a given date increases in value to  $S$  at some later date,

$P$  is called the principal,

$S$  is called the amount ,

And

$I = S - P$  is called the interest.

The interest is usually calculated according to an **annual rate of interest**, denoted by  $r$ , and given as a percentage (6%, for example) or as the equivalent decimal fraction (0.06).

The borrower is also called debtor and the lender is called creditor

## II

## COMPOUND INTEREST

When interest is periodically added to the principal and this new sum is used as the principal for the following time period and this procedure is repeated for a certain number of periods, the final amount is called the compound amount. The difference between the compound amount and the original principal is called the compound interest.

The time period between two consecutive interest computation is called the interest period, or conversion period; and may be any convenient length of time.

The interest rate is usually quoted on a yearly basis and must be changed to the interest rate per interest period for computational purposes.

The conversion period is usually taken as an exact divisor of the year, such as a month, 3 months, 6 months, or a year.

### Compound amount

If  $P$  is the original principal, and  $r$  is the annual rate of the compound interest (or nominal rate) and if  $k$  is the number of interest periods per year, we take  $i = \frac{r}{k}$ .

The amount at the end of the first period is

$$S_1 = P + iP = P(1 + i).$$

The amount at the end of the second period is

$$S_2 = S_1 + iS_1 = P(1+i) + iP(1+i) = (1+i)(P+iP) = P(1+i)^2.$$

By writing  $S_3 = S_2 + iS_2$ , on obtient  $S_3 = S(1+i)^3$ .

and continuing in this manner for  $n$  periods gives the final compound amount as :

$$S_n = P(1+i)^n$$

The equation above is called the fundamental formula of compound interest.

If the conversion period is a year we have  $k = 1$ .

If it is a month then  $k = 12$

If it is a quarter then  $k = 4$

If for example  $r = 12\%$  and if the conversion period is the month

$$\text{then } i = \frac{0.12}{12} = 0.01.$$

This means that the monthly rate of the compound interest is 1%.

This formula allows us to obtain the present value  $P$  of an amount  $S$  due in  $n$  interest period at  $i$  interest per interest period

$$P = \frac{S}{(1+i)^n}$$

### ❖ Example

Find the compound amount of \$10,000 for 3 years at an annual rate of 10% in the following cases:

- 1-The conversion period is one year,
- 2-The conversion period is one quarter,
- 3-The conversion period is one month.

$$1^\circ) r = 10\% = 0.1; \quad k = 1; \quad i = 0.1; \quad n = 3.$$

$$V_3 = 10\,000 (1 + 0.1)^3 = 13\,310 \text{ \$}.$$

$$2^\circ) r = 0.1; \quad k = 4; \quad i = \frac{0.1}{4} = 0.025; \quad n = 3 \times 4 = 12$$

(12 periods in 3 years).

$$V_3 = 10\,000 (1 + 0.025)^{12} \approx 13\,448.89 \text{ \$}.$$

$$3^\circ) r = 0.1; \quad k = 12; \quad i = \frac{0.1}{12}; \quad n = 3 \times 12 = 36.$$

$$V_3 = 10\,000 \left(1 + \frac{1}{120}\right)^{36} \approx 13\,481.82 \text{ \$}.$$

These results show that for the same nominal rate, the compound interest is larger if the period of conversion is smaller.

## Compound interest

The compound interest  $I$  corresponding to a duration of  $n$  periods is the difference between the compound amount  $V_n$  and the principal  $P$ .

$$I = V_n - P$$

## III ANNUITIES

An annuity is a sequence of periodic payments, usually equal, made at equal intervals of time.

The period of time between consecutive payments is called the payment interval and may be any convenient length.

The term of an annuity is the time from the beginning of the first payment interval to the end of the last.

When the payments are due or made at the ends of the payment intervals, the annuity is called an ordinary annuity. When the payments are due or made at the beginnings of the payment intervals, the annuity is called an annuity due.

Familiar examples of annuities are:

- Monthly payments of rent.
- Annual premiums on life insurance.
- Semiannual interest payments.



Hereafter we will use the word annuity to mean an ordinary annuity..

## Amount of an annuity

The amount  $S$  of an annuity is the sum of compound amounts of the several payments each accumulated to the end of the term. Thus the Amount  $S$  of an annuity of  $n$  payments of a fixed amount  $R$  at an interest rate per period  $i$  is given by the formula:

$$S = R \frac{(1+i)^n - 1}{i}$$

When the payment interval and the interest conversion period coincide, the annuity is called a simple annuity; otherwise, a general annuity. In this chapter, we deal only with simple annuities.

### ❖ Example

Sonia deposits \$1500 at the end of each month on a savings account at an interest rate of 10%, the interest being compounded monthly.

- a- How much money is there in her account after 3 years?
- b- Which is the total amount of interest earned after 3 years?

a) We have  $i = \frac{r}{k} = \frac{0.1}{12} \cong 0.0083$ .  $n = tk = 3 \times 12 = 36$ .

Where we have  $S_{36} = 1500 \times \frac{1.0083^{36} - 1}{0.0083} \cong 62\,634 \$$ .

b)  $I = 62\,634 - 36 \times 1500 = 8634 \$$ .

## Present value of any annuity

The present value  $A$  of an annuity is the sum of the present values of the several payments. It represents the amount of money, which it should be placed now so that it has, at the end of the term of the annuity, a present value equal to the total amount of this annuity.

To obtain the present value  $A$  of an annuity we note that:

$$S = A(1+i)^n$$

$$\text{Thus } A = S(1+i)^{-n}$$

This gives :

$$A = R \frac{1 - (1+i)^{-n}}{i}$$



### ❖ Example

Find the present value of an annuity of \$200 per month during 3 years at an annual rate of interest 9% compounded monthly :

$$i = \frac{r}{k} = \frac{0.09}{12} = 0.0075 ; n = 3 \times 12 = 36$$

$$A_{36} = 200 \frac{1 - 1.0075^{-36}}{0.0075} \approx 6289.36 \$.$$

## FOCUS

- ❖ The interest  $I$  is the income from invested money, or it is the fee paid for the use of money.
- The principal  $P$  or present value is the amount of the money borrowed or invested.
- The annual interest rate  $r$  is the interest earned by a principal of amount 1 L.P. (or \$1, or 1 F.F....) for a time period of one year. It is expressed as a percentage or the equivalent decimal.  
(Ex:  $r = 0.12$  or  $r = 12\%$ ).

- ❖ When the time is given in months,  $t$  is the number of months divided by 12; when the time is given in days,  $t$  is the number of days divided by 365.
- The amount  $S$  is the sum of the principal and of the interest:  $S = P + I$
- In the case of a simple interest with annual interest rate  $r$  and a time period  $t$  expressed in years, the amount is  $P = (1 + rt)$ .

- ❖ A loan is said to be at a compound interest if the interest is calculated and added periodically to the principal. If  $P$  is the principal and  $i$  is the interest rate per period then the compound amount at the end of  $n$  interest periods is  $S = P(1+i)^n$  and the compound interest is  $I = S - P$  where  $i = \frac{r}{k}$ ,  $k$  being the number of interest periods per year and  $n = kt$  ( $r$  is the annual rate of the interest and  $t$  is the period in years).

- ❖ An annuity is a sequence of equal periodic payments.
- The amount  $S$  of an annuity of  $n$  payments of a fixed amount  $R$  due at the end of each period at an interest rate  $i$  per period is given by the formula

$$S = R \frac{(1+i)^n - 1}{i}.$$

- The present value  $A$  of an annuity of  $n$  fixed payments  $R$  due at the end of every period at a given interest rate per period  $i$  is given by the following formula:

$$A = R \frac{1 - (1+i)^{-n}}{i}.$$

# EXERCISES

- 1 Calculate the simple interest paid for a loan of \$6,000 for a period of 3 years at an interest rate of 10% per year.
- 2 A capital of \$2,000 is deposited in a bank for 4 years. The interest rate is 7.5% per year. Calculate the simple interest gained.
- 3 Calculate the simple interest on a capital of 5,000,000 L.L. for a period of 3 months with an interest rate of 6.5% per year.
- 4 Calculate the simple interest on a capital of \$60,000 for a 60 days period at an interest rate of 12.25% per year.
- 5 Calculate the simple interest paid for a loan of \$12,000 borrowed the 10 December 1998 and terminated the 16<sup>th</sup> of March of the coming year, at an interest rate of 10.75% per year.
- 6 Leyla borrowed \$2,500 for 6 months at a simple interest rate of 12.5% per year. How much must she refund at the end of the transaction?
- 7 On January 6, 1999, \$5,000 is borrowed, and it is repaid on September 12 of the same year at a simple interest rate of 12.25% per year. Find the amount.
- 8 Find  $k$ ,  $i$ ,  $n$  and  $S$  in each of the following cases:
  - i-  $P = 3000\$$ ,  $r = 5\%$  per year,  $t = 2$  years and the interest is compounded annually;
  - ii-  $P = 4000\$$ ,  $r = 6\%$  per year,  $t = 4$  years and the interest is compounded quarterly;
  - iii-  $P = 4000\$$ ,  $r = 6\%$  per year,  $t = 3$  years and the interest is compounded monthly.
- 9 Maria deposit \$1400 on a savings account of 7% compounded annually. The period of conversion is 1 quarter.
  - i- Determine the amount on her account after 4 years.
  - ii- What is the interest amount gained by that capital.
- 10 Walid invested a capital of \$200 at an annual interest rate of 6% compounded monthly.
  - i- Determine the compound amount after 6 years.
  - ii- What is the interest amount gained by that capital?
- 11 Find the compound amount of \$10,000 for 2 years at an annual interest rate of 10% compounded monthly. Determine the compound interest..

b) The set  $I$  has an identity element, it is the number 1, while the set  $P$  doesn't have any.

c)  $P$  is obviously not a group (it doesn't have an identity element) and  $I$  is not a group either, because apart from the numbers 1 and -1, the other elements of  $I$  do not have an inverse in  $I$ .

$$2. a) \quad 0 * 1 = \frac{0+1}{2} = \frac{1}{2}$$

$$0 * 2 = \frac{0+2}{2} = 1$$

$$0 * (-1) = \frac{0-1}{2} = -\frac{1}{2}$$

$$0 * \left(-\frac{1}{2}\right) = \frac{0-\frac{1}{2}}{2} = -\frac{1}{4}$$

$$1 * 1 = \frac{1+1}{2} = 1$$

$$1 * (-1) = \frac{1-1}{2} = 0$$

$$\frac{1}{2} * \frac{1}{3} = \frac{\frac{1}{2} + \frac{1}{3}}{2} = \frac{\frac{3}{6} + \frac{2}{6}}{2} = \frac{\frac{5}{6}}{2} = \frac{5}{12}.$$

b) The operation  $*$  does not define a binary operation, neither on  $\mathbb{N}$ , nor on  $\mathbb{Z}$ .

c) The operation  $*$  defines on  $\mathbb{Q}$  a binary operation. This operation is commutative, but it is not associative, in fact :

$$(a*b)*c = \frac{(a*b)+c}{2} = \frac{\frac{a+b}{2}+c}{2} = \frac{a+b+2c}{4} \text{ and}$$

$$a*(b*c) = \frac{a+(b*c)}{2} = \frac{a+\frac{b+c}{2}}{2} = \frac{2a+b+c}{4}.$$

If we take  $a = 0$ ,  $b = 0$  and  $c = 1$ , we have  $(a*b)*c = \frac{1}{2}$  and  $a*(b*c) = \frac{1}{4}$ .

The operation  $*$  does not have an identity element, because 0 is not an identity element, and any other element  $a \neq 0$  is not an identity element because we have,

$$a*2a = \frac{a+2a}{2} = \frac{3}{2}a \neq 2a \quad (a \text{ being } \neq 0).$$

## Chapter 3.

1.      a)      False                      b) False                      c) True  
             d)      False                      e) False

7 Riad buys a car which costs \$15,000 . He paid \$3,000 cash and must carry out fixed monthly payments, at the end of each month, during 3 years and a half. What is the amount of this monthly payment knowing that the annual interest rate is 12.5%

8 If \$ 15,000 is used to achieve an annuity made up of equal payments at the end of each year during the three next years. What is the amount of each payment knowing that the annual interest rate is 8%?

9 Suppose that after having made a payment of \$3000 cash to buy a car, Salem must carry out a monthly constant payment of \$350 for four consecutive years. The annual interest rate is 10%  
i- Find the original cost of the car.  
ii- Determine the amount of interest paid

## JUST FOR FUN

Aunt Nadya borrowed to the housing bank a sum of 60,000,000 LP to buy a house. The annually interest rate is 11%. To pay back this loan, she must make a constant monthly payment for 10 years. After 5 years, she has earned the first prize at the Lebanese Loto and she want to pay off this debt. How much must she pay?

# SELF EVALUATION ANSWERS AND INDICATIONS

## Chapter 1.

1. a) and d).

2. ❖ We know  $[(p' \Rightarrow q) \Leftrightarrow (\neg p' \vee q)]$  is a tautology. It's enough to replace  $p'$  by  $\neg p$ , since  $\neg p'$  is  $p$ . we can also use the truth table.

❖ The truth table of this second proposition is written, by denoting it by  $r$ :

| $p$ | $q$ | $(p \Rightarrow q)$ | $\neg(p \Rightarrow q)$ | $\neg q$ | $p \wedge \neg q$ | $r$ |
|-----|-----|---------------------|-------------------------|----------|-------------------|-----|
| T   | T   | T                   | F                       | F        | F                 | T   |
| T   | F   | F                   | T                       | T        | T                 | T   |
| F   | T   | T                   | F                       | F        | F                 | T   |
| F   | F   | T                   | F                       | T        | F                 | T   |

3. a) Let  $p$  be the proposition "I go out a lot" and  $q$  the proposition "I am not studying sufficiently" and  $r$  the proposition "I skipped my exam". We have  $p \Rightarrow q$  (if  $p$  is true then  $q$  is true) and  $q \Rightarrow r$  (if  $q$  is true then  $r$  is true), therefore  $p \Rightarrow r$  (if  $p$  is true then  $r$  is true). But this doesn't mean necessarily that  $r \Rightarrow p$  ( $r$  can be true without  $p$  being). Therefore the reasoning is not valid.

b) Let  $p$ : "Rashid is tired",  $q$ : "Rashid is ill" and  $r$ : "Rashid is not unbearable".

We have,  $(p \vee q)$  is true and  $p \Rightarrow r$  (if  $p$  is true then  $r$  is true) therefore  $\neg r \Rightarrow \neg p$  (if  $\neg r$  is true then  $\neg p$  is true). So if Rashid is unbearable ( $\neg r$  is true), then he is not tired ( $\neg p$  is true) and therefore  $p$  is false.

It means that  $q$  is true because  $(p \vee q)$  is true. Which means that Rashid is ill. The reasoning is then valid.

## Chapter 2.

1. a) Yes, the usual multiplication defines a binary operation on  $I$  and  $P$  (the product of two odd numbers is odd, and that of two even numbers is even).

## SELF-EVALUATION

- 1 A sum of 12,000,000 L.L. is deposited in a bank at a simple interest rate of 10% per year. What is the amount of this principal at the end of 150 days?
- 2 Calculate the compound interest gained by a capital of \$5000 placed at an annual interest rate of 8% for a period of 9 months, the conversion period is one month.?
- 3 Let us suppose that you deposit on your savings account a sum of 100,000 L.L. at the end of each month at an annual rate of 12%. What is the amount in your account after 3 years (the interest is compounded monthly)?

## PROBLEMS

- 1 A capital of 3,000,000 L.L. brings to Carole 45,000 L.P. as interest on the first three months of the year. Find the annual simple interest..
- 2 Suppose that you want to borrow \$1500 at an annual simple rate interest of 15% to buy a computer. If you are going to pay \$225 as interest, find the duration of the loan.?
- 3 How much money must be deposited now in your savings account so that in 9 months the amount of your account rises to \$20,000. The annual simple interest rate is 10%..
- 4 Determine the amount of money which must be placed now by a person 40 years old , to have at the time of retirement at 65 years a sum of \$100,000. The interest is compounded quarterly at an annual rate of 10%.
- 5 A bank pays an interest compounded annually at the rate of 8%. Another bank pays an interest compounded monthly at a rate of 6.5%. Find the difference between the interests gained for a capital of \$25,000 after 6 years.
- 6 What is the most advantageous rate from the following:  
i- 12.8% with a monthly compounded interest  
ii- 13% with a quarterly compounded interest  
iii- 13.5% with a annual compounded interest

- 12 Find the amount for each of the following annuities where we indicate in each case the amount of each payment  $R$ , the period, the annual rate  $r$ , and the term of the annuity.

i-  $R = \$5000$  paid annually,  
 $r = 4\%$ , 5 years

ii-  $R = \$4000$  paid annually,  
 $r = 6\%$ , 4 years

iii-  $R = \$400$  paid monthly,  
 $r = 5\%$ , 2 years.

- 13 To buy a house in four years Fouad deposits a sum of \$1000 at the end of each month in a bank, which pays interest at the annually rate of 4% compounded monthly.

i- Calculate the amount in his account at the end of the fourth year.

ii- Calculate the interest gained by all these payments.

- 14 Suppose that you deposit on your savings account a sum of \$400 at the end of each month for a period of 2 years. The interest is compounded monthly at an annual rate of 8%.

i- Find the amount of the money you would have saved at the end of the 2 years?

ii- Find the amount of the interest gained?

- 15 Find the present value of an annuity of \$500 per month for the 2 coming years at an annual interest rate of 6% (the interest is compounded monthly).

- 16 Calculate the present value of a \$8000 annuity paid annually for the five years to come at a compounded annually interest rate of 5%. (Interest is compounded annually).

- 17 Suppose that you want to buy an annuity, which will gain \$900 per month for the three following years (the interest is compounded monthly at an annual rate of 10%). How much must you pay now to achieve this annuity?

## TRUE OR FALSE

Answer by true or false

i- In the case of simple interest, the principal gains an interest proportional to the time period.

ii- The present value  $P$  and the amount  $S$  are related by the formula  $P = S(1 + rt)$ .

iii- The conversion period is the interval of time located between two successive additions of interest.

iv- If the interest is compounded quarterly, then the conversion period is 4 months.

v- The compound interest is generally used for the long-term loans and the simple interest is generally used for those in the short run.

vi- The compound interest gives back the lender less money than the simple interest when the rate is the same.

vii- The payment of an annuity can be monthly, quarterly, or annually.

viii- The interest calculation in an annuity is carried out using compound interest.

2. It is the median 12 (the average is 16.8 and there is no mode).

3. The mode is 6, the median is 7 and the mean is 8. The highest concentration is corresponding to the mode.

## Chapter 4.

1. a) The function  $x \mapsto f(x) = x^2 - 5x + 4$  is defined on  $\mathbb{R}$  and

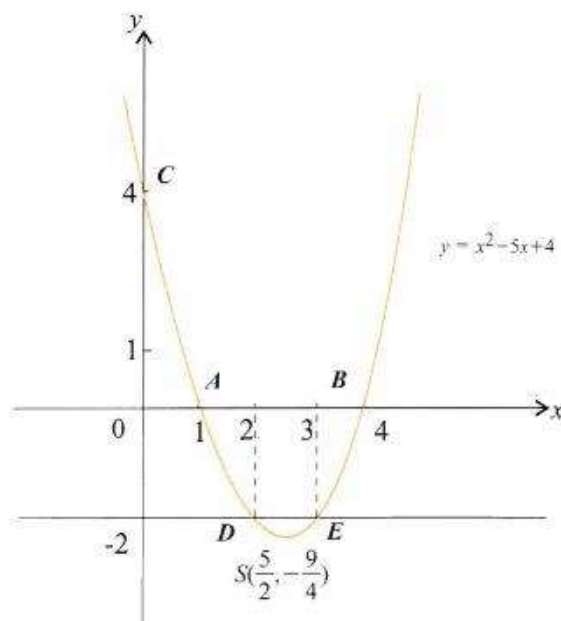
$$\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = +\infty.$$

$$f'(x) = 2x - 5 = 0 \Leftrightarrow 2x = 5 \Leftrightarrow x = \frac{5}{2}.$$

$$f(5/2) = \frac{25}{4} - \frac{25}{2} + 4 = -\frac{9}{4}.$$

$$f'(x) > 0 \text{ for } x > \frac{5}{2} \text{ and } f'(x) < 0 \text{ for } x < \frac{5}{2}.$$

|         |           |                |           |
|---------|-----------|----------------|-----------|
| $x$     | $-\infty$ | $\frac{5}{2}$  | $+\infty$ |
| $f'(x)$ | $-$       | $0$            | $+$       |
| $f$     | $+\infty$ | $-\frac{9}{4}$ | $+\infty$ |



- b) ♦ The points of intersection with the  $x$  axis are  $A(1, 0)$  and  $B(4, 0)$ .  
 ♦ The point of intersection with the  $y$  axis is  $C(0, 4)$ .  
 ♦ The points of intersection with the straight line  $y = -2$  are  $D(2, -2)$  and  $E(3, -2)$ .

c) ♦  $x^2 - 5x + 4 = 0 \Leftrightarrow x = \frac{5 + \sqrt{25-16}}{2}$  or  $x = \frac{5-3}{2} \Leftrightarrow x = 4$  or  $x = 1$ .

♦  $f(0) = 4$ .

♦  $x^2 - 5x + 4 = -2 \Leftrightarrow x^2 - 5x + 6 = 0 \Leftrightarrow x = \frac{5 + \sqrt{25-24}}{2}$  or

$x = \frac{5-1}{2} \Leftrightarrow x=3$  or  $x=2$ .

2. ♦ The function  $x \mapsto f(x) = \frac{x+2}{x-1}$  is defined for  $x \neq 1$ .

♦  $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = 1$ .

$\lim_{\substack{x \rightarrow 1 \\ x > 1}} f(x) = +\infty$

and

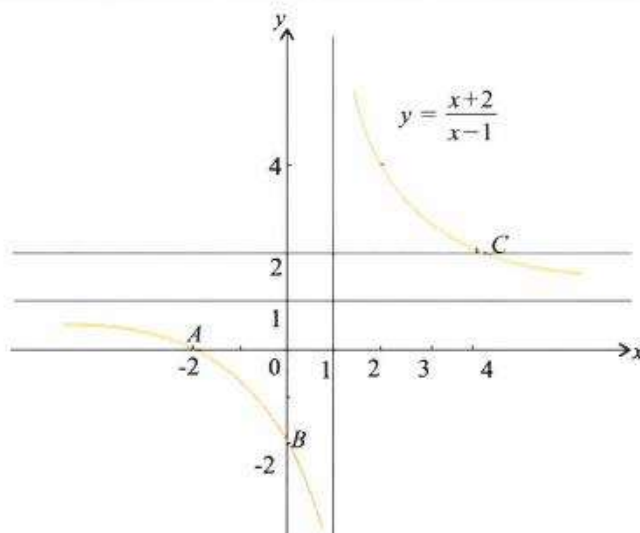
$\lim_{\substack{x \rightarrow 1 \\ x < 1}} f(x) = -\infty$

The straight lines with equations  $y = 1$  and  $x = 1$  are asymptotes to the curve.

- ♦ For  $x \neq 1$ , we have

$$f'(x) = \frac{1 \times (x-1) - (x+2) \times 1}{(x-1)^2} = -\frac{3}{(x-1)^2} < 0.$$

|         |           |            |           |            |           |
|---------|-----------|------------|-----------|------------|-----------|
| $x$     | $-\infty$ |            | $1$       |            | $+\infty$ |
| $f'(x)$ |           | $-$        |           | $-$        |           |
| $f$     | $1$       | $\searrow$ | $-\infty$ | $\nearrow$ | $1$       |



$$2. \bar{x} = 70$$

$$\nu = \frac{4950}{16}$$

$$\sigma = \sqrt{\frac{4950}{16}} \approx 17.6.$$

$$3. \bar{x}_1 = \bar{x}_2 = 10$$

$$\sigma_1 = \sqrt{\frac{112}{7}} = \sqrt{16} = 4$$

$$\sigma_2 = \sqrt{\frac{92}{7}} \approx 625.$$

As long as the standard deviation of  $S_2$  is smaller than that of  $S_1$ , the values of  $S_2$  are less dispersed around their average than the ones of  $S_1$ .

## Chapter 8.

$$1. D_f = [0, 7[.$$

2.  $\diamond f$  is increasing on  $[0, 2]$ , decreasing on  $[2; 4.5[$  and increasing on  $[4.5; 7[$ .

$\diamond$  For  $x = 2$ , there is a local maximum equal to 3.

$\diamond$  For  $x = 4.5$ , there is a local minimum equal to 1.

$\diamond$  The absolute minimum is 0, it is for  $x = 0$ .

$\diamond$  There is no absolute maximum.

$$4. \lim_{x \rightarrow 0} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 7} f(x) = +\infty.$$

The asymptote equation is  $x = 7$ .

5. If  $0 < m < 1$ , There is only one solution.

If  $m = 1$ , There are 2 solutions.

If  $1 < m < 3$ , There are 3 solutions.

If  $m = 3$ , There are 2 solution .

If  $m > 3$ , There is only one solution.

$$6. f(x) < 3 \Leftrightarrow x \in [0, 2[ \cup ]2, 6[.$$

## Chapter 9.

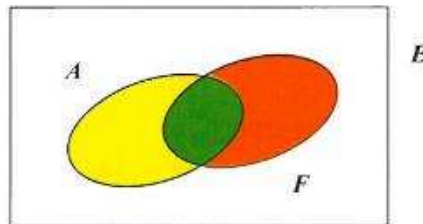
1.

|                             |               |               |                         |   |               |   |   |   |
|-----------------------------|---------------|---------------|-------------------------|---|---------------|---|---|---|
| $x$                         | -6            | -3            | -1                      | 0 | 1             | 3 | 6 | 9 |
| $\frac{x}{2^{\frac{x}{3}}}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{\sqrt[3]{2}}$ | 1 | $\sqrt[3]{2}$ | 2 | 4 | 8 |

## Chapter 6.

1. No, because  $P(M) \times P(H) = 0.6 \times 0.8 = 0.48$  is different then  $0.5 = P(M \cap H)$ .

2. a) Let  $A$  be the set of English studying students,  $F$  the set of French studying students and  $E$  the set of all the students. We have  $E = A \cup F$  and  $\text{card } E = \text{card } A + \text{card } F - \text{card } (A \cap F)$ , then  $\text{card } (A \cap F) = 18 + 24 - 32 = 10$ .



b)  $1^\circ P(A) = \frac{18}{32} = \frac{9}{16}$ .

$2^\circ P(A \cap F) = \frac{10}{32} = \frac{5}{16}$ .

$3^\circ P(F/A) = \frac{P(A \cap F)}{P(A)} = \frac{\frac{10}{32}}{\frac{18}{32}} = \frac{10}{32} \times \frac{32}{18} = \frac{10}{18} = \frac{5}{9}$ .

$4^\circ P(A/F) = \frac{P(A \cap F)}{P(F)} = \frac{\frac{10}{32}}{\frac{24}{32}} = \frac{10}{32} \times \frac{32}{24} = \frac{10}{24} = \frac{5}{12}$ .

3. Let  $F$  be the event "the chosen student is a girl",  $G$  the event "the chosen student is a boy" and  $B$  the event "the student plays Basketball".

We have  $P(F) = 0.6$ ,  $P(B/G) = 0.4$  and  $P(B/F) = 0.25$ .

a)  $P(B \cap G) = P(B/G) \times P(G) = 0.4 \times 0.4 = 0.16$ .

b)  $P(\bar{B} \cap F) = P(\bar{B}/F) \times P(F) = 0.75 \times 0.6 = 0.45$ .

## Chapter 7.

1.  $1^\circ F$   $2^\circ T$   
 $3^\circ F$   $4^\circ T$   
 $5^\circ F$   $6^\circ T$ .

- b) ❖ The point of intersection with the  $x$  axis is  $A(-2, 0)$ .  
 ❖ The point of intersection with the  $y$  axis is  $B(0, -2)$ .  
 ❖ The point of intersection with the straight line  $y = 2$  is  $C(4, 2)$ .

c) ❖  $f(0) = -2$ .

❖  $f(x) = 0 \Leftrightarrow x + 2 = 0 \Leftrightarrow x = -2$ .

❖  $f(x) = 2 \Leftrightarrow \frac{x+2}{x-1} = 2 \Leftrightarrow x + 2 = 2x - 2 \Leftrightarrow x = 4$ .

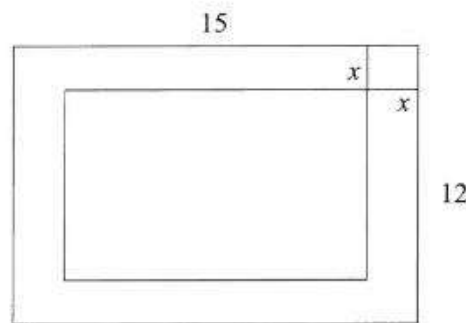
## Chapter 5.

1. If  $x$  is the width of the tape, we must have

$$(15 - 2x)(12 - 2x) = 108 \Leftrightarrow 180 - 30x - 24x + 4x^2 = 108$$

$$\Leftrightarrow 4x^2 - 54x + 72 = 0 \Leftrightarrow x = 1.5 \text{ or } x = 13.5.$$

the only acceptable solution is  $x = 1.5 \text{ m}$ .



2. Let  $x, y, z$  be the numbers of bills of 5000, 10 000 and 20 000 LL. respectively. The problem can be represented by the following linear system of 3 equations in 3 unknowns:

$$\begin{cases} x + y + z = 60. \\ 5000x + 10\,000y + 20\,000z = 500\,000. \\ x = 2(y + z). \end{cases}$$

The system resolution gives  $x = 40, y = 10$  and  $z = 10$ .

3. If  $y = ax^2 + bx + c$  is the equation of this parabola, we have:

$$\begin{cases} 1000 = 2500a + 50b + c \\ 1500 = 10\,000a + 100b + c \\ 3000 = 40\,000a + 200b + c. \end{cases}$$

Which gives  $a = 0.1; b = -5$  and  $c = 1000$  and the equation of the parabola is  $y = 0.1x^2 - 5x + 1000$ .



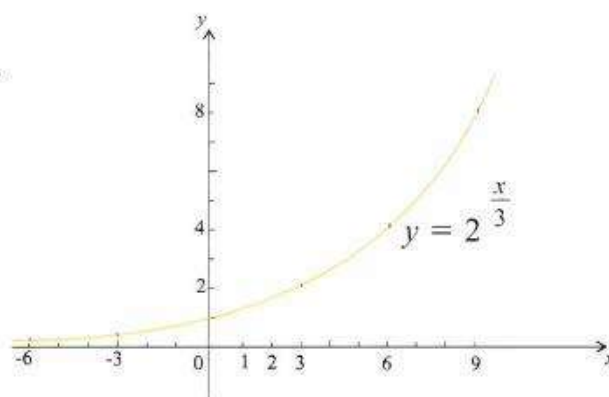
2. a)  $16^x = (2^4)^x = 2^{4x}$  and  $64 = 2^6$ .

$$2^{4x} = 2^6 \Leftrightarrow 4x = 6 \Leftrightarrow x = \frac{6}{4} = 1.5.$$

b)  $81^{-2x} = (3^4)^{-2x} = 3^{-8x}$ .

$$3^{x-1} = 3^{-8x} \Leftrightarrow x-1 = -8x \Leftrightarrow$$

$$9x = 1 \Leftrightarrow x = \frac{1}{9}.$$



3. ♦ On the 1<sup>st</sup> of January 1990, the city counts  $200\,000 \times e^0 = 200\,000$  inhabitants.  
 ♦ On the 1<sup>st</sup> of January 2010, it will count  $200\,000 \times e^{0.03 \times 20} = 200\,000 e^{0.6} = 364\,423$  inhabitants.

## Chapter 10.

1.  $S = P(1+rt)$  where  $C = 12\,000\,000$ ,  $r = 0.1$  and  $t = \frac{150}{365}$ .

$$S = 12\,000\,000(1 + 0.1 \times \frac{150}{365}) = 12,493,150.68 \text{ LL.}$$

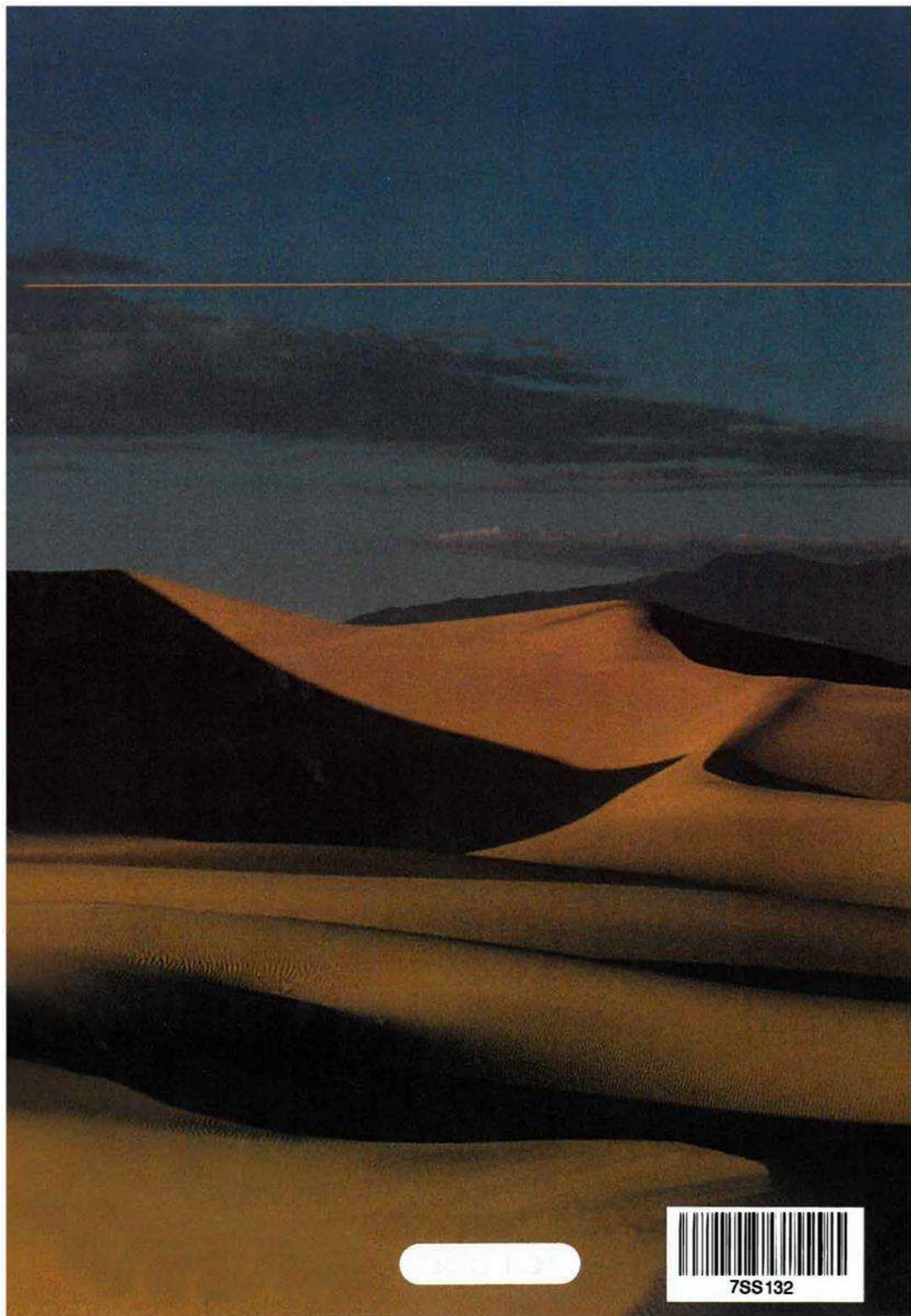
2.  $S_n = P(1+i)^n$  where  $P = 5,000$ ,  $i = \frac{r}{12} = \frac{0.08}{12}$  and  $n = 9$ .

$$S = 5,000(1 + \frac{0.08}{12})^9 = 5,308.1257 \text{ \$}.$$

3.  $S_n = R \frac{(1+i)^n - 1}{i}$ . Here  $R = 100\,000$ ,  $n = 3 \times 12 = 36$  and

$$i = \frac{0.12}{12} = 0.01.$$

$$S = 100\,000 \frac{(1.01)^{36} - 1}{0.01} = 4\,307\,687.84 \text{ LL.}$$



X 2 1 25

